

Multi-scenario optimization of future energy mix for strategic energy planning in Morocco

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ABSTRACT

Morocco seeks to reduce its long-standing dependence on imported fossil fuels, historically above 90.0% of primary energy supply, while exploiting substantial solar and wind resources to support long-term decarbonization. This study develops an integrated optimization framework coupling the Long-range Energy Alternatives Planning system (LEAP) with the North African Electricity Model Optimizer (NEMO) to evaluate alternative electricity and hydrogen transition pathways to 2050. Five scenarios are assessed: Business-as-Usual (BAU), Halfway to Clean Energy, National Energy Plan (two optimization variants), and Net Zero Emissions (NZE). Results reveal strongly differentiated trajectories. Under BAU, fossil fuels remain above 85.0% of generation, with renewable capacity below 20 GW. In contrast, NZE achieves full decarbonization by 2050, requiring approximately 70 GW of renewable capacity and producing 347,200 TJ of hydrogen annually. Intermediate policy-driven scenarios reduce fossil dependence to 33.0–38.0%, reach renewable shares above 80.0%, and deliver 134,400–173,600 TJ of hydrogen. Total system investment in NZE rises to 2,456 billion MAD, corresponding to about 3.1% of projected 2050 gross domestic product, while land requirements expand to 47,437 ha for solar photovoltaics and 34,720 ha for hydrogen infrastructure. Multi-criteria performance scores increase from 35/100 (BAU) to 98/100 (NZE), reflecting higher capital intensity but substantial environmental gains. All reported results represent model-based projections under specific assumptions, providing evidence that ambitious decarbonization pathways are achievable with supportive policies, strategic investment, and coordinated planning.

Introduction

Background

The strategic geography and ambitious energy goals of Morocco form a critical backdrop for this research. The country borders both the Atlantic Ocean and the Mediterranean Sea and lies only 14 kilometers from Europe across the Strait of Gibraltar, creating strategic opportunities for cross regional energy exchange [1]. Historically, the national energy mix has been dominated by imported fossil fuels, which have accounted for more than 90.0% of total energy use [2]. In response, the government introduced the National Energy Strategy in 2009, setting a target for renewables to supply 52.0% of installed electricity capacity by

2030 [3]. This transformation is central not only to environmental commitments but also to strengthening energy independence and easing economic pressures, as energy imports have represented more than 95.0% of the national trade deficit in previous years [4].

This domestic transition unfolds within a rapidly growing global renewable energy sector. Supported by climate policies and sustained cost reductions, worldwide installed renewable capacity reached approximately 4,448 GW by the end of 2024. Solar photovoltaic capacity accounted for about 2,200 GW, wind energy reached roughly 1,021 GW, and biomass-based electricity totaled nearly 97 GW. Between 2020 and 2024, global renewable capacity expanded by around 50.0%, indicating a structural redirection of investment toward low carbon technologies and confirming the financial competitiveness of large-scale renewable deployment [5].

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Nomenclature**Abbreviation**

BAU	Business-as-Usual
CSP	Concentrated Solar Power
GHG	Greenhouse Gas
HTCE	Halfway to Clean Energy
LEAP	Long-Range Energy Alternatives Planning System
MILP	Mixed-Integer Linear Programming
NDC	Nationally Determined Contribution
NEMO	Next Energy Modeling System for Optimization
NEP-NEMO	National Energy Plan - NEMO Optimized Mix
NEP-PULP	National Energy Plan -PULP Optimized Mix
NZE	Net Zero Emissions
PV	Photovoltaic

Greek Symbols

δ	Discount rate [%]
ρ	Degradation rate [%]
$\Delta_{i,t,h,n,s}^{CF}$	Improvement coefficient [-]
$\alpha_{i,t}$	Annuity factor for technology i [-]
β_i	Learning exponent [-]
$\epsilon_{f,t,s}$	Random error term representing stochastic fuel price shock for fuel type f at time t and scenario s [MAD/unit]
q_f	Emission factor [gCO _{2e} /kWh]
κ_i	Heat-rate conversion for thermal technology i [GJ/MWh]
$\{\eta_k^{chg}, \eta_k^{dis}\}$	Storage charge/discharge efficiencies [%]
π_s	Scenario probability [%]
$\mu_{f,s}$	Long-term mean fuel price for fuel type f under scenario s [MAD/unit]

Symbols

Δh	Duration of time-slice h [hours]
f	Index for fuel type [-]
i	Index for generation technology [-]
h	Index for intra-year time slice [-]
k	Index for generation technology [-]
l	Index for transmission line [-]
n	Spatial node [-]
r	Index for renewable resource type [-]
s	Scenarios index [-]
t	Time [calendar year]
t_0	Base time [calendar year]
$build_limit_{i,t,n,s}$	Maximum buildable capacity per site and year [MW/unit]
$c_{i,t}^{cap}$	Overnight capital cost [MAD/MW]
$c_{i,t}^{fix}$	Fixed O&M cost [MAD/MW]
$c_{i,t}^{var}$	Variable O&M cost [MAD/MW]
$exp_{t,h,n,s}$	Exported power [MW]
$f_{i,t,h,s}$	Power flow on line l [MW]
$g_{i,t,h,n,s}$	Power generation from source i [MW]

$imp_{t,h,n,s}$	Imported/exported power [MW]
$land_factor_i$	Land required per technology i [ha/MW]
$loss_n^{T\&D}$	Energy loss fraction from electric transmission and distribution at node n [%]
$national_build_limit_{i,t,s}$	National max build capacity [MW]
$p_{f,t,s}$	Fuel price of fuel f at year t , scenario s [MAD/unit]
$\{p_{k,t,h,n,s}^{chg}, p_{k,t,h,n,s}^{dis}\}$	Storage charge/discharge power of storage k [MW]
$\{p_{k,t,h,n,s}^{chg,max}, p_{k,t,h,n,s}^{dis,max}\}$	Maximum storage charge/discharge power of storage k [MW]
$pot_{r,n,t,s}$	Renewable resource potential [MWh/year]
$unmet_{t,h,n,s}$	Unmet load [MW]
$\{x_{i,t,n,s}^{add}, x_{k,t,n,s}^{rem}\}$	Added/retired capacity of technology i in year t , node n , and scenario s [MW]
M	Large penalty cost for unmet demand [MAD/MWh]
$\tilde{C}_{i,t,n,s}$	Derated available capacity for technology i at year t , node n , scenario s [MW]
$C_{i,t,n,s}$	Installed capacity stock [MW]
$CAPEX_{i,t,n,s}^{ann}$	Annualized capital expenditure for technology i [MAD]
$CF_{i,t,h,n,s}$	Capacity availability factor [%]
$CUM_{i,t,s}$	Cumulative installed capacity of technology i up to year t under scenario s [MW]
$D_{d,t,h,n,s}$	Electricity power demand for sector d at year t , slice h , node n , and scenario s [MW]
$D_{t,h,n,s}^{el}$	Total electricity power demand [MW]
$E_{t,s}$	Emissions in year t for scenario s [tCO _{2e}]
$E_{k,t,n,s}^{max}$	Storage energy capacity [MWh]
$Emissions^s$	Discounted emissions [tCO _{2e}]
F_l^{max}	Line thermal limit [MW]
$\widetilde{Fuel}_{t,s}^{cost}$	Total fuel-related cost at year t , scenario s [MAD]
$Fuel_{i,t,h,n,s}$	Fuel consumed by technology i [GJ or tonne]
$G_{i,t_0,n}^{exist}$	Existing capacity at base year [MW]
$Generation_Mix_{i,t}$	Annual generation by technology i [MWh/year]
$Installed_Capacity_Mix_{i,t}$	Installed capacity mix per technology i [MW]
L_i	Technical lifetime of technology i [years]
LR_i	Learning rate of technology i [-]
$Land_{i,t,n,s}$	Land used by technology i [ha]
$Land_{t,s}$	Total land used for renewable generation at year t , scenario s [ha]
$Land^s$	Discounted land requirement per scenario s [ha]
$LCOE_{i,t,s}$	Levelized cost of energy by technology i [MAD/MWh]
$\{O_i^{norm}, O_i^{max}, O_i^{min}\}$	Normalization objectives [-]
NPC^s	Net present cost per scenario s [MAD]
$P_{t,n,s}^{peak}$	Peak load [MW]
$RM_{t,s}$	Reserve margin requirement [%]
$SOC_{k,t,h,n,s}$	State-of-charge of storage k [MWh]

International momentum further underscores the urgency of decarbonization. Commitments under frameworks such as the Paris Agreement [6] have intensified national mitigation efforts. Morocco's Nationally Determined Contributions aim to reduce greenhouse gas emissions by 40.0% by 2030 and by up to 74.0% by 2050 [7]. At the same time, renewable technologies have become increasingly affordable, with utility scale solar costs falling by approximately 85.0% since 2010. Coupled with solar irradiation levels exceeding 2,500 kWh/m²/year in several regions and strong wind resources [8], these conditions position Morocco to satisfy domestic demand while developing a

competitive renewable electricity and green hydrogen sector.

Energy situation in Morocco

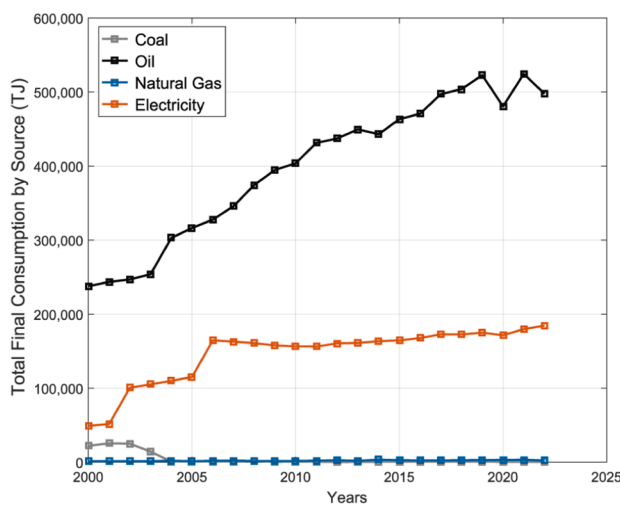
The energy situation in Morocco reflects a system at a critical turning point, characterized by persistent fossil fuel dependence alongside an accelerating low-carbon transition. Imported oil and coal still dominate the national energy mix. These fuels supported an almost threefold increase in electricity generation between 2000 and 2022 and were associated with a 127.0% rise in national greenhouse gas emissions

during the same period. At the same time, renewable electricity production expanded steadily, with wind and solar capacity increasing at an average annual rate of about 11.3%. This parallel evolution highlights the structural challenge of meeting rapidly growing energy demand while advancing toward the objectives of the Low Carbon Strategy, which envisions emission reductions of up to 74.0% by 2050.

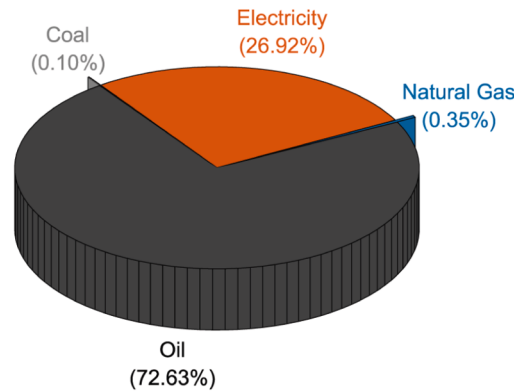
All historical datasets on energy use, electricity generation, and emissions were standardized before modeling. Source data were converted into consistent accounting frameworks, common energy units in joules (J), electricity units in watt hours (Wh), and emission indicators expressed as carbon dioxide equivalent (CO_2e). This harmonization ensured comparability over time across fuels, sectors, and performance indicators. To enhance reliability, national energy balances were cross checked against electricity generation records and greenhouse gas inventories, reducing inconsistencies related to system boundaries or methodological variation.

Current context

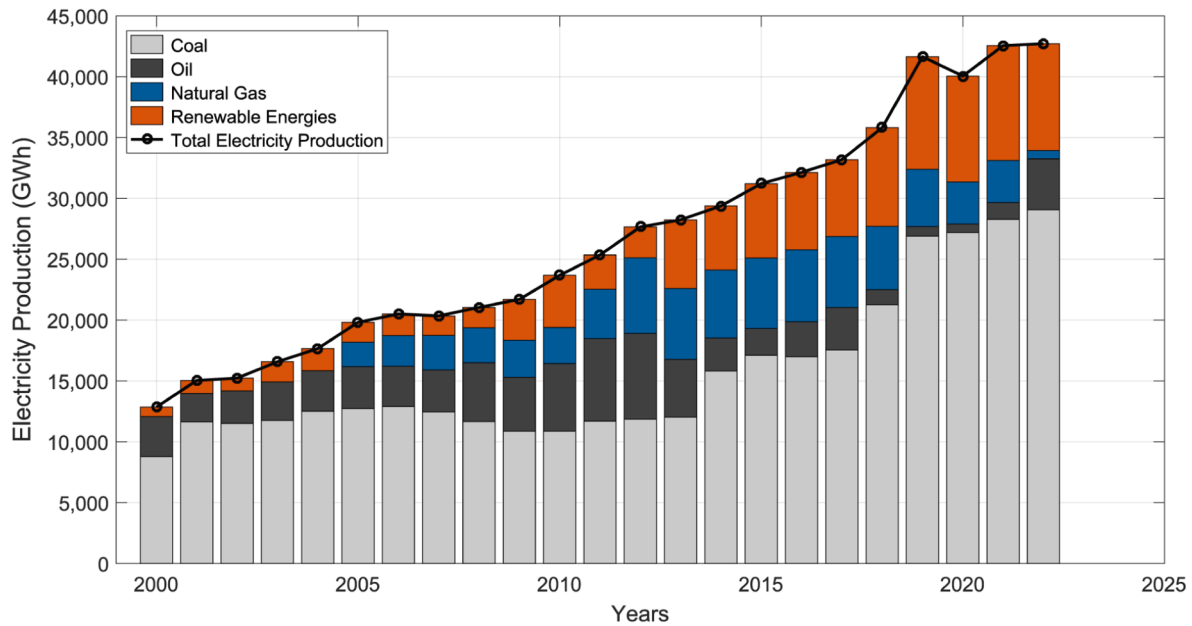
The current energy context of Morocco continues to be shaped by a strong dependence on fossil fuels, although renewable integration is progressing gradually. As illustrated in Fig. 1a, oil has consistently represented the largest share of final energy consumption between 2000 and 2022. Total oil use rose from 237,801 TJ in 2000 to more than 497,702 TJ in 2022, with temporary variations linked to international price volatility and shifts in demand. Electricity consumption increased markedly over the same period, climbing from 49,251 TJ in 2000 to a maximum of 184,483 TJ in 2022. This expansion reflects continued electrification across both urban and rural regions, together with industrial development [9]. Coal consumption remained comparatively modest, reaching about 653 TJ in 2022, and is mainly associated with industrial applications and thermal power generation. Natural gas use has been constrained and variable, ranging between 1,400 TJ and 3,400 TJ per year, largely because of limited domestic resources and dependence on imports [10]. The compound annual growth rate for the



(a) Historical Energy Consumption by Source from 2000 to 2022



(b) Total Energy Consumption in 2022



(c) Evolution Pattern of Electricity Production by Source from 2000 to 2022

Fig. 1. Evolution pattern of energy consumption by energy source in Morocco.

indicators analyzed in this period is calculated using the standard expression presented in (1), where $V_{initial}$ and V_{final} represent the starting and ending values, and n denotes the number of years considered. This calculation method is applied consistently throughout the manuscript for energy consumption, electricity generation, and emissions variables.

$$CAGR = \left[\left(\frac{V_{initial}}{V_{final}} \right)^{\frac{1}{n}} - 1 \right] \times 100\% \quad (1)$$

Fig. 1b further indicates that total final energy consumption in 2022 remained heavily concentrated in oil, representing 78.9% of the overall mix. Electricity accounted for 20.7%, while coal and natural gas held very limited shares of 0.2% and 0.3%, respectively. Fig. 1c illustrates the evolution of the electricity generation structure between 2000 and 2022. Total output increased from 12,863 GWh in 2000 to 42,719 GWh in 2022, corresponding to a compound annual growth rate of 5.5% over a 22-year period, accompanied by noticeable changes in the composition of generation sources. Electricity produced from coal expanded from 8,783 GWh to 29,064 GWh, with a compound annual growth rate of 5.6%, maintaining its position as the largest contributor in 2022 at nearly 68.0% of total generation. In contrast, electricity derived from oil rose modestly from 3,298 GWh in 2000 to 4,186 GWh in 2022, representing nearly 9.8% of total output in that year, indicating a partial transition away from oil in power production. Natural gas, which was virtually absent in the initial years, displayed intermittent development and reached 686 GWh in 2022, accounting for about 1.6% of total generation. When 2005 is considered as a reference point at 2,003 GWh, gas-based generation reflects an overall decline, largely associated with supply interruptions following the cessation of the Maghreb Europe pipeline. Renewable electricity experienced the most pronounced expansion, rising from 782 GWh, approximately 6.1% of total generation in 2000, to 8,783 GWh, around 20.6% of the total in 2022, with an average annual growth rate of roughly 11.3%. Overall, these indicators show a power sector that remains strongly reliant on coal while simultaneously experiencing accelerated renewable deployment and gradual substitution of oil with gas and renewable sources [11].

The trajectory of renewable electricity generation between 2000 and 2022 reflects a profound transformation of the power sector, characterized by technological expansion and increasing geographic diversification through large scale solar, wind, and hydro initiatives. As illustrated in Fig. 1c, total renewable output expanded more than thirteen times, rising from 782 GWh in 2000 to 8,783 GWh in 2022. This growth was largely driven by the rapid deployment of wind and solar capacity, while hydropower output declined in response to climatic variability. Hydropower, which previously represented the principal renewable source, reached a maximum of 3,631 GWh in 2010 but decreased sharply to 679 GWh in 2022, corresponding to a reduction of 81.3%. This contraction is mainly attributed to repeated drought episodes and lower reservoir inflows that constrained dam-based generation. Wind energy recorded the most significant expansion, increasing from 64 GWh in 2000 to 5,355 GWh in 2022 and accounting for approximately 61.0% of total renewable generation. This progression was supported by major projects such as Tafaya with 301 MW, Tangier II with 150 MW, and Midelt with 210 MW. Solar electricity, almost absent before 2015, expanded rapidly following implementation of the Noor Program. Concentrated solar power generation rose from 4 GWh in 2015 to 863 GWh in 2022, while photovoltaic generation increased from 1 GWh to 584 GWh during the same period. Together, solar technologies became the second largest renewable contributor, representing about 16.0% of renewable output. In parallel, bioenergy including biofuels and biomass has delivered a stable contribution of 1,302 GWh annually since 2013, reflecting the use of agricultural and agro-industrial residues in combined heat and power systems [10].

Greenhouse gas emissions are examined progressively, beginning with total national emissions, followed by per capita values, and concluding with fuel and sector specific contributions. Although the

global share remains limited at approximately 0.2%, equivalent to 68 MtCO_{2e} per year, the domestic emissions pathway reveals substantial internal pressure linked mainly to energy system transformation. At the aggregate level, Fig. 2a indicates that total emissions rose by 127.0% between 2000 and 2022, reflecting sustained increases in energy demand associated with economic growth, urban expansion, and demographic change [7].

When expressed on a per capita basis, the upward pattern remains evident. Emissions per inhabitant increased by 73.0%, rising from 1.035 to 1.793 tCO_{2e} per capita over the same period, placing the country ninth among African nations. This increase demonstrates that emissions growth cannot be attributed solely to population expansion, but also to higher energy intensity and evolving consumption structures, particularly in urban, industrial, and transport activities [9].

The composition of emissions provides further structural insight. As shown in Fig. 2b, the emissions profile is dominated by fossil fuels, with oil contributing 57.1% corresponding to 38.2 MtCO_{2e} and coal accounting for 42.2% or 28.1 MtCO_{2e}, while natural gas represents only 0.7%. This distribution highlights the continued carbon intensity of the energy mix despite expanding renewable deployment. From a sectoral standpoint, electricity generation constitutes the largest source, responsible for 47.6% corresponding to 32.0 MtCO_{2e} of national emissions in 2022, followed by transport at 26.1% or 17.6 MtCO_{2e} [7]. Together, these findings indicate that substantial emissions reduction depends primarily on accelerating decarbonization in power generation [12] and transport systems [13–15], where structural dependencies remain most pronounced.

In summary, the national energy and environmental landscape is shaped by the combined effects of increasing demand, strong reliance on fossil fuels, and constrained domestic resource availability, which together intensify the sustainability challenge. Annual energy consumption is approximately 936,000 TJ, placing the country within the intermediate range of energy users at both regional and international levels. Owing to limited indigenous fossil reserves, about 94.0% of primary energy requirements are satisfied through imports. In 2022, energy expenditures reached 11.5 billion USD, maintaining pressure on foreign currency reserves and affecting energy security. Electricity represents roughly 27.0% of final energy use, with demand continuing to expand alongside progressive electrification across end use sectors. Total electricity generation in 2022 was about 43 TWh, a level still associated with substantial carbon dioxide emissions despite significant renewable growth over the last twenty years. The simultaneous expansion of renewable capacity and the 127.0% increase in greenhouse gas emissions since 2000 reveal persistent structural dependence on fossil fuels within the energy system. This combination of environmental stress and import reliance underscores the necessity of advancing low carbon strategies that diversify the energy portfolio, improve trade balance stability, and reinforce long term energy autonomy.

Low carbon strategy

In June 2021, Morocco revised its climate pledge under the United Nations Framework Convention on Climate Change, updating its Nationally Determined Contribution to reflect enhanced ambitions. The revised NDC establishes a target of reducing greenhouse gas emissions by 40.0% by 2030 compared with a no-policy baseline and envisions a long-term reduction of 74.0% by 2050 under strengthened climate measures, as shown in Fig. 3. These objectives are structured through three overarching narratives: Business-as-Usual, Increased Ambition, and Green Development. These narratives function as strategic policy pathways, providing qualitative guidance and indicative investment priorities for steering the national transition to a low-carbon economy, rather than representing fully optimized energy system scenarios.

Under the NDC-defined Business-as-Usual (BAU) pathway, the energy system of Morocco evolves under current policy and investment trends without structural reform in fossil fuel dependency, energy pricing, or demand-side efficiency. Total greenhouse gas emissions rise

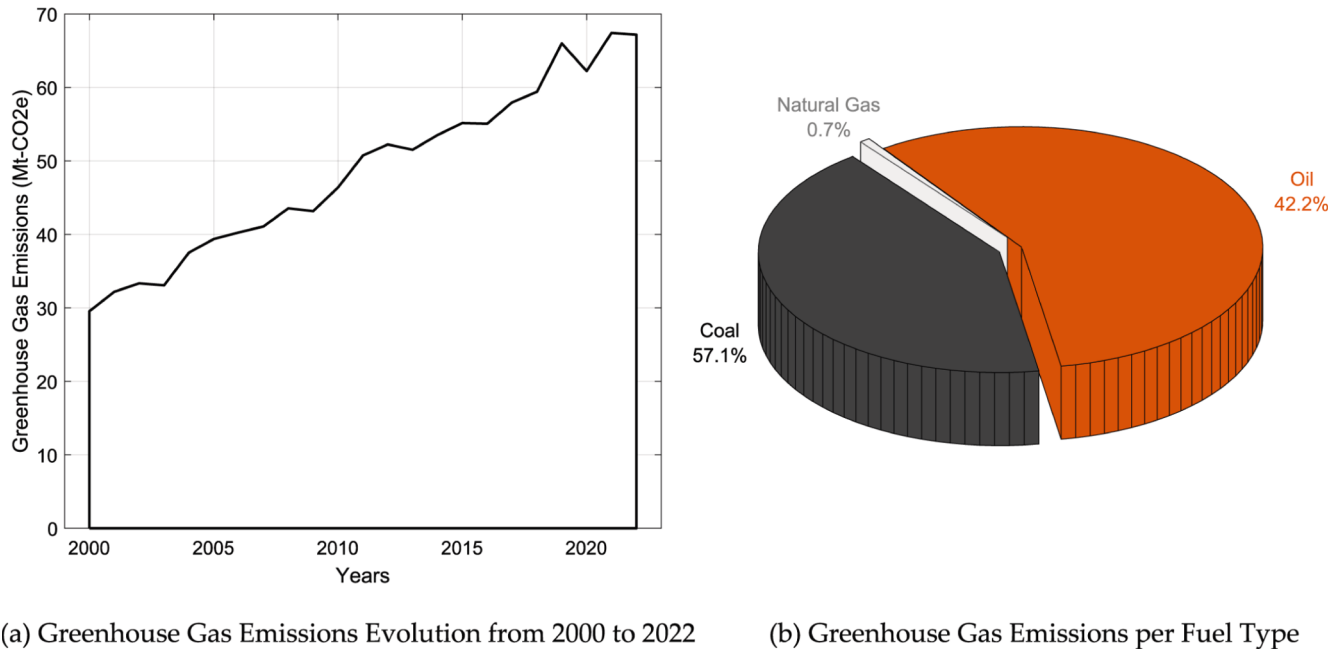


Fig. 2. Energy consumption data of Morocco.

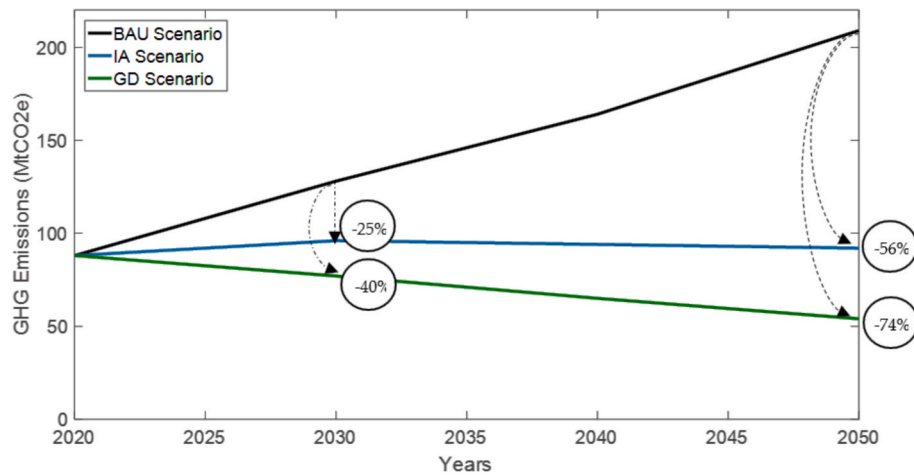


Fig. 3. Decarbonization scenarios of Morocco.

steadily from roughly 90 MtCO_{2e} in 2020 to 190 MtCO_{2e} by 2050, representing an average annual growth rate of 2.6%. The primary energy supply remains largely fossil-based, with coal, oil, and gas collectively accounting for about 76.0% of total energy in 2030. Electricity generation continues to depend heavily on coal-fired plants, maintaining an average emission intensity of 0.6 tCO_{2e} per MWh, while renewable capacity remains below 40.0%. This pathway establishes the baseline against which more ambitious mitigation scenarios are evaluated.

The Increased Ambition (IA) pathway implements mitigation measures exceeding current policies, reflecting an intermediate climate ambition. It envisions the addition of 10 to 12 GW of solar and wind capacity by 2030, alongside reforms to liquified-petroleum-gas (LPG) subsidies, stronger energy efficiency standards for buildings and industry, and accelerated electrification of transport. By 2030, renewable sources are projected to constitute 63.0% of installed power capacity. Compared with BAU, emissions are reduced by 25.0% by 2030, equivalent to roughly 31 MtCO_{2e} avoided, and by 56.0% by 2050, lowering total emissions to around 84 MtCO_{2e}. This pathway represents a moderate-to-rapid energy transition and informs intermediate modeled

scenarios, specifically the Halfway to Clean Energy (HTCE) and the National Energy Plan (NEP).

The Green Development (GD) pathway corresponds to the highest ambition level, aligned with a 1.5 °C-compatible trajectory. It integrates nearly full renewable electricity deployment, deep energy efficiency improvements, widespread sectoral electrification, green hydrogen systems, and carbon capture and storage technologies. By 2050, renewables are expected to supply 80.0–85.0% of electricity, with the remainder covered by green hydrogen and gas with CCS. Emissions decline by 40.0% by 2030 and reach approximately 50 MtCO_{2e} by 2050, achieving cumulative avoidance exceeding 340 MtCO_{2e} relative to BAU. Energy intensity falls below 2.1 MJ per USD GDP, indicating structural decoupling of economic growth from carbon emissions, forming the strategic basis for the Net Zero scenario developed later.

Motivation and scope

The main motivation of this research arises from pressing energy security and economic challenges in Morocco, driven by heavy

dependence on imported fossil fuels, which accounted for nearly 90.0% of energy demand and over 11.5 billion USD in expenditures in 2022 [9]. This reliance exposes the economy to fluctuations in global energy markets, exerts continuous pressure on foreign currency reserves, and conflicts with long-term climate objectives, as demonstrated by a 127.0% increase in GHG emissions since 2000. Despite an ambitious low-carbon vision targeting 52.0% renewable installed capacity by 2030 and progressive deep decarbonization pathways, there remains a gap in identifying cost-effective, robust energy transition strategies that balance economic affordability, energy security, environmental performance, and land-use constraints under uncertainty. This study therefore develops a quantitative, integrative decision-support framework to guide strategic investment and policy decisions within this complex energy landscape.

While energy trends and climate commitments are well described in national reports, a key quantitative question remains unanswered: what configuration of future energy infrastructure achieves cost-optimal, spatially feasible, and emission-consistent outcomes under varying policy ambition levels when economic, environmental, and land constraints are optimized simultaneously rather than sequentially? The central research question is whether Morocco can reach deep decarbonization by 2050 in a manner that is cost-efficient, land-feasible, and energy-secure, evaluated through integrated multi-objective optimization rather than narrative projections. The working hypothesis is that integrated multi-objective optimization will reveal pathways that differ substantially from descriptive policy scenarios, particularly in technology mix, hydrogen deployment, and spatial allocation, due to endogenous cost-emission-land trade-offs absent in conventional projections. The contribution of this work is therefore:

- Methodological innovation through integrated LEAP-NEMO multi-objective coupling;
- Endogenous optimization of energy system structure under policy-constrained scenarios;
- Quantitative policy evaluation grounded in techno-economic feasibility rather than narrative ambition.

The scope covers development and application of a multi-scenario optimization framework to determine the optimal energy mix of Morocco over 2030–2050. The analysis includes the electricity system and its interactions with major demand sectors, considering technology deployment, resource availability, land-use requirements, and policy constraints. Multiple scenarios capture different levels of decarbonization ambition, technology uptake, and policy enforcement. The model is formulated as a simultaneous multi-objective problem, optimizing total discounted system cost, cumulative GHG emissions, and land requirements jointly, rather than sequentially or with post-hoc constraints. Scenario-specific targets enforce minimum emissions reductions and renewable penetration, while emissions and land use remain endogenous objectives. Outputs include optimal installed capacity trajectories, energy balances, renewable deployment pathways, and land requirements, providing a transparent basis for comparing alternative long-term transition strategies.

Novelty and outline

The novelty of this research lies in the development of the first fully integrated multi-scenario optimization framework dedicated to long-term strategic energy planning in Morocco, addressing gaps in previous studies that largely relied on deterministic projections or technology-specific analyses. This work implements a hybrid LEAP-NEMO optimization architecture that simultaneously models demand evolution, multi-vector energy flows, renewable resource potential, land requirements, and techno-economic constraints to identify cost- and emission-efficient transition pathways from 2022 to 2050. Unlike earlier approaches, the framework jointly minimizes net present system cost

and GHG emissions while respecting sectoral, spatial, and resource limitations, producing results that are internally consistent across cost, emissions, and system structure. The methodology captures uncertainties in demand growth, renewable intermittency, technology learning, import dependency, and carbon pricing, supporting a dynamic evaluation of five transition pathways: Business-as-Usual (BAU), Halfway to Clean Energy (HTCE), National Energy Plan (NEP-PULP and NEP-NEMO), and Net Zero Emissions (NZE).

It is important to clarify that the NDC pathways (BAU, IA, GD) are narrative policy constructs. They describe qualitative ambition levels, investment intentions, and mitigation targets, but they do not result from endogenous energy system optimization. They therefore represent exogenous policy frameworks rather than mathematically optimized system configurations. To eliminate conceptual ambiguity between narrative NDC pathways and optimized model scenarios, Table 1 explicitly maps policy narratives to modeled scenarios, distinguishing exogenous inputs from endogenous decision variables. The conceptual distinction is that the NDC pathways define exogenous policy ambition levels and regulatory constraints, while the LEAP-NEMO scenarios determine the endogenous configuration of the energy system that satisfies those imposed constraints.

The objectives of the study are to determine the cost-optimal, spatially feasible, and emission-consistent configuration of the future energy system of Morocco under varying policy ambition levels - to quantify trade-offs among economic, environmental, and spatial goals - and to provide policymakers with a transparent, actionable framework for strategic planning. Key quantitative outputs include renewable installed capacity (20 GW under BAU to 70 GW under NZE by 2050), system-wide emissions (58.9 MtCO_{2e} under BAU to zero under NZE), land requirements for solar PV (6,458 ha under BAU in 2030 to 47,437 ha under NZE in 2050), and hydrogen deployment (from zero under BAU to 34,720 ha under NZE). Multi-criteria evaluation further distinguishes scenario performance, with scores ranging from 35/100 for BAU to 98/100 for NZE, clearly reflecting the relative robustness and sustainability of each transition pathway.

The remainder of this paper is structured as follows. Following this introduction, Materials and Methods (Section 2) details the multi-scenario optimization methodology, including the key energy parameters, the mathematical formulation of the objective function and constraints, and the simulation software employed. It also describes the scenario designs, database, and key modeling assumptions. Results and Analysis (Section 3) presents the core findings, analyzing the optimized installed capacity mix, energy balance, land use requirements, and comprehensive evaluation (in terms of cost, emission, and final score) under each scenario, culminating in a critical discussion of the key takeaways. Finally, Conclusion and Perspectives (Section 4) synthesizes the major conclusions from the study, articulates policy recommendations derived from the optimized pathways, and suggests directions for future research to build upon the current work.

Materials and methods

This section presents the methodological framework used to construct and assess the prospective energy mix of Morocco through a multi scenario optimization approach. The framework combines qualitative insights with quantitative modeling in order to provide a comprehensive evaluation of national energy transition pathways. The first part surveys existing multi scenario optimization techniques employed in national and regional energy planning, thereby establishing the analytical basis and methodological rationale for the study. The second part explains the model architecture, including the specification of key energy parameters, the mathematical formulations adopted, and the simulation instruments applied for system optimization. The final part describes the overall research design and anticipated outputs, covering data sources, scenario structures, modeling assumptions, and the interpretation of optimization results. Together, these components

Table 1

Mapping of NDC narrative pathways to optimized LEAP-NEMO scenarios.

NDC Narrative Pathway	Corresponding Modeled Scenario	Exogenous (Imposed) Elements	Endogenous (Optimized) Elements
Business-as-Usual (BAU)	Business-as-Usual (BAU)	Demand growth trajectory; absence of additional carbon constraints; current policy continuation	Technology mix; capacity expansion; dispatch; land allocation; cost minimization
Increased Ambition (IA)	Halfway to Clean Energy (HTCE) National Energy Plan (NEP)	Emission reduction targets; renewable penetration floors; electrification targets	Optimal renewable mix; hydrogen scaling; storage sizing; cost–land–emission trade-offs
Green Development (GD)	Net Zero Emissions (NZE)	Net-zero constraint by 2050; carbon neutrality requirement	Hydrogen penetration; CCS deployment; renewable overcapacity; spatial allocation

establish a coherent and systematic structure for examining long term energy transition strategies in Morocco under alternative policy, technological, and resource conditions.

Literature review

Multi-scenario optimization methods

Multi scenario optimization has become a central methodological pillar for long term energy system planning under structural uncertainty in technology costs, demand growth, fuel price volatility, renewable intermittency, and climate policy constraints. Modern formulations extend beyond deterministic capacity expansion models toward multi objective evolutionary algorithms, stochastic and robust optimization, and hybrid artificial intelligence integrated mixed integer programming frameworks. These approaches typically minimize total discounted system cost subject to energy balance, capacity adequacy, emission limits, and technical feasibility constraints, while incorporating scenario dependent parameters.

At the methodological frontier, Tangi and Amaranto [16] applied Epsilon Multi Objective Evolutionary Algorithm (ε-MOEA) and Multi Objective Particle Swarm Optimization (MOPSO) to optimize integrated multi energy systems in Italy. Their formulation treated investment cost, operating cost, and CO₂ emissions as simultaneous objective functions. Decision variables included installed capacities of photovoltaic, wind, combined heat and power units, and storage systems, while constraints enforced load balance, technical limits, and operational feasibility. Results showed that ε-MOEA achieved broader Pareto front dispersion, enabling improved diversity in non-dominated solutions, whereas MOPSO exhibited faster convergence rates and lower computational time. The comparison highlights a technical tradeoff between exploration capability (solution diversity) and convergence speed, which is critical in large scale multi objective planning problems.

Iseri et al. [17] introduced a hybrid framework integrating Bayesian Neural Networks (BNN) with two stage stochastic mixed integer linear programming (MILP). In the first stage, investment decisions were determined; in the second stage, recourse variables adjusted operations under demand uncertainty scenarios generated by the BNN probabilistic forecasts. The stochastic MILP formulation included binary capacity expansion variables, continuous dispatch variables, and scenario weighted expected cost minimization. Compared with deterministic baselines, the hybrid model improved total system cost efficiency by 8.0–12.0% and enhanced demand coverage reliability by 15.0%. These quantified gains demonstrate the measurable economic and reliability benefits of embedding probabilistic forecasting within optimization.

Liu et al. [18] developed a two-stage stochastic robust optimization framework for rural integrated energy systems incorporating both source side (renewable output variability) and load side uncertainties. Their model combined scenario based stochastic programming with uncertainty budget sets derived from robust optimization theory. The objective minimized total lifecycle cost while enforcing emission constraints and power balance. Results showed reductions in both total system cost and emissions under structured uncertainty budgets, alongside sensitivity analyses demonstrating stability of optimal solutions across uncertainty intervals. The study confirms that explicit

uncertainty modeling materially alters optimal infrastructure portfolios compared with deterministic formulations.

Beyond bespoke algorithmic models, several established planning platforms structure national and regional energy analysis. EnergyPLAN [19] is a deterministic simulation tool operating at hourly resolution. It models electricity, heating, transport, and industry sectors with detailed operational constraints, including ramp rates, storage dynamics, and cross sectoral coupling. Although not an optimization solver by default, it enables scenario comparison of renewable penetration levels, grid stability, and curtailment under fixed capacity assumptions, making it widely used for high renewable integration assessments. OSeMOSYS [20,21] is a linear optimization framework based on long term cost minimization. It uses a bottom up technology representation with capacity expansion variables, activity levels, resource constraints, and emission accounting. The model is typically formulated as a large-scale linear programming problem minimizing net present value of system cost over multi period horizons. The systematic review by Plazas Niño et al. [20] documents its global application in national decarbonization strategies, least cost electrification planning, and energy access assessments, highlighting its scalability and transparency for developing country contexts.

Artificial intelligence based operational optimization is also expanding. Wang et al. [22] implemented reinforcement learning for multi market electricity procurement strategies. Using a Markov decision process formulation, the agent learned optimal bidding strategies through iterative policy updates based on reward maximization. Their results show improved adaptive decision-making performance compared with rule based strategies, demonstrating the potential of reinforcement learning to handle dynamic and stochastic market environments.

Collectively, these studies indicate three major methodological directions in contemporary research:

- Multi objective optimization balancing cost, emissions, and reliability;
- Explicit modeling of uncertainty through stochastic and robust formulations;
- Integration of artificial intelligence methods to enhance predictive and operational performance.

However, most applications remain limited to either operational dispatch optimization, sector specific capacity planning, or scenario comparison without fully endogenous treatment of land allocation, hydrogen deployment, and spatial resource constraints within a unified national scale framework. The integration of multi objective optimization, spatial constraints, hydrogen infrastructure scaling, and quantified uncertainty propagation remains comparatively underdeveloped in the Moroccan context, motivating the methodological development presented in this study.

Regional and North African context

In the Arab and North African region, energy transition research has increasingly combined techno economic modeling, hybrid system simulation, spatial resource assessment, and sustainability evaluation

frameworks adapted to regional climatic and socio institutional conditions.

Moumani [23] analyzed sustainable development management within Arab and international cooperation frameworks using institutional policy analysis and governance benchmarking methods. The study emphasized that alignment between regional energy strategies and international climate cooperation mechanisms significantly determines implementation feasibility. While not optimization based, the methodological contribution lies in multi-level governance evaluation linking institutional capacity indicators with sectoral sustainability targets, demonstrating that policy misalignment can materially delay renewable deployment trajectories.

Hybrid renewable configurations have been extensively assessed under real climatic datasets in Palestine and Libya. Aqila et al. [24] designed a PV-Wind-Battery hybrid system for residential applications in southern Libya using techno economic optimization with simulation tools such as HOMER. The model minimized net present cost (NPC) while satisfying hourly load balance constraints under solar irradiance and wind speed time series data. Sensitivity analyses were conducted on fuel prices and battery replacement cycles. The study confirmed techno economic feasibility under high solar resource availability, demonstrating competitive levelized cost of energy relative to diesel-based alternatives. Similarly, El Khozondar et al. [25] performed a technical, economic, and environmental evaluation of grid connected PV-Wind hybrid systems in Gaza. Their assessment incorporated capital cost, operational cost, emission reductions, and renewable fraction metrics. Using multi scenario simulations, they demonstrated that hybridization improves renewable penetration and reduces dependence on imported electricity, while lowering lifecycle emissions compared to conventional supply configurations. The methodology integrated resource variability modeling, grid interconnection constraints, and emission accounting.

Hydrogen integration has also been evaluated regionally. Al-Hamamre et al. [26] investigated renewable powered hydrogen production through electrolysis under regional solar resource conditions. The study modeled electrolyzer efficiency, capacity factors derived from solar generation profiles, and hydrogen production cost components including capital expenditure and electricity input cost. Results highlighted the sensitivity of levelized hydrogen cost to renewable capacity factor and electrolyzer utilization rates, demonstrating the importance of high renewable penetration for cost competitiveness.

Urban and residential decarbonization strategies have been examined in Libya and neighboring contexts. Nassar et al. [27] conducted cost benefit analysis of electricity sector decarbonization pathways in Libya. Their financial modeling incorporated discounted cash flow analysis, environmental externality valuation, and scenario based renewable penetration targets. Results demonstrated that incorporating environmental externality costs materially improves the economic attractiveness of renewable expansion relative to fossil fuel dominated scenarios.

Hesri et al. [28] analyzed grid integration and electric vehicle deployment strategies using policy evaluation and infrastructure readiness assessment. Their framework incorporated charging demand projections, grid capacity constraints, and regulatory standards. The findings indicate that coordinated policy, infrastructure standardization, and grid reinforcement are necessary to avoid operational bottlenecks under high EV penetration scenarios.

Importantly, recent research emphasizes that renewable energy deployment is not environmentally neutral. Al Maghalseh [29] evaluated large scale photovoltaic installations in Palestine using environmental impact assessment methodology. The study documented land use competition, visual intrusion, ecosystem disturbance, and community acceptance challenges. These findings demonstrate that renewable expansion requires spatial planning integration rather than purely techno economic optimization. In Libya, Amhimmid et al. [30] assessed wind farm development in urban and peri urban zones using integrated financial and environmental modeling. Their approach internalized social and environmental externalities within cost calculations, showing

that inclusion of these factors can significantly alter project viability rankings compared with purely financial assessments.

Spatial resource and storage assessments using Geographic Information Systems (GIS) have also been conducted in North Africa. Al-Naas et al. [31,32] mapped potential pumped hydro energy storage sites in Libya by combining topographic elevation data, water resource availability, and proximity to transmission infrastructure. Their methodology applied spatial filtering criteria (slope thresholds, elevation differences, land exclusion zones) to identify technically feasible storage locations. Such GIS based multi criteria analysis demonstrates how spatial constraints materially shape feasible energy storage capacity.

In Morocco, resource specific assessments have been performed. Bougrine et al. [33] and Maatallaoui et al. [34] conducted GIS based evaluations of solar and wind potential for green hydrogen production. Their studies combined high resolution solar irradiance maps, wind speed datasets, land suitability filters, and electrolyzer efficiency parameters to estimate hydrogen production potential and spatial distribution. Results confirm the high renewable resource endowment of Morocco but also show spatial heterogeneity requiring region specific infrastructure planning.

Identified gap and contribution

Despite the discussed methodological advances, applications of integrated multi-scenario optimization frameworks at the national scale remain limited in Morocco. Existing Moroccan studies predominantly assess the techno-economic feasibility of isolated renewable technologies, such as utility-scale PV, wind farms, concentrated solar power (CSP), or green hydrogen pilot systems [35–39]. These analyses typically employ deterministic simulation tools or single-objective cost-minimization approaches (e.g., levelized cost of electricity estimation, net present cost comparison, or static capacity factor assessment). While such studies provide valuable insights into resource potential and project-level viability, they rarely embed these technologies within a fully integrated national energy system model that simultaneously optimizes investment, operation, emissions, land allocation, and energy security under multiple future scenarios.

Methodologically, most national-level assessments rely on sequential modeling workflows: demand is projected exogenously; renewable potential is estimated via GIS screening; and system expansion is simulated under predefined policy targets. This sequential structure limits endogenous feedback between demand evolution, technology deployment, spatial constraints, and policy instruments. Furthermore, uncertainty treatment is often limited to sensitivity analysis on a small number of parameters (e.g., capital cost or fuel price variation), rather than formal stochastic or robust optimization. As a result, key uncertainties, such as demand growth trajectories, renewable intermittency profiles, technology learning curves, import dependency, and carbon pricing evolution, are not jointly quantified within a coherent optimization framework, thereby constraining the robustness of policy recommendations. The literature hence reveals three structural gaps:

- **Lack of endogenous land-use integration:** Renewable siting studies [33,34] identify spatial potential using GIS-based exclusion criteria, yet land occupation is not treated as a binding optimization constraint in national planning models. Consequently, trade-offs between cost minimization and land scarcity remain underexplored.
- **Limited hydrogen system integration:** Hydrogen assessments [26,34] typically evaluate production cost under given renewable inputs but do not endogenously optimize electrolyzer capacity, storage, and downstream utilization within a national multi-vector energy balance.
- **Absence of multi-objective national optimization:** While global studies employ Pareto-based or stochastic programming techniques [16–18], Moroccan applications have not simultaneously optimized total discounted system cost, cumulative GHG emissions, and land-

use requirements within a unified national-scale decision framework.

To address these gaps, this study introduces the LEAP-NEMO framework, an endogenous national energy system optimization model tailored to Morocco. Unlike conventional single-framework applications of LEAP, TIMES, or MESSAGE, LEAP-NEMO integrates simulation-based demand modeling with endogenous multi-objective optimization. Technically, the framework:

- Formulates a multi-period mixed-integer optimization problem minimizing total discounted system cost while simultaneously constraining cumulative GHG emissions and renewable penetration targets.
- Treats technology deployment (generation, storage, hydrogen production), spatial land allocation, and infrastructure scaling as decision variables rather than exogenous inputs.
- Treats technology deployment (generation, storage, hydrogen production), spatial land allocation, and infrastructure scaling as decision variables rather than exogenous inputs.
- Embeds scenario-specific policy constraints, including emissions ceilings, renewable share mandates, and import dependency limits.
- Explicitly models land-use intensity as binding constraints within the optimization.
- Captures uncertainty through structured scenario sets representing demand growth variability, renewable intermittency, technology cost learning, and carbon pricing evolution.

This integrated formulation enables the simultaneous optimization of optimal installed capacity expansion pathways, technology-specific generation mix trajectories, hydrogen production and storage scaling strategies, cumulative emissions profiles, and national land occupation requirements. Unlike deterministic or sequential approaches, LEAP-NEMO generates endogenous trade-off surfaces between cost, emissions, and land-use intensity, thereby revealing system-level compromises not observable through isolated techno-economic analysis.

By synthesizing global methodological advances [16–22], regional spatial and techno-economic evidence [24–34,40], and Morocco-specific renewable assessments [33,34], the framework delivers quantitatively grounded, policy-relevant outputs. It provides decision-makers with robust insights into cost-effective, land-feasible, and emission-consistent transition pathways under structural uncertainty—thereby advancing national-scale energy planning beyond the current state of the Moroccan literature [35–39].

Multi-scenario optimization modeling

The present research adopts an alternative structure for long term energy planning through an integrated modeling architecture referred to as LEAP NEMO. This framework combines the analytical capabilities of LEAP and NEMO in order to capitalize on their complementary strengths. Within this structure, LEAP functions as the scenario development and accounting environment. It defines projections of final energy demand, technology transition pathways, fuel supply balances, and baseline emission trajectories under different policy and socio-economic assumptions. Scenario narratives are translated into quantitative inputs, including technology costs, efficiency parameters, resource availability, and emission factors. These structured datasets are then transferred to NEMO for formal optimization. NEMO operates as the mathematical optimization core of the framework. The problem is formulated as a Mixed Integer Linear Programming model over a multiyear planning horizon. Decision variables include technology investment choices, generation levels, and operational scheduling parameters. The objective function minimizes total system cost or, in alternative formulations, cumulative emissions, subject to energy balance equations, installed capacity constraints, technical performance limits, and policy

requirements. Binary variables govern discrete investment decisions, while continuous variables represent production and flow levels across the energy system. The optimization outputs generated by NEMO include optimal capacity expansion trajectories, technology specific generation schedules, and aggregated cost indicators. These results are subsequently reintroduced into LEAP, where they are processed for detailed energy accounting, graphical representation, and emission evaluation across sectors. Through this iterative interaction, LEAP enables structured scenario interpretation and transparent reporting, whereas NEMO performs rigorous mathematical optimization. The combined framework therefore links narrative scenario design with endogenous system wide cost minimization, ensuring internal consistency between policy assumptions, technology deployment, and long-term energy balances.

To ensure consistency and convergence between the scenario narratives defined in LEAP and the optimized solutions generated by NEMO, the coupling follows a controlled iterative workflow in which initial demand projections, technology bounds, and policy constraints are first specified in LEAP and passed to NEMO for optimization, after which the optimized outputs (installed capacities, generation profiles, and system costs) are re-imported into LEAP and evaluated for energy balance closure, emission consistency, and compliance with predefined scenario assumptions. Iterations are repeated only if deviations exceed predefined tolerance thresholds ($\pm 1.0\%$ for energy balance and $\pm 0.5\%$ for emissions), where these levels are grounded in standard numerical convergence practice for large-scale MILP-based energy system models, balancing computational tractability with solution precision. The

$\pm 1.0\%$ threshold for energy balance reflects the typical statistical uncertainty range of national energy data (on the order of 1.0–2.0%), ensuring that convergence requirements do not exceed underlying data accuracy, while the tighter $\pm 0.5\%$ threshold for emissions accounts for the cumulative sensitivity of long-term decarbonization trajectories to marginal changes in generation mix. Mathematically, convergence is defined through relative residual errors between successive iterations: letting $E^{(k)}$ denote total primary energy supply and $G^{(k)}$ total GHG emissions at iteration k , convergence requires (2). These conditions correspond to relative L_1 –norm tolerances [7,8] commonly applied in fixed-point iterative coupling schemes for large-scale linear optimization models, ensuring that the solution approaches a numerically stable fixed point without inducing oscillatory feedback. To verify that the selected thresholds do not materially influence the optimized outcomes, additional sensitivity tests were conducted using stricter tolerances ($\pm 0.5\%$ for energy balance and $\pm 0.25\%$ for emissions), resulting in variations below 0.3% in total discounted system cost and below 0.2% in installed renewable capacity by 2050, and thereby confirming robustness with respect to convergence parameter selection.

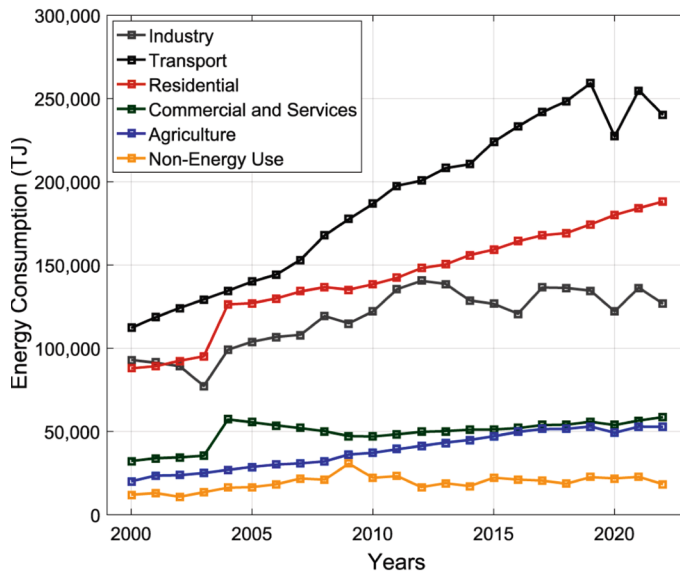
$$\begin{cases} \left| \frac{E^{(k)} - E^{(k-1)}}{E^{(k-1)}} \right| \leq 0.01 \\ \left| \frac{G^{(k)} - G^{(k-1)}}{G^{(k-1)}} \right| \leq 0.005 \end{cases} \quad (2)$$

Energy input parameters

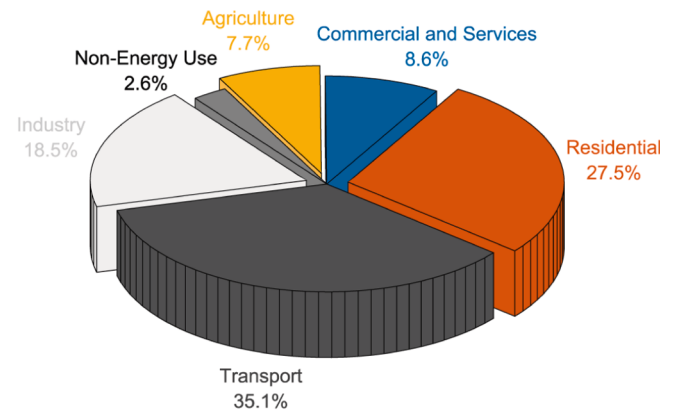
The LEAP–NEMO framework relies on a comprehensive set of energy system input parameters describing both the current state and future evolution of the energy sector in Morocco. These include sectoral energy demand (residential, industrial, commercial, and transport), peak load and growth rates, fuel consumption by source (coal, oil, natural gas, and renewables), installed generation capacities and technology shares, renewable resource availability (solar, wind, hydro, and biomass), as well as technical parameters such as efficiencies, capacity factors, storage characteristics, and transmission and distribution losses. Economic inputs comprise domestic and international fuel price trajectories and leveled energy costs by technology. These harmonized datasets form

the empirical basis for scenario construction and iterative coupling between LEAP and NEMO, with identical temporal resolution and system boundaries to avoid structural inconsistencies. In this framework, demand trajectories and policy constraints are specified exogenously in LEAP, while supply-side investment and dispatch decisions are optimized endogenously in NEMO, ensuring a well-posed and reproducible optimization problem. To ensure convergence between LEAP-generated scenarios and NEMO optimization outcomes, an explicit iterative procedure is implemented: (i) LEAP first produces demand projections and policy-constrained trajectories; (ii) these trajectories are transferred to NEMO, which solves for the least-cost capacity expansion and dispatch

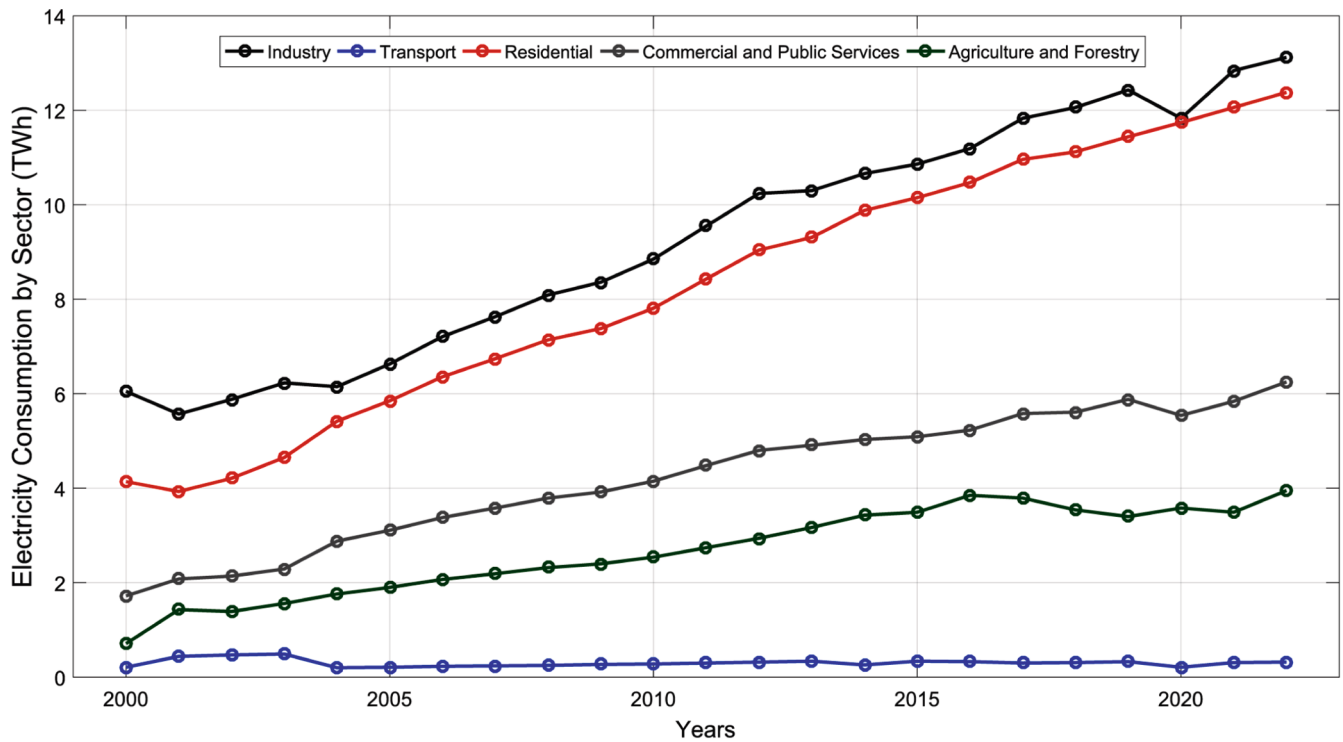
solution; (iii) key outputs from NEMO are then fed back into LEAP to update demand-side assumptions and consistency checks; and (iv) this exchange is repeated until successive iterations produce variations below a predefined tolerance threshold. Convergence is formally achieved when relative changes in total discounted system cost and cumulative CO₂ emissions between two consecutive iterations fall below 1.0%, and no structural inconsistencies in capacity adequacy or fuel balances remain. Although stochastic formulations are used upstream to generate alternative trajectories for sectoral demand growth and fuel prices within bounded uncertainty ranges (± 10.0 – 15.0% for demand growth and $\pm 20.0\%$ for international fuel prices), the optimization



(a) Historical Energy Consumption from 2000 to 2022



(b) Energy Consumption Ratio in 2022



(c) Historical Electricity Consumption (2000-2022)

Fig. 4. Evolution pattern of energy consumption by sector in Morocco.

results reported in this study are obtained deterministically using representative expected (mean) trajectories derived from these stochastic samples. The stochastic realizations are employed solely to verify the stability of long-term trends prior to optimization, while convergence of the deterministic LEAP–NEMO iterations is achieved when variations in total system cost and cumulative emissions fall below 1.0%. Consequently, the robustness of the results should be interpreted in terms of structural consistency and relative scenario performance rather than probabilistic optimality across multiple uncertainty realizations.

Beginning with energy demand by sector, Fig. 4 presents the historical progression of final energy consumption by sector in Morocco over the period 2000 to 2022. Throughout these 22-years, aggregate final energy demand increased consistently across the principal sectors, a pattern associated with economic expansion and broader access to electricity. As illustrated in Fig. 4a, energy use in industry displayed moderate year to year variation but followed a clear upward direction, rising from 93,011 TJ in 2000 to 126,743 TJ in 2022, which corresponds to an average annual growth rate of approximately 1.4%. The transport sector recorded the strongest expansion, increasing from 112,207 TJ in 2000 to 240,157 TJ in 2022, equivalent to growth exceeding 114.0%, largely linked to accelerated vehicle ownership and higher fuel consumption. Residential demand evolved along a sustained increasing path, moving from 87,977 TJ to 188,111 TJ over the same interval, reflecting an increase close to 114.0% driven primarily by demographic growth and urban development. Demand in commercial and service activities advanced from 32,347 TJ to 58,664 TJ, while agricultural consumption rose from 20,067 TJ to 52,829 TJ, nearly tripling as a consequence of expanded irrigation systems and agro-industrial modernization. Non energy applications, including feedstock uses, increased from 12,066 TJ to 18,375 TJ during the study horizon. The sectoral distribution in 2022, shown in Fig. 4b, indicates that transport represented 35.1% of total final consumption, followed by residential at 27.5%, industry at 18.5%, commercial and services at 8.6%, agriculture at 7.7%, and non-energy uses at 2.6%, demonstrating a diversified yet transport oriented structure. Regarding electricity demand in Fig. 4c, total consumption reached 36.0 TWh in 2022. Industry accounted for the largest portion with 13.1 TWh or 36.0%, followed by residential use at 12.38 TWh or 34.0%, commercial and public services at 6.3 TWh or 17.0%, and agriculture and forestry at 4.0 TWh or 11.0%, whereas

transport represented a limited share of 0.3 TWh or 1.0%.

Fig. 5 presents the evolution of peak load and annual growth rates in Morocco over the period 2000–2022, forming a dynamic and seasonally responsive time series that incorporates both macroeconomic drivers and stochastic components affecting electricity demand. The underlying data were compiled, inferred, and modeled using several official national reports [11,41–44]. Annual peak demand increased from 2,780 MW in 2000 to 6,940 MW in 2022, corresponding to an average annual growth rate of 4.6%. On this basis, a dataset with monthly resolution was generated to represent variations within each year. To reproduce representative load patterns, seasonal baseline coefficients were introduced, showing higher demand during summer due to cooling requirements and moderate increases during winter months. A Gaussian random disturbance term was added to reflect short term daily variability. In addition, rare extreme conditions such as heat waves or cold spells were introduced with a probability of 15.0%, producing deviations of plus or minus 12.0% to emulate stress situations in grid operation. The formulation also integrates macroeconomic growth tendencies, weekend consumption effects, and demand reductions associated with holidays in order to reproduce realistic temporal load dynamics.

The simulation results are displayed using a dual axis time series format, where the left vertical axis indicates monthly peak demand in MW and the right vertical axis represents the associated monthly growth rate in percent. Two distinct operational phases are highlighted through shaded areas: A Pre-Renewable Era covering 2000 to 2009 and a Renewable Expansion Era spanning 2010 to 2022, corresponding to major transition milestones in the national energy system. A marked irregularity appears in 2020, reflecting the contraction in demand linked to the COVID-19 pandemic, as shown by a visible decline in the growth trajectory. Statistical evaluation indicates an average monthly peak demand close to 4,970 MW and a standard deviation of about 1,150 MW, pointing to increasing variability associated with renewable penetration and economic fluctuations. To evaluate the impact of stochastic parameters on expansion planning outcomes, a sensitivity assessment was conducted by adjusting Gaussian disturbance amplitudes within a range of 5.0–15.0% and modifying the probability of extreme events between 10.0 and 20.0%, then examining the corresponding changes in optimized installed capacities. The findings show that uncertainty in peak demand results in variations of up to $\pm 3.0\%$ in

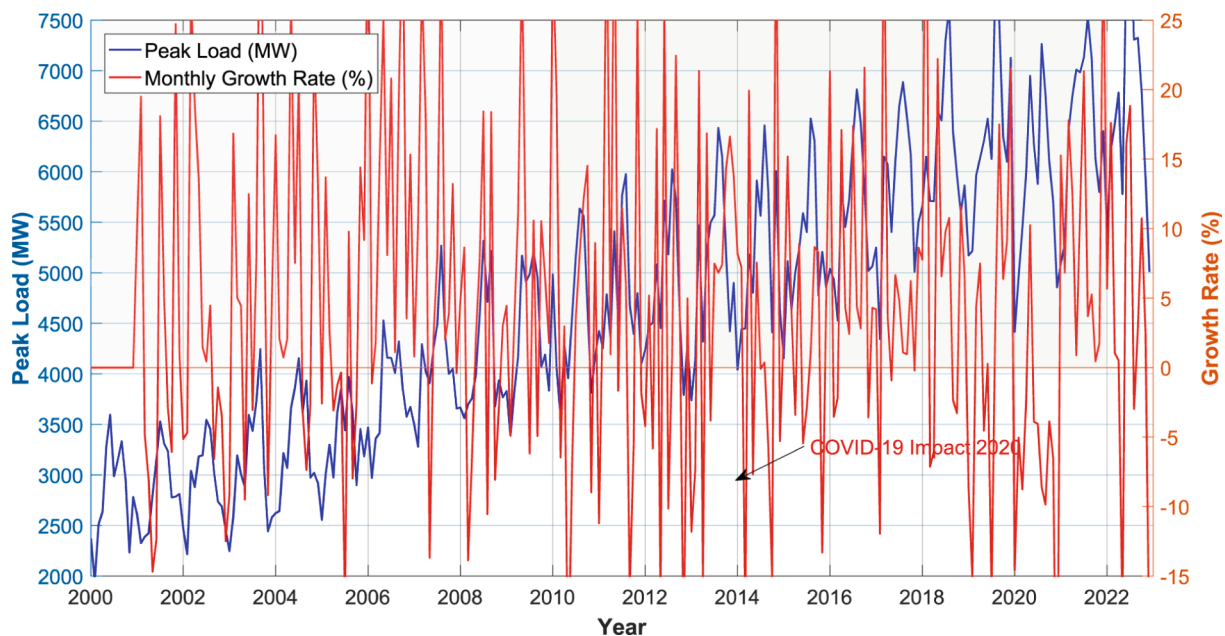


Fig. 5. Morocco monthly peak load and growth rate (2000–2022).

total generation capacity and $\pm 5.0\%$ in sectoral dispatch levels, indicating that the optimization outcomes remain structurally stable under credible peak load uncertainty conditions.

As depicted in Fig. 1, energy consumption by source in Morocco illustrates a system historically reliant on fossil fuels while gradually integrating renewable technologies. Between 2000 and 2022, oil remained the dominant source of final energy, rising from 237,801 TJ to 497,702 TJ and representing nearly 79.0% of total consumption in 2022 (Fig. 1a-b). Electricity consumption showed a marked increase over the same period, from 49,251 TJ to 184,483 TJ, driven by accelerated electrification, industrial expansion, and rural access initiatives. Coal and natural gas continued to play minor roles, contributing only 0.17% and 0.23% of total final energy, respectively, reflecting limited gas supply and constrained industrial use of coal. Total electricity generation increased from 12,863 GWh in 2000 to 42,719 GWh in 2022 (CAGR = 5.5%), with coal-based generation maintaining a dominant share at 29,064 GWh ($\approx 68.0\%$). Oil-fired generation contributed 9.8%, while gas output declined due to interruptions following the Maghreb-Europe pipeline closure. Renewable generation exhibited the most significant growth, rising from 782 GWh (6.1%) to 8,783 GWh (20.6%). Hydropower, once predominant, decreased from 3,631 GWh in 2010 to 679 GWh in 2022, whereas wind generation expanded from 64 GWh to 5,355 GWh, accounting for roughly 61.0% of renewable output. Solar production accelerated after 2015 with the Noor CSP and PV projects, reaching a combined 1,447 GWh in 2022, while bioenergy maintained a consistent contribution of approximately 1,302 GWh annually since 2013.

Fig. 6a presents installed generation capacity from 2000 to 2022 [8,41–44], showing an increase from 3,905 MW to 11,030 MW, a growth of roughly 183.0% over 22 years. Early capacity expansion was dominated by thermal plants, with coal rising modestly from 1,355 MW to 2,210 MW, oil and diesel units remaining near 1,920 MW after 2014, and natural-gas capacity stabilizing around 950 MW after its introduction in 2005. In contrast, renewable technologies displayed exponential growth (Fig. 6b). Hydropower remained near 700 MW, wind capacity increased from 0 MW in 2000 to 1,750 MW in 2022, placing Morocco among Africa's leading wind producers. Solar PV and CSP capacities emerged after 2013, reaching 500 MW and 800 MW, respectively, by 2022, while biomass and waste-to-energy installations grew to 1,950 MW equivalent. Collectively, renewable sources represented approximately 5,000 MW or 45.0% of total installed capacity in 2022. Although

renewable capacity accounts for 45.0% of the total, the corresponding generation share of 20.6% aligns with capacity factor considerations. Hydropower plants operate with 35.0–40.0% capacity factors, wind farms around 25.0–28.0%, and solar PV and CSP units between 18.0–22.0%, reflecting daily and seasonal variability. Applying these utilization rates produces expected renewable generation closely matching the reported 8,783 GWh in 2022, with deviations under 5.0% relative to observed data. These capacity factor ranges are based on empirical Moroccan operational statistics and resource assessments [8], which report long-term average wind capacity factors of 30.0–40.0% at high-resource Atlantic sites and solar PV performance ratios yielding effective capacity factors near 20.0% under local irradiance conditions.

Fig. 7 provides a detailed depiction of the techno-operational characteristics of the Moroccan power sector, encompassing generation, storage, and grid performance indicators. In Fig. 7a, thermal and renewable generation technologies are compared in terms of efficiency, capacity factor, availability, and operational flexibility. Coal and combined-cycle gas turbines achieve the highest conversion efficiencies, ranging from 45.0% to 58.0%, reflecting mature thermodynamic cycles and dispatchable operation [9,45]. Solar PV and biomass show lower efficiencies (20.0–35.0%) due to intermittency and inherent conversion limits, while wind and hydropower attain very high operational availability (95.0–100.0%) and moderate-to-high capacity factors (35.0–52.0%), demonstrating reliability under Moroccan climatic conditions [8,11,46]. Seasonal complementarity between solar and wind, with peak solar generation in summer and peak wind in winter, highlights the technical potential for hybridized operation in a diversified energy mix. Fig. 7b presents key storage system specifications for grid integration, including pumped hydro storage (PHS), lithium-ion batteries, and hydrogen-based systems. PHS achieves the highest round-trip efficiency at nearly 80.0% and offers large-scale energy capacity in the GWh range, making it suitable for daily load shifting and frequency regulation. Lithium-ion batteries provide rapid response (<1 s) and round-trip efficiencies of 90.0–95.0%, optimal for short-term smoothing of solar and wind fluctuations, though with limited discharge durations of 2–6 hours. Hydrogen storage, while less efficient (30.0–45.0%), enables long-duration discharge and strategic seasonal storage or sector coupling [7,47]. The comparison underscores the importance of a multi-layered storage strategy - short-term batteries, medium-term PHS, and long-term hydrogen - to maintain grid stability under high renewable penetration. Fig. 7c highlights transmission and distribution

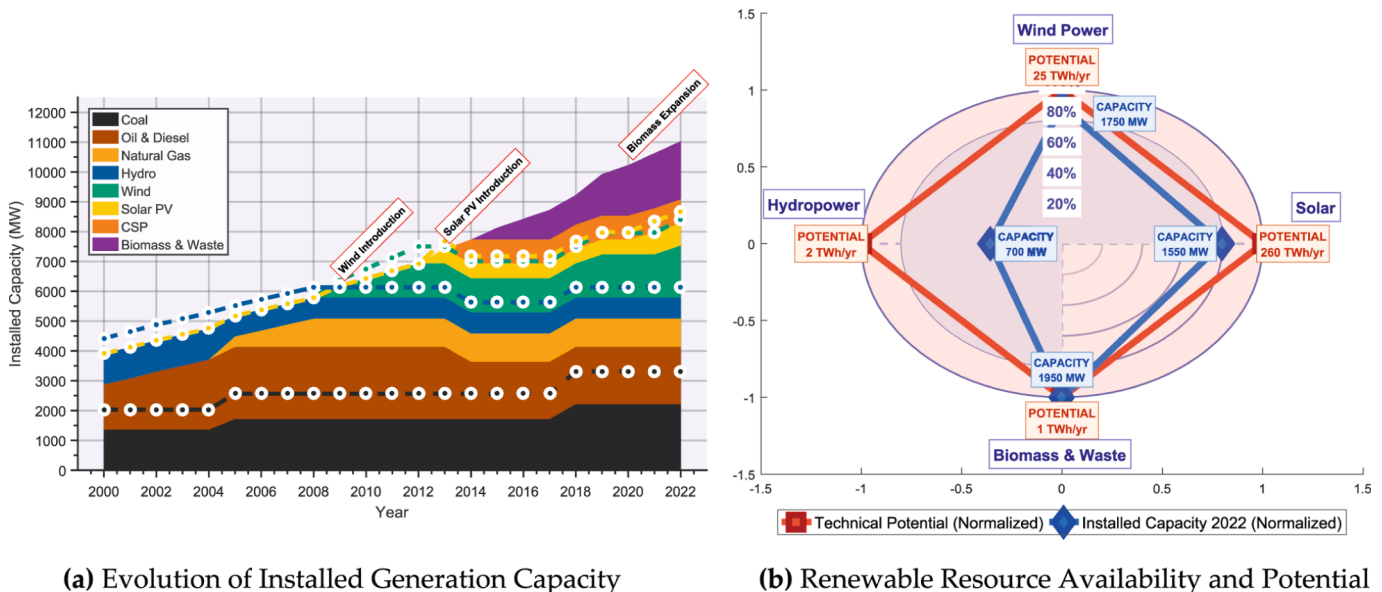
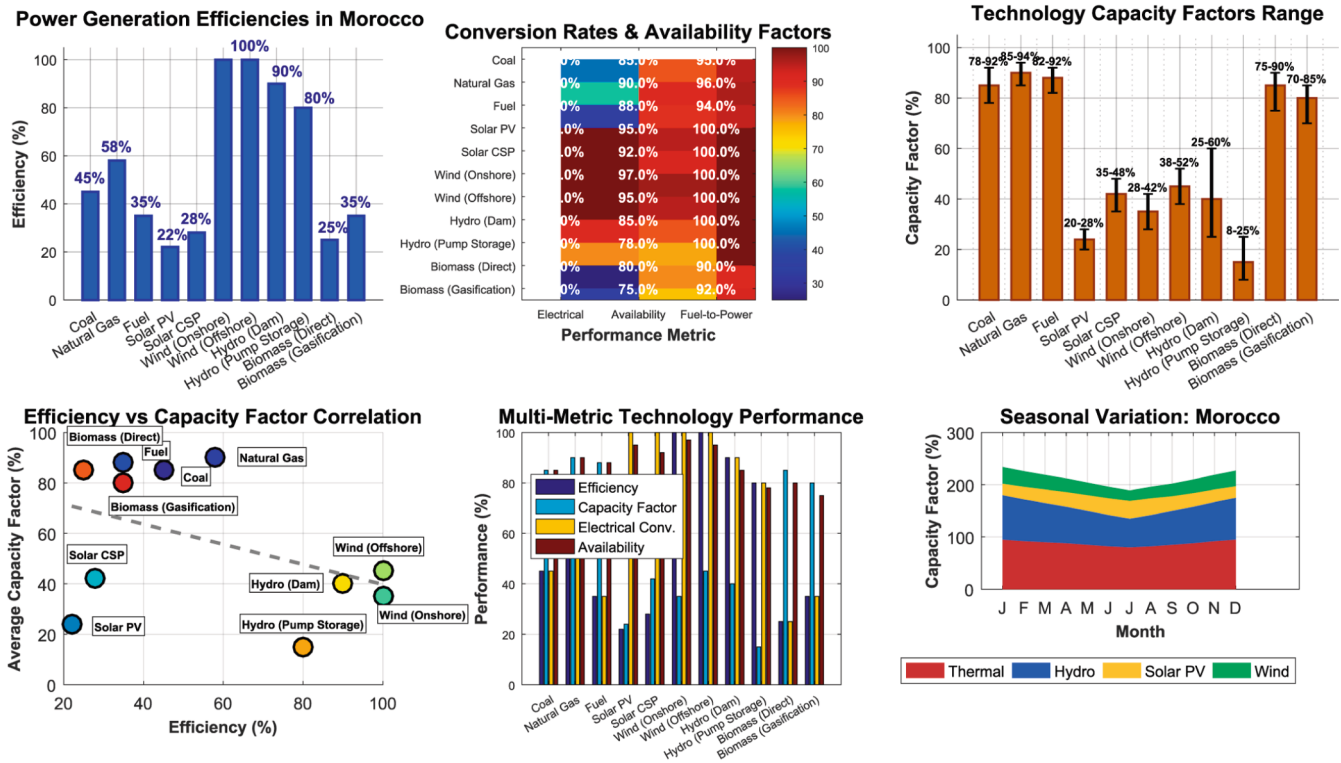
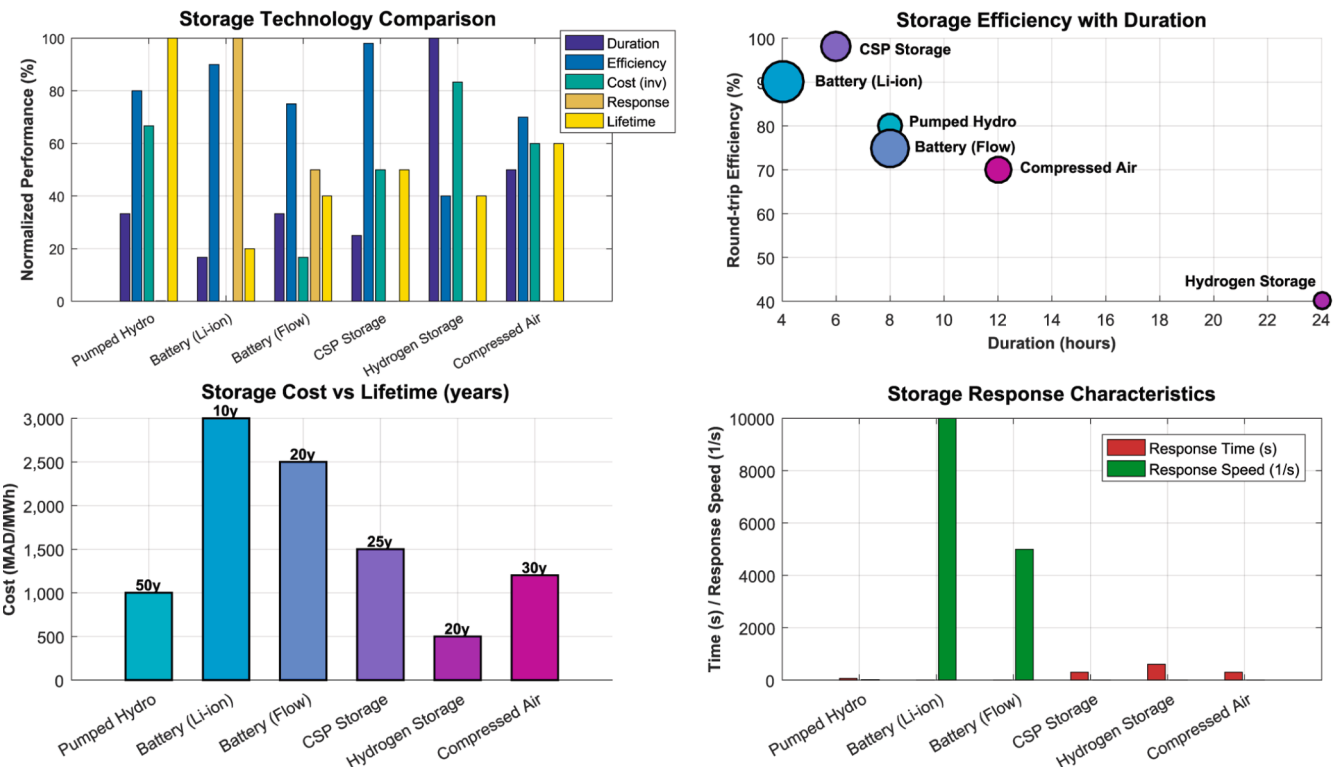


Fig. 6. Generation capacity and potential data.

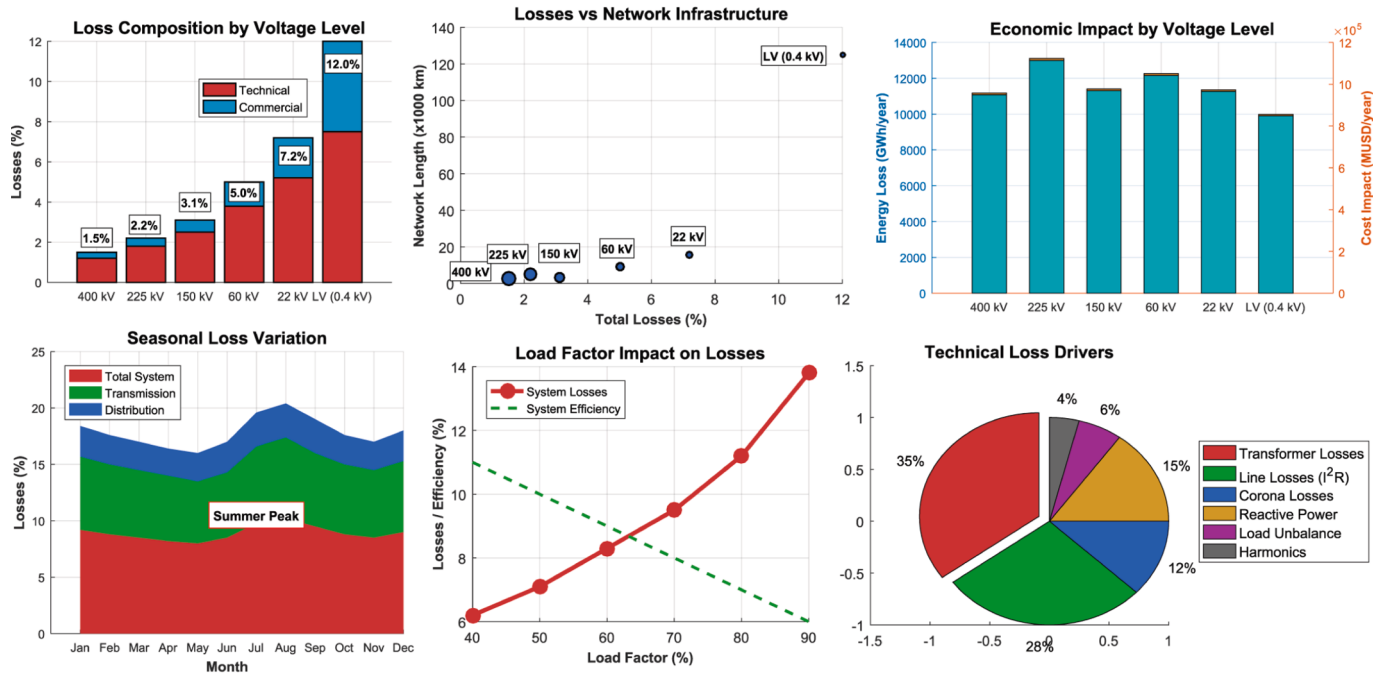


(a) Technical Characteristics of Generation Technologies



(b) Technical Characteristics of Storage Technologies

Fig. 7. Technical characteristics of generation, storage, and grid systems in the Moroccan power sector.



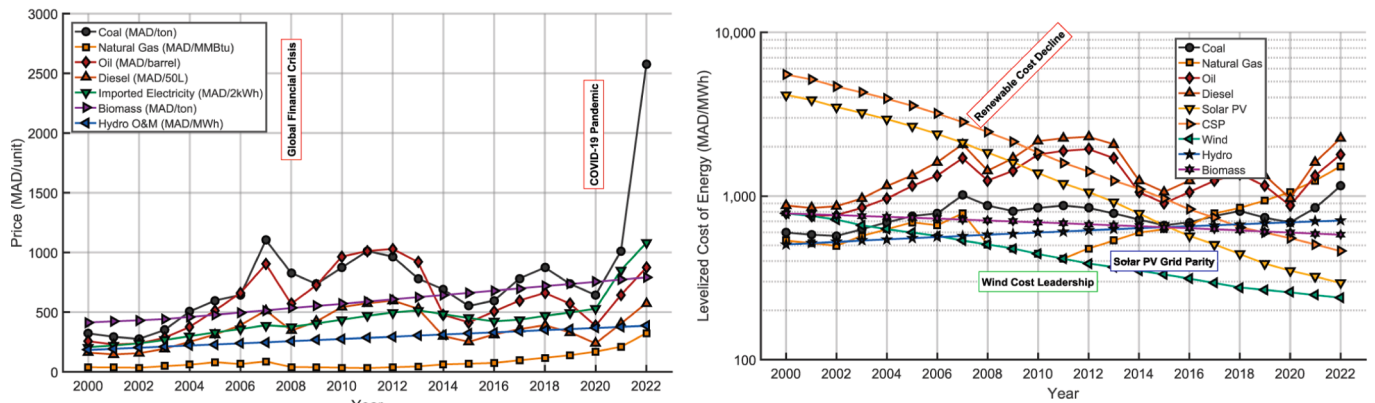
(c) Transmission and Distribution Losses

Fig. 7. (continued).

performance in Morocco from 2000 to 2022, showing a reduction in total system losses from approximately 17.0% to 11.0% [8]. Transmission losses remained low and stable at 3.0–4.0%, while the bulk of losses occurred in distribution networks due to aging infrastructure and rural network extensions. The overall decline reflects modernization efforts by ONEE, including high-voltage reinforcement, smart-meter deployment, and improved power-factor correction. Nonetheless, rural regions continue to experience elevated losses exceeding 13.0%, indicating the need for ongoing grid upgrades and digital monitoring to support decentralized renewable integration [44,48,49]. Collectively, Fig. 7 illustrates the progressive technical maturation of the Moroccan power sector, characterized by a transition from efficiency-focused conventional generation toward flexibility-oriented renewable and storage systems, accompanied by measurable improvements in grid performance.

Fig. 8 depicts the long-term evolution of fuel price trajectories (Fig. 8a) and levelized cost of energy (LCOE) by technology (Fig. 8b), highlighting structural and economic transformations in the Moroccan

power sector over the past two decades. Fig. 8a shows that fossil fuel prices underwent substantial volatility due to international market shocks, currency fluctuations, and import dependence [50]. Coal prices rose from 322 MAD/ton in 2000 to 2,576 MAD/ton in 2022, reflecting an eightfold increase, largely linked to the 2022 global energy crisis. Imported natural gas increased from 39 MAD/MMBtu to 324 MAD/MMBtu, and crude oil climbed from 257 MAD/barrel to 874 MAD/barrel over the same period. Diesel prices mirrored these dynamics, rising from 3.2 MAD/L to 11.5 MAD/L, while imported electricity tariffs increased fivefold, from 0.4 MAD/kWh to 2.2 MAD/kWh, illustrating the sensitivity of Morocco to imported energy price fluctuations [10]. By contrast, domestic renewable inputs such as biomass and hydropower showed greater price stability, with biomass feedstock rising moderately from 414 MAD/ton to 791 MAD/ton, and hydro O&M costs increasing gradually from 184 MAD/MWh to 386 MAD/MWh. Fig. 8b emphasizes LCOE trends and the divergence between conventional and renewable technologies. Thermal generation costs escalated with fuel prices: coal-fired LCOE increased from 598 MAD/MWh in 2000 to 1,150 MAD/MWh



(a) Fuel Price Trajectories of Energy Sources

(b) Levelized Cost of Energy (LCOE) by Energy Source

Fig. 8. Economic information of energy sources in Morocco (2000–2022).

in 2022, oil-based generation rose from 782 MAD/MWh to 1,794 MAD/MWh, and gas-fired generation showed wider volatility, ranging from 534 MAD/MWh to 1,518 MAD/MWh depending on import conditions. Renewable technologies experienced sharp cost reductions due to technological learning and deployment scale. Solar PV LCOE decreased by over 90.0%, from 4,140 MAD/MWh in 2000 to 294 MAD/MWh in 2022, while concentrated solar power (CSP) fell from 5,520 MAD/MWh to 460 MAD/MWh, driven largely by the Noor Program [8,51]. Wind generation also declined, from 782 MAD/MWh to 239 MAD/MWh, making it the least-cost renewable option in 2022. Hydropower and biomass maintained relatively stable costs of approximately 600–700 MAD/MWh, reflecting their operational maturity and reliability. These trends illustrate the growing economic competitiveness of renewables and the increasing financial exposure of conventional generation to global fuel price fluctuations.

Mathematical modeling

The mathematical formulation of the LEAP-NEMO hybrid framework establishes the quantitative core of the multi-scenario optimization process for the future energy mix of Morocco. It integrates a comprehensive set of input parameters encompassing demand-side characteristics (sectoral electricity demand, peak load, and annual growth rates), supply-side technical attributes (installed generation capacities, technology efficiencies, capacity factors, conversion rates, and renewable resource availability), and economic variables (fuel price trajectories and levelized cost of energy). These inputs collectively define the techno-economic and environmental boundaries within which the optimization problem operates. The model outputs include the optimal installed capacity mix, energy balance and distribution across technologies and regions, renewable energy pathways over time, and the associated land requirements for different generation technologies. The optimization is structured as a multi-objective MILP problem that minimizes total discounted system cost (NPC), GHG emissions, and land footprint, subject to energy balance, resource availability, capacity, reserve margin, and policy constraints. By integrating the scenario-based narrative management capabilities of LEAP with the rigorous mathematical optimization engine of NEMO, the framework ensures a robust, data-driven evaluation of Morocco's long-term energy transition under multiple policy and technological futures.

The state equations govern the temporal evolution of generation, storage, and network operations. Capital stock dynamics are defined in (3), where total available capacity $C_{i,t,n,s}$ for technology i at node n under scenario s at time t is calculated as the sum of existing capacity, newly added units $x_{i,t,n,s}^{add}$, and retired units $x_{i,t,n,s}^{rem}$. The initial condition in (4) anchors the model to the base-year generation stock $G_{i,t_0,n}^{exist}$ at the base year t_0 . Retirement enforcement in (5) ensures decommissioning after reaching operational lifetime $L_{i,n,s}$, maintaining realistic long-term projections. Available derated capacity $\tilde{C}_{i,t,h,n,s}$ in (6) adjusts installed capacity by hourly capacity factor $CF_{i,t,h,n,s}$ reflecting operational limits and renewable intermittency. Hourly dispatch $g_{i,t,h,n,s}$ is bounded between zero and derated capacity, as per (7), ensuring technical feasibility. Fuel consumption for thermal units is expressed in (8), linking dispatched output to fuel use through heat-rate coefficients κ_i . Annual emissions $E_{t,s}$ are computed in (9) by aggregating fuel consumption weighted by emission factors $\varphi_{f(i)}$ across all technologies, nodes, and time slices. Storage dynamics are modeled in (10), tracking state of charge $SOC_{k,t,h,n,s}$ for each storage technology k , incorporating charging and discharging efficiencies η_k^{chg} and η_k^{dis} . Bound constraints in (11) maintain storage operation within technical limits. Nodal energy balance is enforced in (12), ensuring total generation, discharging, imports, and inflows match demand plus storage charging, exports, outflows, and losses. Transmission flows $f_{l,t,h,s}$ are constrained by line capacities in (13), while reserve adequacy in (14) requires derated generation to exceed peak load $P_{t,n,s}^{peak}$ by reserve margin $RM_{t,s}$. Renewable siting and

resource constraints in (15)–(17) restrict capacity expansion to feasible regional and national potentials. Finally, Equation (18) enforces non-negativity and integrality of decision variables, preserving logical consistency across all modeled dimensions.

$$C_{i,t,n,s} = C_{i,t-1,n,s} + x_{i,t,n,s}^{add} - x_{i,t,n,s}^{rem}, \forall (i, t > t_0, n, s) \in \mathbb{N}^4 \quad (3)$$

$$C_{i,t_0,n,s} = G_{i,t_0,n}^{exist} \quad (4)$$

$$x_{i,t,n,s}^{rem} \geq x_{i,t-L_i,n,s}^{add}, \forall (i, t > L_i, n, s) \in \mathbb{N}^4 \quad (5)$$

$$\tilde{C}_{i,t,h,n,s} = C_{i,t,n,s} \cdot CF_{i,t,h,n,s} \quad (6)$$

$$0 \leq g_{i,t,h,n,s} \leq \tilde{C}_{i,t,h,n,s}, \forall (i, t, h, n, s) \in \mathbb{N}^5 \quad (7)$$

$$Fuel_{i,t,h,n,s} = \kappa_i \cdot g_{i,t,h,n,s} \quad (8)$$

$$E_{t,s} = \sum_n \sum_{h \in H} \sum_i Fuel_{i,t,h,n,s} \cdot \varphi_{f(i)} \quad (9)$$

$$SOC_{k,t,h,n,s} = SOC_{k,t,h-1,n,s} + \eta_k^{chg} P_{k,t,h,n,s}^{chg} \Delta h - \frac{P_{k,t,h,n,s}^{dis}}{\eta_k^{dis}} \Delta h \quad (10)$$

$$\begin{cases} 0 \leq SOC_{k,t,h,n,s} \leq E_{k,t,n,s}^{max} \\ 0 \leq P_{k,t,h,n,s}^{chg} \leq P_{k,t,n,s}^{chg,max} \\ 0 \leq P_{k,t,h,n,s}^{dis} \leq P_{k,t,n,s}^{dis,max} \end{cases} \quad (11)$$

$$\begin{aligned} \sum_i g_{i,t,h,n,s} + \sum_k P_{k,t,h,n,s}^{dis} + imp_{t,h,n,s} + \sum_{l \in L_n} f_{l,t,h,s} \\ = D_{t,h,n,s}^{el} (1 + loss_n^{T\&D}) + \sum_k P_{k,t,h,n,s}^{chg} + exp_{t,h,n,s} + \sum_{l \in L_n} f_{l,t,h,s} + unmet_{t,h,n,s} \end{aligned} \quad (12)$$

$$-F_l^{max} \leq f_{l,t,h,s} \leq F_l^{max} \quad (13)$$

$$\sum_i C_{i,t,n,s} \cdot derate_i \geq P_{t,n,s}^{peak} (1 + RM_{t,s}) \quad (14)$$

$$0 \leq x_{i,t,n,s}^{add} \leq build_limit_{i,t,n,s} \quad (15)$$

$$\sum_n x_{i,t,n,s}^{add} \leq national_build_limit_{i,t,s} \quad (16)$$

$$\begin{cases} Land_{t,n,s} = \sum_{i \in RE} land_factor_i \cdot C_{i,t,n,s} \\ Land_{t,s} = \sum_n Land_{t,n,s} \end{cases} \quad (17)$$

$$\begin{cases} \{x^{add}, x^{rem}, C, g, p, Fuel, imp, unmet\} \geq 0 \\ y_{i,t,h,n,s} \in \{0, 1\} \end{cases} \quad (18)$$

The projection sub-models within the LEAP-NEMO framework establish forward-looking trajectories for energy system inputs that directly influence optimization outcomes under diverse policy and resource scenarios. These equations govern the temporal evolution of sectoral demand, fuel prices, technology costs, and renewable resource availability, ensuring alignment between historical baselines and future projections. Sectoral electricity demand for residential, industrial, commercial, and transport sectors is modeled through discrete multiplicative growth in (19) or as continuous exponential growth in (20). Peak load projections follow a similar multiplicative formulation in (21), capturing macroeconomic and demographic effects on annual maximum demand. In the baseline, sectoral demand trajectories are treated as exogenous scenario drivers derived from population and economic projections, with stochastic perturbations applied to annual growth rates to represent bounded uncertainty. The framework does not endogenously capture short-term price elasticity or feedbacks between electricity prices, electrification processes (power-to-hydrogen or

transport electrification), and final consumption. To evaluate the potential impact of this assumption, a post-processing analysis applied a constant price elasticity parameter ($\varepsilon_d \in [-0.1, -0.3]$) to deviations in average system LCOE relative to the baseline. Re-optimization results indicate that high-renewable and hydrogen-intensive scenarios induce only minor demand rebound or contraction (within $\pm 4.0\%$ of projected 2040 electricity consumption), leading to adjustments in optimal capacity expansion below 5.0% and variations in NPC below 3.0%. These results suggest that endogenous demand–price feedbacks slightly modify investment levels but do not alter overall scenario rankings or qualitative trade-offs. Fuel price projections can be introduced as deterministic scenario paths or stochastically modeled via an autoregressive process in (22), linking future fuel price $p_{f,t,s}$ to its lagged value and long-run mean while incorporating random market shocks. Technology costs evolve dynamically according to learning-by-doing, where cumulative installed capacity drives cost reductions through a power-law learning curve in (23), adjusting capital costs $c_{i,t,s}^{cap}$ based on the learning exponent β_i derived from the technology's empirical learning rate. Capacity factors are updated in (24) with a time- and scenario-dependent coefficient $\Delta_{i,t,h,n,s}^{CF}$ to reflect anticipated technological or climatic improvements. Finally, renewable resource availability $\text{pot}_{r,n,t,s}$, evolves in (25), scaling baseline potentials by projected growth or degradation rates $\rho_{r,t,s}$.

$$D_{i,t,h,n,s} = D_{i,t_0,h,n,s} \cdot \prod_{\tau=t_0+1}^t (1 + g_{d,\tau,n,s}) \quad (19)$$

$$D_{i,t,h,n,s} = D_{i,t_0,h,n,s} \cdot \exp\left(\sum_{\tau=t_0+1}^t \gamma_{d,\tau,n,s}\right) \quad (20)$$

$$p_{t,n,s}^{peak} = p_{t_0,n,s}^{peak} \cdot \prod_{\tau=t_0+1}^t (1 + g_{\tau,n,s}^{peak}) \quad (21)$$

$$p_{f,t,s} = \mu_{f,s} + p_f(p_{f,t-1,s} - \mu_{f,s}) + \epsilon_{f,t,s} \quad (22)$$

$$\begin{cases} c_{i,t,s}^{cap} = c_{i,t_0}^{cap} \left(\frac{\text{CUM}_{i,t,s}}{\text{CUM}_{i,t_0}} \right)^{\beta_i} \\ \text{CUM}_{i,t,s} = \sum_{\tau \leq t} \sum_n x_{i,\tau,n,s}^{add} + C_{i,t_0,n} \\ \beta_i = \frac{\ln(1 - LR_i)}{\ln 2} \end{cases} \quad (23)$$

$$CF_{i,t,h,n,s} = CF_{i,t_0,h,n,s} \cdot (1 + \Delta_{i,t,h,n,s}^{CF}) \quad (24)$$

$$\text{pot}_{r,n,t,s} = \text{pot}_{r,n,t_0,s} \cdot (1 + \rho_{r,t,s}) \quad (25)$$

The economic evaluation within the LEAP-NEMO framework relies on a set of cost and sustainability accounting equations that systematically quantify investment, operational, and environmental outcomes across technologies and scenarios. Annualized capital expenditure (CAPEX) for each technology i in node n and scenario s , given in (26), converts the total capital cost $c_{i,t,s}^{cap}$ into an equivalent yearly payment using the capital recovery factor $\alpha_{i,t}$, which is a function of the discount rate δ and the technology lifetime L_i . Fixed operation and maintenance (O&M) costs, shown in (27), scale linearly with installed capacity $C_{i,t,n,s}$ through the fixed O&M coefficient $c_{i,t}^{fix}$. Variable costs, including fuel expenses, are captured in (28), where total fuel-related costs $\widetilde{\text{Fuel}}_{t,s}^{cost}$ are computed across all regions, time slices, and technologies by combining technology-specific variable cost $c_{i,t}^{var}$ with fuel consumption $p_{f,t,s}$ multiplied by fuel prices $\text{Fuel}_{i,t,h,n,s}$. Total system cost, represented in (29), aggregates annualized CAPEX, O&M, and fuel costs across all technologies and regions, while including transmission expansion costs $C_{t,s}^{trans}$ and penalties for unmet demand weighted by a large factor M to enforce

reliability. The expected NPC in (30) is calculated as the discounted sum of total system costs over the planning horizon, weighted by the probability π_s of each scenario, providing a comprehensive long-term financial metric. The LCOE, defined in (31), is the ratio of total discounted lifecycle costs (CAPEX, O&M, variable, and fuel costs) to total electricity generation $\sum_{h,n} g_{i,t,h,n,s}$, with a small constant ϵ to avoid division by zero, representing marginal cost per megawatt-hour. Environmental and spatial impacts are quantified through discounted emissions in (32) and land requirements in (33). Total annual emissions $E_{t,s}$ and land occupation $\text{Land}_{t,s}$ are summed over time and discounted to the base year t_0 , enabling simultaneous evaluation of economic, environmental, and spatial performance within the multi-scenario optimization framework.

$$\begin{cases} \text{CAPEX}_{i,t,n,s}^{ann} = \alpha_{i,t} c_{i,t,n,s}^{cap} x_{i,t,n,s}^{add} \\ \alpha_{i,t} = \frac{\delta(1 + \delta)^{L_i}}{(1 + \delta)^{L_i} - 1} \end{cases} \quad (26)$$

$$\text{O\&M}_{i,t,h,n,s} = c_{i,t}^{fix} \cdot C_{i,t,n,s} \quad (27)$$

$$\widetilde{\text{Fuel}}_{t,s}^{cost} = \sum_n \sum_h \sum_i \left(c_{i,t}^{var} g_{i,t,h,n,s} + \sum_f (p_{f,t,s} \cdot \mathbf{1}_{f(i)=f} \text{Fuel}_{i,t,h,n,s}) \right) \quad (28)$$

$$\begin{aligned} \text{Cost}_{t,s} = & \sum_n \sum_i \left(\text{CAPEX}_{i,t,n,s}^{ann} + \text{O\&M}_{i,t,h,n,s} \right) + \widetilde{\text{Fuel}}_{t,s}^{cost} \\ & + C_{t,s}^{trans} + M \sum_{n,h} \text{unmet}_{t,h,n,s} \end{aligned} \quad (29)$$

$$\begin{cases} \text{NPC}_s^{\text{expected}} = \sum_{s \in S} (\pi_s \cdot \text{NPC}_s) \\ \text{NPC}_s = \sum_{t \in T} \frac{\text{Cost}_{t,s}}{(1 + \delta)^{t-t_0}} \end{cases} \quad (30)$$

$$\text{LCOE}_{i,t,s} = \frac{\text{CAPEX}_{i,t,s}^{ann} + \text{O\&M}_{i,t,s} + \sum_{h,n} (c_{i,t}^{var} g_{i,t,h,n,s} + p_{f(i),t,s} \text{Fuel}_{i,t,h,n,s})}{\sum_{h,n} g_{i,t,h,n,s} + \epsilon} \quad (31)$$

$$\text{Emissions}_s^e = \sum_t \frac{E_{t,s}}{(1 + \delta)^{t-t_0}} \quad (32)$$

$$\text{Land}_s^e = \sum_t \frac{\text{Land}_{t,s}}{(1 + \delta)^{t-t_0}} \quad (33)$$

The hybrid LEAP-NEMO optimization framework employs a multi-objective MILP formulation to identify the optimal long-term energy mix of Morocco across multiple policy and technology scenarios. The optimization simultaneously minimizes the expected Net Present Cost (NPC), cumulative discounted emissions, and land requirements as expressed in (34), where scenario-weighted objectives (π_s) represent probabilistic expectations over uncertain socio-economic and policy trajectories. To clarify the treatment of multiple objectives, the tri-objective problem in (34) is not solved through simple weighted aggregation nor by converting secondary objectives into fixed constraints; instead, trade-offs are explicitly explored through the construction of a Pareto frontier. In practice, the model is solved iteratively using an ϵ -constraint approach, where one objective (NPC) is minimized while the remaining objectives (discounted emissions and land use) are sequentially tightened through parametric bounds. By varying these bounds across feasible ranges, a discrete set of Pareto-efficient solutions is generated, each representing a non-dominated configuration in the objective space. This procedure ensures that emissions and land indicators are treated as true competing objectives rather than merely binding constraints, and it allows explicit visualization and quantitative assessment of marginal trade-offs. Consequently, the reported results correspond to selected Pareto-optimal points, consequently, the reported results correspond to selected Pareto-optimal points. This tri-

objective formulation captures the trade-offs between economic efficiency, environmental performance, and spatial sustainability, with the Pareto-optimal frontier reflecting the feasible transition pathways of Morocco toward decarbonization. The resulting Installed Capacity Mix in (35) aggregates total installed capacities $C_{i,t,n,s}$ across regions and scenarios, yielding the expected capacity configuration by technology i and year t , effectively describing national-scale infrastructure expansion. Similarly, the Generation Mix in (36) integrates hourly or time-slice generation $g_{i,t,h,n,s}$ across all nodes, time periods, and scenarios, reflecting the contribution of each technology to Morocco's evolving power generation portfolio. The Energy Balance constraint set in (37) defines the equilibrium between generation, imports, exports, and storage dispatch at each time step, ensuring supply-demand consistency within each node and scenario, core to system feasibility in the MILP formulation of NEMO. The Energy Pathways expression in (38) cumulatively sums technology additions $x_{i,t,n,s}^{add}$ up to time t , providing a chronological trace of infrastructure evolution, expansion sequencing, and technological diffusion patterns, key for temporal investment analysis. Finally, the Land Requirement function in (39) quantifies the total land footprint of renewable technologies, where κ_i denotes land-use intensity (ha/MW) and $C_{i,t,n,s}$ represents installed renewable capacity, capturing spatial trade-offs between energy expansion and land availability.

$$\min \left(\text{NPC}_s^{\text{expected}}, \sum_s \pi_s \text{Emissions}_s^s, \sum_s \pi_s \text{Land}_s^s \right) \quad (34)$$

$$\text{Installed_Capacity_Mix}_{i,t} = \sum_n \sum_s \pi_s C_{i,t,n,s} \quad (35)$$

$$\text{Generation_Mix}_{i,t} = \sum_n \sum_s \pi_s \sum_h g_{i,t,h,n,s} \quad (36)$$

$$\text{Energy_Balance} = \left\{ g_{i,t,h,n,s}, \text{imp}, \text{exp} \right\}_{t,h,n,s} \quad (37)$$

$$\text{Energy_Pathways} = \left\{ \sum_{\tau \leq t} x_{i,\tau,n,s}^{add} \right\}_{i,t,n,s} \quad (38)$$

$$\text{Land_Requirement}_t = \sum_{i \in \text{RE}} \kappa_i C_{i,t,n,s} \quad (39)$$

To ensure transparency regarding the derivation of land-use coefficients, the parameter κ_i is defined as the total gross site area per unit of installed capacity and explicitly includes (i) inter-turbine or inter-row spacing requirements (for wind and solar PV, respectively), (ii) auxiliary and balance-of-plant infrastructure such as access roads, inverter stations, substations, and maintenance areas, and (iii) mandatory environmental setback distances and exclusion buffers (e.g., for protected areas or safety corridors). The coefficients were derived from a synthesis of empirical project-level data and spatial planning guidelines, using reported project footprints divided by installed capacity to obtain technology-specific averages. For wind energy, the effective land envelope reflects rotor-diameter-based spacing (typically 5–10D in the prevailing wind direction and 3–5D laterally), resulting in a gross allocation that exceeds the physically sealed surface but represents the spatial exclusion zone relevant for land-use planning. For solar PV, the coefficients incorporate row spacing adjusted for tilt angle and shading constraints, including internal service roads and transformer pads. Hydropower and biomass coefficients account for reservoir surface area or plant-site boundaries, respectively. Sensitivity testing with $\pm 20.0\%$ variation in κ_i indicates proportional shifts in total land outcomes but negligible changes in the cost-optimal technology mix ($< 2.0\%$ variation in installed capacity), confirming that while absolute land estimates depend on spatial design assumptions, the comparative trade-offs along the Pareto frontier remain robust.

Simulation software

The execution of the multi-scenario optimization framework for the future energy mix of Morocco relied on a coordinated use of complementary software platforms, each performing a specific function within the data-model integration workflow. The combined Excel-MATLAB-LEAP-NEMO setup ensures analytical precision alongside modeling flexibility. Excel and MATLAB handle empirical data processing and computational tasks, while LEAP and NEMO provide the scenario management and optimization layers.

Microsoft Excel served as the primary environment for data organization and preprocessing, enabling structured compilation, cleaning, and harmonization of datasets spanning the 2000–2022 historical period. These datasets included sectoral energy demand, fuel consumption, installed generation capacities, technology-specific performance parameters, cost trajectories, and renewable resource potentials. The analytical tools of MS Excel allowed temporal alignment and normalization of heterogeneous sources into consistent time-series inputs for simulation, while embedded VBA functions facilitated automatic cross-checks and updates under varying policy or macroeconomic scenarios.

MATLAB functioned as a scientific computation and visualization platform to support both pre-optimization and post-optimization stages. Prior to optimization, MATLAB scripts generated synthetic projections, including peak load evolution, sectoral demand growth, and fuel price trajectories, employing stochastic simulations and autoregressive models to capture volatility and scenario uncertainty. Post-optimization, MATLAB enabled detailed analysis and visualization of outputs such as generation profiles, capacity expansion trajectories, and sensitivity studies.

The Long-range Energy Alternatives Planning System (LEAP) acted as the scenario management and energy accounting platform within the hybrid framework. Its hierarchical tree-based structure allowed the definition and comparison of alternative pathways reflecting energy policy targets, socioeconomic projections, and technology adoption strategies in Morocco. Sectoral demand projections were treated as scenario-dependent exogenous variables, while the supply module represented conversion and transformation processes for each technology option. LEAP also performed emission accounting and energy balance validation, leveraging its integrated environmental database to quantify GHG emissions across sectors under each scenario.

The Next Energy Modeling System for Optimization (NEMO) served as the computational engine for MILP-based optimization. Using techno-economic inputs exported from LEAP, NEMO conducts multi-period, scenario-based optimization to determine optimal capacity expansion, generation dispatch, and investment decisions over the 2030–2050 planning horizon. The solver incorporates endogenous technology learning, stochastic fuel price projections, and renewable resource constraints, ensuring convergence to globally optimal solutions consistent with the strategic policy objectives of Morocco. Optimized outputs - including installed capacity mix, generation profiles, cost breakdowns, and emissions - are then re-imported into LEAP for visualization, comparative analysis, and scenario evaluation.

Approach and outcomes

This section summarizes the practical implementation of the multi-scenario optimization framework, describing how the modeling structure, input datasets, and optimization procedures are integrated to produce coherent, policy-relevant projections. It presents the workflow through which historical data, scenario specifications, and model constraints are processed within the LEAP-NEMO platform to project the evolution of the future energy mix in Morocco. The section specifies the development of the underlying database and the procedures for data standardization that ensure interoperability, accuracy, and quality control (Section 2.3.1), the construction and parameterization of policy and technology scenarios used to guide the simulations (Section 2.3.2),

the primary modeling assumptions that define system boundaries, temporal resolution, and operational dynamics (Section 2.3.3), and the synthesis of model outputs in terms of optimal installed capacities, generation portfolio, land footprint, and nodal system balance under each energy scenario (Section 2.3.4). Serving as a core component of the Materials and Methods, this section operationalizes the mathematical formulation into a reproducible analytical framework, connecting quantitative data inputs with scenario-based optimization results.

Database and standards

The database framework supporting the multi-scenario optimization of Moroccan future energy mix was developed using a set of authoritative, multi-source datasets and internationally recognized standards to ensure both analytical consistency and model interoperability.

The International Energy Agency (IEA) supplied macro-level energy balances, national statistics, and long-term projections, which were critical for historical calibration and verification of sectoral and fuel-specific energy consumption. In parallel, the Ministry of Energy Transition and Sustainable Development (MTEDD) provided official reports on national energy and climate strategies, forming the foundation for scenario design and policy constraint definition. Technical and operational datasets, including installed generation capacity, grid expansion, and electricity dispatch, were obtained from the National Office of Electricity and Drinking Water (ONEE), while ANRE supplied official information on market structure, tariffs, and regulatory frameworks influencing the development of Morocco's liberalized power sector. At the modeling stage, the LEAP Model Database functioned as the primary repository for sectoral demand breakdown, baseline energy accounting, and scenario management, whereas the NEMO Model Repository contained high-resolution techno-economic and operational parameters required for mixed-integer linear programming optimization.

All data were standardized and cross-checked following ISO 15600:1997 guidelines on Technical Energy Systems and Basic Concepts, ensuring consistent system boundaries and accurate energy flow representation. Additionally, LEAP-NEMO interoperability protocols were implemented to preserve data integrity during the transfer of time-series inputs, technology characteristics, and scenario configurations between the two platforms. This comprehensive data management and standardization approach ensured reproducible results and reinforced the technical reliability of the multi-scenario optimization framework.

Scenario designs

Developing multiple, technically consistent scenarios was a key methodological step, enabling systematic evaluation of alternative energy transition pathways for Morocco under varying policy, economic, and technological conditions. Given the rapid growth in demand, emerging renewable sectors, and ambitious national decarbonization objectives, distinct scenarios ensured that the modeling framework could explore a wide range of potential futures. Each scenario represents a specific combination of policy measures and technology assumptions, providing a structured basis for analyzing trade-offs across cost, emissions, energy security, and land requirements. The scenario set - Business-as-Usual (BAU), Net Zero Emissions (NZE), Halfway to Clean Energy (HTCE), National Energy Plan - NEMO Optimized Mix (NEP-NEMO), and National Energy Plan - PULP Optimized Mix (NEP-PULP) - covers the spectrum from conservative, fossil-intensive pathways to fully decarbonized, renewable-dominant trajectories. These scenarios are explicitly aligned with Moroccan NDC targets to avoid conceptual overlap: BAU corresponds to the BAU-NDC reference trajectory, NZE reflects the Green Development (GD-NDC) pathway, and the Increased Ambition (IA-NDC) trajectory is represented through two optimization variants (HTCE and NEP), which share similar emission reduction objectives but differ in technological assumptions, policy enforcement, and transition pace. This scenario architecture provides comprehensive coverage of national energy strategies while enabling sensitivity and uncertainty analysis for long-term planning.

The BAU scenario models the inertial evolution of the Moroccan energy system, assuming continuation of current policies, investment patterns, and consumption trends. It maintains the existing fuel mix, with only gradual renewable deployment, limited electrification in transport and industry, and ongoing dependence on fossil imports. BAU functions as the reference case against which policy interventions and technological advancements are evaluated.

The NEP scenarios align with official national strategies, including NDCs and the National Energy Strategy 2030–2050. Two complementary optimization approaches are applied. The NEP-PULP Optimal Mix uses linear programming through the PULP Python library to minimize total system cost under policy constraints, ensuring renewable penetration targets of 52.0% by 2030, 70.0% by 2040, and 80.0% by 2050. Technologies include solar PV, CSP, onshore wind, hydropower, and Power-to-X, producing a deterministic least-cost representation of Morocco's energy roadmap. In contrast, the NEP-NEMO Optimal Mix integrates dynamic dispatch and reliability assessment within the LEAP-NEMO hybrid platform. It simulates temporal system evolution, including demand growth, technology retirements, and capacity additions, optimizing cost, emissions, and system adequacy over time. This approach provides a more operationally constrained perspective, emphasizing grid stability, dispatchability, and resource complementarity.

The HTCE scenario represents a transitional pathway, targeting a 50.0% reduction in fossil fuel use and capacity by 2050 relative to BAU. It balances emissions mitigation with energy security, reflecting a pragmatic strategy that accounts for infrastructure inertia and technological uncertainty. HTCE assumes accelerated renewable deployment, moderate electrification in transport and industry, and partial integration of storage and demand-side flexibility.

The NZE scenario represents the most ambitious pathway, aiming for complete decarbonization of the energy system by 2050. It assumes full fossil fuel phase-out, 100.0% renewable electricity generation, and deep electrification across all sectors. Large-scale solar PV and wind deployment is complemented by CSP with thermal storage, pumped hydro, and long-duration energy storage systems to maintain reliability. Carbon-neutral fuels and land-use optimization are incorporated to support renewable expansion and operational feasibility.

Together, these five scenarios establish a multi-dimensional framework that integrates policy realism, technological feasibility, and economic optimization. Using LEAP and NEMO, each scenario generates quantitative projections for capacity expansion, emissions reduction, and land-use dynamics. Table 2 provides a technical summary of the modeling scenarios for future energy mix optimization.

Modeling assumptions

The modeling assumptions for this study establish the structural, temporal, and technological constraints within which Morocco's future energy system is optimized. These assumptions ensure that multi-scenario simulations remain internally consistent, comparable, and reflective of plausible national and international developments. They cover demographic, techno-economic, and policy dimensions, encompassing energy demand evolution, technology cost trajectories, renewable integration rates, and sectoral transformation pathways. The methodological goal is to maintain alignment between the LEAP accounting framework, which manages demand and scenario evolution, and the NEMO optimization engine, which determines cost-optimal capacity expansion and dispatch, thereby ensuring coherence between empirical observations and modeled outcomes.

The first assumption relates to demand and electrification. Energy demand growth is disaggregated into four end-use sectors—residential, industrial, commercial, and transport—based on historical trends and projected growth rates. Electricity demand rises according to sector-specific elasticities, with total national consumption increasing from roughly 35 TWh in 2020 to 57 TWh by 2030, in line with NREL projections [52]. The transport sector incorporates an annual electric

Table 2

Overview of modeling scenarios for future energy mix optimization of Morocco.

Scenario ID	Scenario Name	Renewable Energy Targets	Optimization Objective	Policy Alignment	Time Horizon	Expected Outcomes
S1	Business-as-Usual (BAU)	None (incremental growth)	Demand-supply balance	Current national policies	2022–2050	Baseline comparison
S2	Halfway to Clean Energy (HTCE)	~50.0% renewable share	Cost-emissions trade-off	Transitional/mitigation strategy	2022–2050	Moderate decarbonization
S3	National Energy Plan - PULP Optimized Mix (NEP-PULP)	52.0% by 2030, 70.0% by 2040, 80.0% by 2050	Minimize system cost	Aligned with NDCs	2022–2050	Least-cost policy compliance
S4	National Energy Plan - NEMO Optimized Mix (NEP-NEMO)	Same as NEP-PULP	Minimize cost, emissions, and land	Aligned with national energy strategy	2022–2050	Optimal dynamic mix
S5	Net Zero Emissions (NZE)	100.0% renewable by 2050	Minimize emissions & land	Beyond NDCs (2050 target)	2022–2050	Net-zero system design

vehicle adoption rate of 10.8% [53,54], gradually replacing fossil fuels in final energy use. Electrification of industry [55] and services [56] is modeled through linear scaling linked to GDP and population growth.

The second assumption concerns generation capacity and technology. Installed capacities evolve according to technology lifetimes, availability factors, and policy objectives. Fossil-based units operate until the end of life, with limited reinvestment under BAU and phased decommissioning in HTCE, NEP, and NZE scenarios. Renewable technologies—including solar PV, CSP, onshore wind, hydro, and biomass—follow capacity expansion trajectories aligned with national targets: 52.0% share of renewable capacity by 2030, 70.0% by 2040, and 80.0% by 2050. Technology efficiencies and derating factors are applied dynamically to capture degradation and operational constraints. A system reserve margin of 15.0% is imposed across all scenarios, supported by Moroccan grid stability studies [8,38] and consistent with regional reliability standards [57,58]. This margin ensures adequacy under peak demand and accounts for renewable variability and forced outages.

Third assumption is in liaison with economic and cost. Fuel price trajectories for coal, oil, natural gas, diesel, and imported electricity are taken from 2000–2022 historical data and projected using scenario-based or autoregressive models. Capital costs (CAPEX) and operating costs (O&M) evolve according to technology-specific learning rates, represented through power-law functions with learning exponents derived from global datasets (IEA, IRENA). While NPC is reported as an aggregate metric, the model internally decomposes total system cost into annualized CAPEX, fixed O&M, variable O&M, fuel expenditures, transmission expansion costs, and penalty costs for unmet demand. Technology-level cost contributions are extracted from (26)–(29), enabling disaggregated analysis of cost drivers across scenarios. In BAU, fuel expenditures account for approximately 45.0–55.0% of total discounted costs, whereas in NZE, CAPEX dominates (60.0–70.0%) due to renewable and hydrogen infrastructure expansion, with fuel costs declining below 10.0%. This structural shift highlights the transition from fuel-intensive to capital-intensive system design and improves policy interpretability beyond aggregate NPC reporting. Furthermore, the model incorporates discount rates of 4.0–6.0% depending on scenario risk, with technology lifetimes between 20 and 40 years. The choice of a 4.0–6.0% real discount rate reflects the blended public-private financing structure of Morocco for large-scale energy infrastructure, consistent with international development bank lending conditions and national energy planning practice. Lower values ($\approx 4.0\%$) represent concessional financing conditions typical for renewable megaprojects [8], while higher values ($\approx 6.0\%$) approximate weighted average cost of capital for mixed public-private investments [7,50]. Because long-lived renewable and hydrogen assets are sensitive to discounting, a sensitivity test was performed at 3.0%, 5.0%, and 8.0%. Increasing the discount rate to 8.0% reduces optimal cumulative renewable capacity by approximately 6.0–9.0% in NZE and delays hydrogen scale-up by 3–5 years, confirming the structural importance of capital cost assumptions. However, relative scenario rankings along the Pareto frontier remain unchanged, indicating robustness of comparative conclusions.

Fourth assumption is in liaison with green hydrogen integration. As Morocco plans to gradually integrate green hydrogen in the national energy mix, hydrogen production is optimized given optimal end-use application and export demands. Hydrogen production relies primarily on surplus renewable electricity from PV and wind, constrained by electrolyzer capacity limits and water availability factors. Electrolyzer efficiency is modeled as 70.0% based on the lower heating value (LHV) of hydrogen, corresponding to an electrical-to-chemical energy conversion of 50–55 kWh/kg H₂. Stack efficiency is assumed at 72.0% for PEM electrolyzers and 68.0% for alkaline electrolyzers, with auxiliary system losses (pumps, compressors, and power electronics) reducing overall system efficiency to 70.0%. Electrolyzer degradation over time is modeled at 0.5% per year, and operational flexibility allows ramping between 0–100.0% of nominal capacity with a minimum stable load of 20.0%, enabling responsive operation to renewable generation fluctuations. Annual capacity factors are assumed to range from 3,000 to 4,000 full-load hours depending on renewable availability, storage interactions, and grid constraints. The model ensures dispatch coordination with the power system to avoid curtailment of variable renewables, prioritizing hydrogen production during periods of excess generation while respecting grid stability and reserve margin requirements [59–61]. Given technological uncertainty in electrolyzer development, fixed parameter assumptions are justified using current commercial performance benchmarks (IEA and IRENA datasets 2022–2024) reflecting mature alkaline and PEM systems [47]. However, to assess robustness, a structured sensitivity analysis was conducted varying (i) system efficiency between 65.0–75.0%, (ii) stack lifetime between 60,000–90,000 operating hours (equivalent to 15–25 years depending on capacity factor), and (iii) degradation rates between 0.3–1.0% annually. Results indicate that hydrogen deployment volumes in the NZE scenario vary by less than 8.0% across this parameter space, while total system NPC shifts by less than 3.0%, suggesting that hydrogen penetration levels are primarily driven by renewable surplus availability and carbon constraints rather than marginal electrolyzer performance improvements [47,62]. Additionally, water availability is incorporated explicitly as a quantitative constraint in the hydrogen module. For each kilogram of H₂ produced, 9 liters of deionized water are required as stoichiometric input, with additional process losses bringing total freshwater demand to approximately 10–12 liters/kg [62]. Regional water availability limits are introduced as annual caps on hydrogen production per node, derived from non-agricultural allocable water resources. In water-stressed regions, desalination-powered supply is permitted but incurs additional electricity demand and cost penalties, thereby internalizing water scarcity into system optimization. Model results indicate that water constraints become binding only in the high-export NZE configuration, where up to 6.0–9.0% of renewable electricity is diverted to desalination [8], slightly increasing LCOE but not altering the overall renewable-dominant structure of the optimal mix.

Fifth assumption is related to system and environment. Transmission and distribution (T&D) losses are fixed at 10.0–12.0% of total electricity output, with possible reductions under advanced grid modernization in NEP and NZE scenarios. Storage technologies (batteries, CSP thermal

storage, and pumped hydro) are modeled with round-trip efficiencies between 75.0% and 90.0%. Carbon emission factors are based on IPCC guidelines, while land-use coefficients are defined per technology (ha/MW), enabling integrated estimation of land requirements for renewables. Land-use coefficients are defined per technology (ha/MW) and calibrated for the context of Morocco, enabling integrated estimation of spatial requirements for renewable energy expansion. Given solar PV: 2–3 ha/MW, CSP with thermal storage: 5–7 ha/MW, onshore wind: 0.05–0.1 ha/MW (footprint only, excluding spacing), biomass: 1–2 ha/MW, and small hydro: 0.5–1 ha/MW, the coefficients account for the direct land occupation by infrastructure, with additional buffers for access roads, maintenance areas, and spacing where relevant. Spatial constraints are incorporated to respect potential land availability conflicts, protected natural areas (national parks and reserves covering around 1.5% of Moroccan territory), and competing land uses including agriculture, urban expansion, and water resources. To operationalize spatial exclusions in a non-aggregated manner, land constraints are implemented at the regional-node level within NEMO rather than as a single national scalar limit. For each node n , the maximum buildable renewable capacity is bounded by a technology-specific spatial availability parameter derived from GIS-based exclusion mapping. Protected areas (national parks, reserves, and Ramsar wetlands), high-productivity agricultural zones, urbanized land, and water bodies are removed from the gross technical potential prior to optimization. The remaining “available land envelope” defines the regional build limit $limit_{i,t,n,s}$ in constraint (15), thereby transforming spatial exclusions into explicit capacity bounds within the MILP formulation. This ensures that land-use conflicts are not treated abstractly but directly constrain siting decisions and renewable deployment density at the nodal level. Consequently,

renewable expansion is geographically feasible and respects protected-area coverage (~1.5% of Moroccan territory) as well as competing land uses. Scenario-specific land-use planning is applied: BAU and HTCE scenarios allow incremental expansion within unconstrained areas; NEP-PULP and NEP-NEMO optimize renewable siting under explicit land availability limits; NZE maximizes renewable deployment while respecting spatial restrictions and trade-offs with agricultural and urban areas.

Final assumption, policy and scenario constraints, all modeled pathways are consistent with the NDCs of Morocco and the National Energy Strategy (2030–2050). Scenarios differ by the intensity of policy enforcement and technological uptake. NEP-NEMO and NEP-PULP are constrained by renewable share mandates, while NZE enforces a strict fossil fuel phase-out constraint by 2050. HTCE introduces a 50.0% fossil reduction benchmark, serving as a transitional scenario between BAU and NZE.

The modeling framework for the future energy mix optimization of Morocco relies on a structured set of quantitative and qualitative assumptions that define the techno-economic and policy landscape of each scenario. Electricity demand is projected to increase from approximately 55 TWh in 2020 to 77 TWh by 2030, reflecting a sustained annual growth of 2.5–4.0%, driven by industrial expansion, urbanization, and the electrification of transport. Renewable capacity targets, 52.0% by 2030, 70.0% by 2040, and 80.0% by 2050, serve as core policy constraints in both the NEMO and PULP optimization frameworks. Technology lifetimes range from 20 years for solar PV to 40 years for hydropower, while capacity factors vary between 15.0% and 45.0% depending on resource availability and technology type. The discount rate is set between 4.0% and 6.0%, capturing scenario-dependent

Table 3
Summary of key modeling assumptions.

Category	Parameter / Variable	Assumption / Description	Unit / Value Range	Source / Basis
Demand and Electrification	Sector energy demand growth	Disaggregated by sector (residential, industrial, commercial, transport) following GDP and population trends	+2.5–4.0% per year	[7,10]
	National electricity demand	Projected to grow from 35 TWh (2020) to 57 TWh (2030)	TWh	[53]
Generation Capacity and Technology	EV adoption rate	Annual increase in electric vehicle stock	10.8% per year	[54]
	Renewable capacity targets	52.0% by 2030, 70.0% by 2040, 80.0% by 2050	% of total installed capacity	[7,11]
	Technology lifetimes	Average plant lifetime by technology	20–40 years	[8]
	Capacity factors	Technology-specific, dynamic with resource and degradation	15.0–45.0%	[8,63]
	Storage efficiency	Round-trip efficiency of battery, CSP thermal, and pumped hydro	75.0–90.0%	[8]
Economic and Cost Parameters	Reserve margin	System reserve margin for all scenarios to ensure reliability under peak load and renewable variability	15.0% of peak load	[58,59]
	Discount rate	Scenario-dependent cost of capital	4.0–6.0%	[64,65]
	CAPEX and O&M evolution	Learning curve applied to each technology (β_i based on LR_i)	Dynamic (–1.5% to –5.0% per doubling)	[8]
	Fuel price projection	Scenario-based and/or autoregressive model	MAD/unit	[51]
Green Hydrogen Integration	LCOE benchmarks	Levelized cost by technology (updated yearly)	MAD/MWh	[8]
	H ₂ demand (moderate)	Domestic and export demand for moderate uptake	14–155 PJ	[6,66]
	H ₂ demand (high)	Domestic and export demand for high uptake	30–310 PJ	[6,67]
	Electrolyzer efficiency	Assumed efficiency for all electrolyzers	70.0%	[60,61]
	Electrolyzer capacity factor	Annual operating hours depending on renewable availability	3,000–4,000 h/year	[61,62]
System and Environmental	Integration pathways	Five: All Domestic, Mostly Domestic, Half Export, Mostly Export (H _{2g}), All Export	-	[6,66–68]
	Transmission and distribution losses	Fraction of generated electricity lost in transmission	10.0–12.0%	[45,49,50]
	Emission factors	Technology-specific, IPCC 2021 default values	tCO _{2e} /MWh	[69,70]
	Land-use coefficients	Land occupation per installed capacity (RE technologies), accounting for protected areas, competing land uses, and spatial restrictions	ha/MW	+ LEAP-NEMO Dataset LEAP-NEMO Dataset + spatial land constraints
Policy and Scenario Constraints	BAU	Continuation of current mix and fossil reliance	-	Baseline reference
	HTCE	50.0% reduction in fossil fuel use by 2050	-	Scenario definition
	NEP (PULP and NEMO)	Renewable targets (52.0–80.0%) with cost minimization and dispatch reliability	-	[6,8,11]
	NZE	100.0% renewable, zero-emission system by 2050	-	[6]

financing conditions, and learning rates of 1.5–5.0% are applied to account for cost reductions through cumulative deployment. Transmission and distribution losses are assumed to decline gradually to 10.0–12.0%, consistent with ONEE's modernization trajectory. For green hydrogen, two market tiers are modeled: a moderate uptake of 14–155 PJ and a high uptake of 30–310 PJ, distributed across domestic consumption and export configurations. Electrolyzer efficiency is assumed at 70.0% and annual capacity factors of 3,000–4,000 hours, ensuring integration with the power system. Land-use limitations account for competing agricultural, urban, and protected areas, restricting deployment of utility-scale renewables in constrained zones. Collectively, the assumptions in Table 3 define the operational, economic, and policy boundaries of the multi-scenario optimization, ensuring realistic yet forward-looking projections of the Moroccan energy transition trajectory.

To ensure transparency and methodological rigor in evaluating trade-offs among competing objectives, the MILP optimization in NEMO employs a multi-criteria scoring framework where economic efficiency (total system cost), environmental performance (cumulative discounted emissions), demand satisfaction (load coverage), and spatial sustainability (land-use requirements) are treated as distinct but equally important objectives. Each criterion is normalized to a dimensionless scale between 0 and 1 using min-max normalization in (40), where O_i is the value of objective i for a given solution, and O_i^{min}, O_i^{max} are the observed minimum and maximum values across all scenarios and iterations. Equal weighting is applied across the four criteria ($w_i = 0.25$ for each), ensuring that no single objective dominates the optimization and that the resulting scenario scores reflect a balanced evaluation of demand fulfillment, cost efficiency, emission reduction, and land-use sustainability. While equal weighting is a neutral assumption reflecting no a priori preference among objectives, alternative weighting schemes reflecting stakeholder priorities (greater emphasis on cost efficiency or environmental performance) could shift scenario rankings; this sensitivity was tested $\pm 10.0\%$, acknowledging that context-specific stakeholder perspectives may yield different relative preferences. The aggregate multi-criteria score for each scenario is then computed as a weighted sum of normalized objectives, as in (41). This transparent procedure allows the high scores observed for the Net Zero scenario to be interpreted as a balanced outcome across all four objectives, rather than as an artifact of subjective weighting or inconsistent normalization. Additionally, sensitivity analyses were conducted by varying weights $\pm 10.0\%$ to confirm that scenario rankings remain robust.

$$O_i^{norm} = \frac{O_i - O_i^{min}}{O_i^{max} - O_i^{min}} \quad (40)$$

$$score_{scenario} = \sum_{i=1}^4 w_i \cdot O_i^{norm} \quad (41)$$

Modeling Outcomes

The outputs generated by the integrated LEAP-NEMO optimization framework constitute the computational consolidation of all input datasets, projections, and scenario constraints into consistent and quantifiable system trajectories. These outputs do not represent the interpretative results themselves; rather, they are the structured deliverables of the optimization phase that provide the analytical basis for subsequent evaluation. Four principal categories of outputs are produced - Installed Capacity Mix, Energy Balance and Distribution, Land Requirements, and Optimal Energy Pathways - each reflecting a specific analytical perspective of the future energy system in Morocco.

The Installed Capacity Mix aggregates the optimized configuration of generation assets by technology, year, and scenario. It is derived from the multi-objective minimization of net present cost, emissions, and land footprint as previously formulated. This output determines the cost-optimal and technically feasible expansion path $C_{i,t,h,n,s}$ that satisfies sectoral energy demand $D_{d,t,h,n,s}$, reliability conditions, and renewable penetration targets. The resulting capacity trajectories describe the

structural transformation of the national generation portfolio under both deterministic linear programming through PULP and mixed-integer linear programming through NEMO.

The Energy Balance and Distribution outputs provide a detailed accounting structure that guarantees temporal and spatial equilibrium for each time slice (t, h, n, s) . They represent the interactions among generation $g_{i,t,h,n,s}$, storage operations $p_{k,t,h,n,s}^{chg/dis}$, electricity imports $imp_{t,h,n,s}$, and exports $exp_{t,h,n,s}$, while incorporating conversion and transmission losses. This accounting structure enables LEAP to reconstruct Sankey-based energy flow diagrams for each scenario, tracing energy from primary supply through transformation stages to final consumption. As a result, the framework supports cross-scenario comparison of efficiency levels, loss patterns, and conversion intensities under varying policy and technology assumptions.

The Land Requirements output evaluates the spatial dimension of renewable deployment using (16); $Land_t$ captures total land occupation associated with installed renewable capacity. This indicator links expansion trajectories with land availability and suitability considerations, integrating spatial constraints directly into the system modeling process.

The Optimal Energy Pathways output consolidates the preceding components into time-resolved trajectories that reflect both physical system development and multi-criteria performance. In addition to tracking cumulative capacity additions $\sum_{\tau \leq t} x_{i,\tau,n,s}^{add}$, these pathways translate infrastructure evolution into quantitative indicators of economic performance, fossil fuel dependence, and decarbonization effectiveness. This enables direct comparison of scenario outcomes by connecting deployment patterns with evaluation metrics including NPC, LCOE, fossil fuel reliance, GHG emissions, and emission intensity factors. This integrated formulation makes it possible to identify the divergence between low-cost but high-emission trajectories (BAU), transitional mid-cost pathways with accelerated decarbonization (HTCE and NEP variants), and high-investment scenarios that deliver full decarbonization (NZE).

The methodology presented in this study therefore establishes a comprehensive analytical structure for modeling, optimizing, and assessing the future energy mix of Morocco under multiple policy and technology scenarios. As illustrated in Fig. 9, the LEAP-NEMO hybrid configuration operates as a closed-loop system that combines scenario management with mathematical optimization. The workflow begins with data compilation and harmonization from national and international institutions including IEA, ONEE, ANRE, and MTEDD, followed by scenario definition and assumption specification that shape the simulation input space. Within this structure, LEAP performs long-term demand projection, sectoral accounting, and emissions baseline estimation, while NEMO formulates a multi-objective MILP problem that minimizes total system cost, emissions, and land occupation subject to technical and policy constraints. Standardized interoperability protocols enable consistent data exchange and iterative feedback between scenario evolution and optimization results. The process yields four categories of outputs: optimized installed capacity configuration, detailed energy balance and distribution, land requirement indicators, and comparative evaluation of energy scenarios.

Results and analysis

This section reports the quantitative outputs derived from the multi-scenario optimization framework, converting the methodological structure of the LEAP-NEMO model into a comprehensive description of potential energy transition trajectories in Morocco. The analysis reflects the techno-economic and environmental progression of the system over the modeling period 2022–2050, highlighting the transformation of the national energy mix under varying policy settings, technological developments, and demand assumptions. Each scenario - Business-as-Usual (BAU), Halfway to Clean Energy (HTCE), National Energy Plan

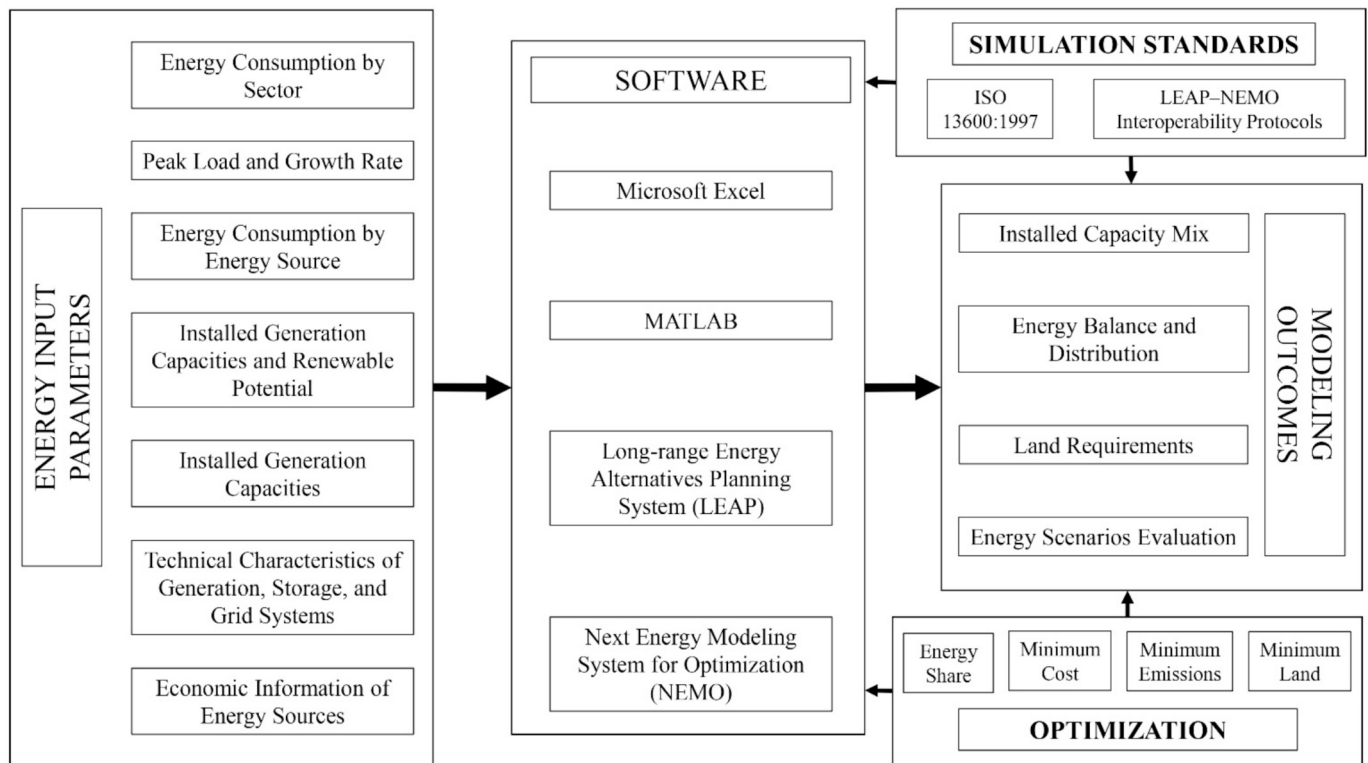


Fig. 9. Concise diagram of the methodology section

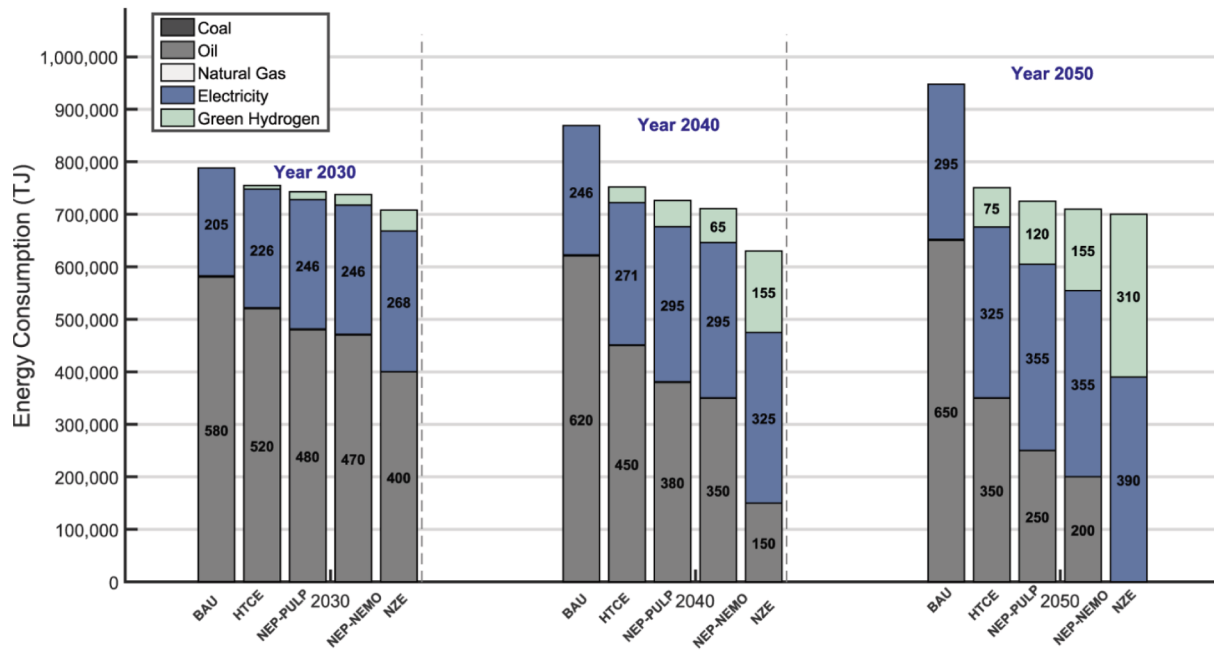
variants (NEP-PULP and NEP-NEMO), and Net Zero Emissions (NZE) - is evaluated with respect to optimized installed capacity configuration, sectoral energy balance, land requirements associated with renewable deployment, and overall scenario performance metrics. The emphasis is placed on the temporal interaction of these elements, clarifying the trade-offs among economic efficiency, emission mitigation, and land-use intensity.

Installed capacity mix

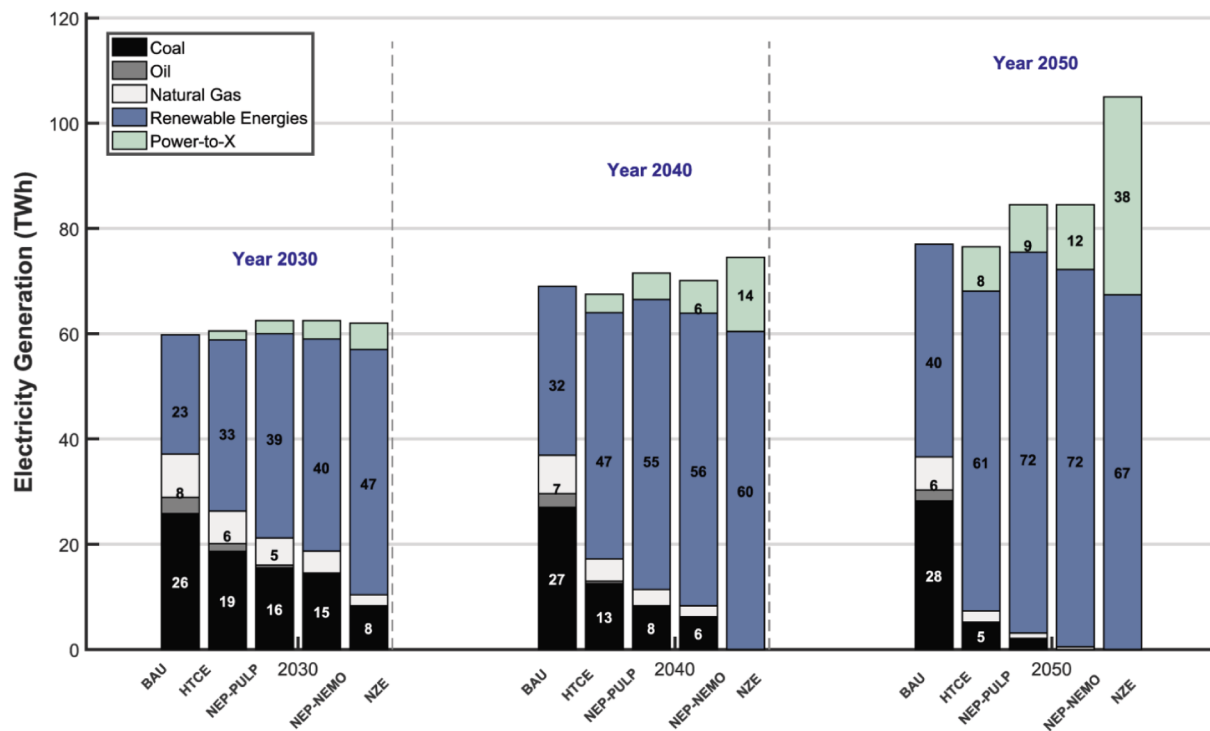
The transformation of the installed capacity structure across scenarios results directly from the LEAP-NEMO multi-scenario optimization procedure, governed by the techno-economic parameters, policy constraints, and resource availability conditions defined in the methodological framework. Within the model formulation, where system cost minimization, emissions mitigation, and land-use limitations are activated either jointly or selectively depending on the scenario, the optimization engine distributes new capacity among fossil technologies such as coal, oil, and natural gas, renewable technologies including solar PV, CSP, onshore wind, and hydropower, as well as emerging Power-to-X options, in alignment with scenario-specific policy and technology assumptions. The final configuration reflects the interaction among demand growth, projected fuel prices, renewable resource characteristics, derating coefficients, construction limits, lifetime-driven retirement dynamics, and the multi-objective MILP structure. The electricity generation mix illustrated in Fig. 10b represents the operational expression of the installed capacity configuration, constrained by hourly availability factors incorporated through derated capacity modeling. Although hourly derated capacity and Power-to-X dispatch are explicitly simulated, grid integration aspects such as operational flexibility, curtailment, ramping capability, and reserve allocation are represented in simplified form. These operational dimensions may affect practical feasibility, indicating that complementary analyses including unit commitment, frequency control, and storage sufficiency assessments would enhance interpretation of the optimization outcomes. The

renewable share of electricity generation presented in Fig. 10c, which includes solar PV, wind, CSP, hydropower, and a minor contribution from other sources such as biomass and prospective tidal technologies, arises from the combined effect of renewable capacity deployment, technology-specific capacity factors, and phased reduction of fossil generation. In parallel, the allocation of renewable technologies for green hydrogen production shown in Fig. 10d follows the optimization principle whereby Power-to-X production selects the most cost-effective and land-efficient renewable resources under the defined hydrogen demand trajectory. Collectively, Figs. 10a through 10d convert methodological inputs - such as LCOE evolution, renewable resource supply curves, mandated coal and oil retirement schedules, build-rate constraints, and hydrogen scaling parameters - into system-level capacity results that demonstrate the internal consistency and technical configuration of each scenario pathway.

The projected final energy consumption trajectories illustrated in Fig. 10a indicate clearly differentiated structural developments across scenarios, determined by varying electrification speeds, fossil fuel retirement strategies, and hydrogen deployment levels. Under the BAU configuration, total final energy demand increases steadily from 788,300 TJ in 2030 to 947,788 TJ in 2050, with oil products remaining predominant, rising from 580,000 TJ to 650,000 TJ, while electricity consumption grows moderately from 205,200 TJ to 295,488 TJ. This pathway reflects limited electrification efforts and no substantial hydrogen integration. By comparison, the HTCE configuration exhibits a markedly stronger fossil substitution pattern, where oil consumption declines from 520,000 TJ in 2030 to 350,000 TJ in 2050, electricity demand expands from 225,720 TJ to 325,037 TJ, and hydrogen reaches 75,000 TJ by 2050, representing an order of magnitude increase relative to BAU. The NEP based configurations, namely NEP PULP and NEP NEMO, reinforce this trend, leading to pronounced reductions in fossil consumption: oil decreases to 250,000 TJ and 200,000 TJ by 2050 under NEP PULP and NEP NEMO, respectively. Electricity demand rises correspondingly to 354,586 TJ in both cases, consistent with accelerated electrification and renewable expansion. Green hydrogen demand



(a) Projected Energy Mix by Scenario



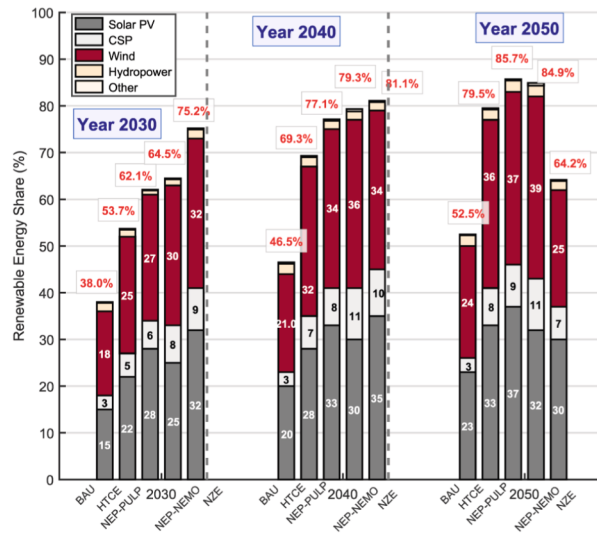
(b) Projected Electricity Mix by Scenario

Fig. 10. Projected multi-scenario installed capacity mix for future energy planning of Morocco.

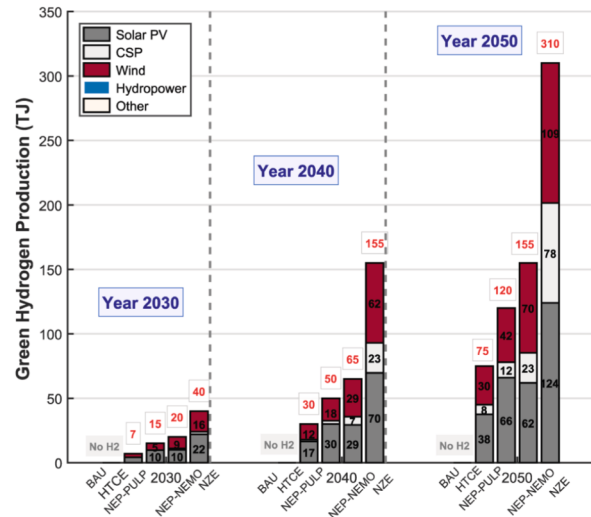
increases significantly to between 120,000 TJ and 155,000 TJ by 2050, indicating an optimization driven transition toward synthetic fuels in sectors that are difficult to decarbonize. The NZE configuration represents the most extensive transformation, with fossil fuels fully phased out by 2050, electricity consumption increasing to 390,044 TJ, and hydrogen demand reaching 310,000 TJ, forming a combined electrification and hydrogen-based structure. Across all configurations, Fig. 10a demonstrates a monotonic decline in fossil energy use as policy ambition

strengthens, while electricity and hydrogen progressively dominate final demand, with NZE showing the most substantial structural shift.

Electricity generation results in Fig. 10b further emphasize the divergence among configurations, highlighting the gradual replacement of fossil-based dispatch by renewable and Power to X generation as policy ambition intensifies. In BAU, coal remains central to electricity supply, increasing from 25,800 GWh in 2030 to 28,200 GWh in 2050, while renewable generation grows moderately from 22,700 GWh to



(c) Renewable Energy Share in Electricity Generation



(d) Technology Portion in Green Hydrogen Production

Fig. 10. (continued).

40,400 GWh, maintaining fossil dominance. Under HTCE, coal generation decreases from 18,600 GWh to 5,200 GWh, renewable output nearly doubles from 32,500 GWh to 60,800 GWh, and Power to X reaches 8,400 GWh by 2050, reflecting a system increasingly supported by synthetic fuel balancing. The NEP configurations accelerate this transformation. NEP PULP reduces coal generation to 2,100 GWh by 2050, increases renewables to 72,400 GWh, and adds 9,000 GWh from Power to X. NEP NEMO advances fossil displacement further, fully removing coal by 2050 and reducing gas generation to 500 GWh, while renewables reach 71,700 GWh and Power to X increases to 12,300 GWh, the highest level among all configurations except NZE. The NZE pathway demonstrates a comprehensive restructuring of supply, with fossil generation eliminated by 2040 and, by 2050, reliance on 67,400 GWh of renewable generation supplemented by 37,600 GWh of Power to X, resulting in the highest total electricity production of 105,000 GWh, driven by electrification and hydrogen requirements. Fig. 10b therefore captures the interaction between capacity expansion, availability constraints, and fossil retirement policies embedded in the modeling structure, with NZE representing the most extensive transition and BAU preserving fossil intensive dispatch.

The trajectories presented in Fig. 10c illustrate how renewable penetration evolves under differing policy ambitions and system constraints. In the BAU scenario, the renewable share increases gradually from 38.0% in 2030 to 52.5% in 2050, primarily driven by incremental deployment of solar PV and onshore wind. The HTCE pathway accelerates this transition, with renewable penetration rising from 53.7% to 79.5% over the same period. Solar PV expands from 22.0% to 33.0% of total generation, while wind increases from 25.0% to 36.0%, supplemented by modest but stable contributions from CSP and hydropower. The NEP scenarios exhibit even stronger renewable expansion. Under NEP-PULP, renewables reach 85.7% of total electricity supply by 2050, supported by a balanced structure comprising 37.0% PV, 37.0% wind, and 9.0% CSP. NEP-NEMO follows a comparable trajectory, achieving 84.9% renewable share by 2050, though with a slightly greater reliance on CSP (11.0%), reflecting different optimization outcomes in technology allocation. In contrast, the NZE scenario displays a distinct dynamic: renewable share peaks at 81.1% in 2040 but declines to 64.2% by 2050. This reduction does not indicate a contraction of renewable capacity; rather, it results from the rapid expansion of Power-to-X generation, which is accounted for outside the “renewables” category in the model’s accounting framework and substantially increases total electricity

output. Across all scenarios, “Other” technologies contribute only 0.2–0.6% of generation, representing limited biomass utilization and early-stage tidal initiatives included within model resource constraints. Thus, Fig. 10c reflects how renewable expansion is moderated by variability limits, hydrogen-induced electricity demand growth, and differing fossil phase-out trajectories embedded in each scenario.

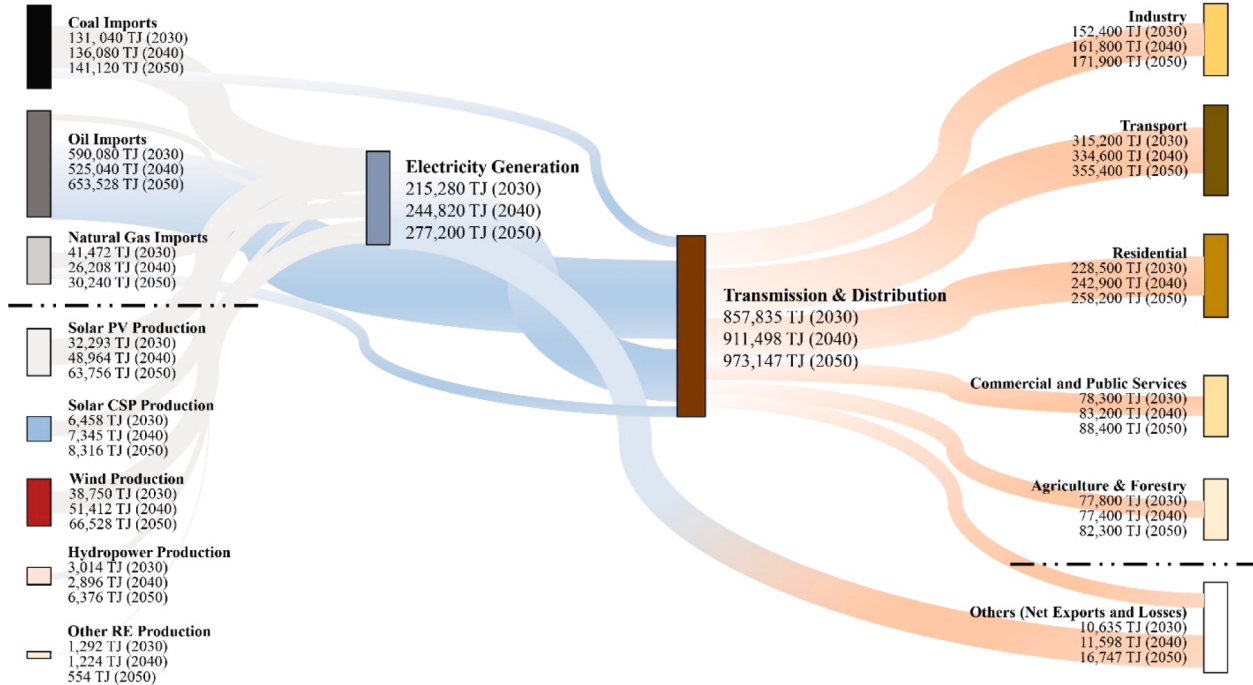
Fig. 10d further clarifies the allocation of renewable electricity to hydrogen production within scenarios that incorporate Power-to-X. Hydrogen supply scales substantially in HTCE, NEP-PULP, NEP-NEMO, and NZE, and is derived almost entirely from solar PV, onshore wind, and CSP, consistent with their comparatively low leveled costs and favorable capacity factors specified in the modeling framework. In HTCE, hydrogen production increases from 7,000 TJ in 2030 to 75,000 TJ in 2050, with approximately 37,500 TJ supplied by PV and 30,000 TJ by wind. NEP-PULP expands further to 120,000 TJ by 2050 (66,000 TJ PV; 42,000 TJ wind). NEP-NEMO reaches 155,000 TJ, where wind slightly exceeds PV (69,750 TJ versus 62,000 TJ), reflecting the model’s cost-optimization preference for wind’s higher effective capacity factor. The NZE pathway exhibits the most rapid scaling, reaching 310,000 TJ of hydrogen by 2050, supplied by 124,000 TJ PV, 108,500 TJ wind, and 77,500 TJ CSP. This broader diversification reflects the need to satisfy very high hydrogen demand while balancing land constraints and solar intermittency. Hydropower and “Other” technologies do not contribute to hydrogen production in any scenario due to limited scalability and binding resource ceilings.

Taken together, Figs. 10a–10d describe a consistent system transformation under progressively stronger policy ambition. BAU preserves a fossil-intensive structure with moderate renewable growth and no hydrogen integration. HTCE initiates partial decarbonization, combining substantial renewable expansion with moderate hydrogen deployment. The NEP pathways achieve deeper structural change, reducing fossil fuels to marginal roles by 2050, raising renewable penetration above 80.0%, and positioning hydrogen (120,000–155,000 TJ) as a central energy carrier. NEP-NEMO achieves slightly stronger integration outcomes than NEP-PULP due to more aggressive cost and resource optimization within the modeling framework. The NZE scenario represents the most transformative pathway, eliminating fossil generation, maximizing electricity output, and deploying hydrogen at unprecedented scale. Importantly, the observed decline in renewable share in NZE by 2050 reflects accounting structure rather than reduced renewable deployment, as electrolysis-related electricity flows alter the

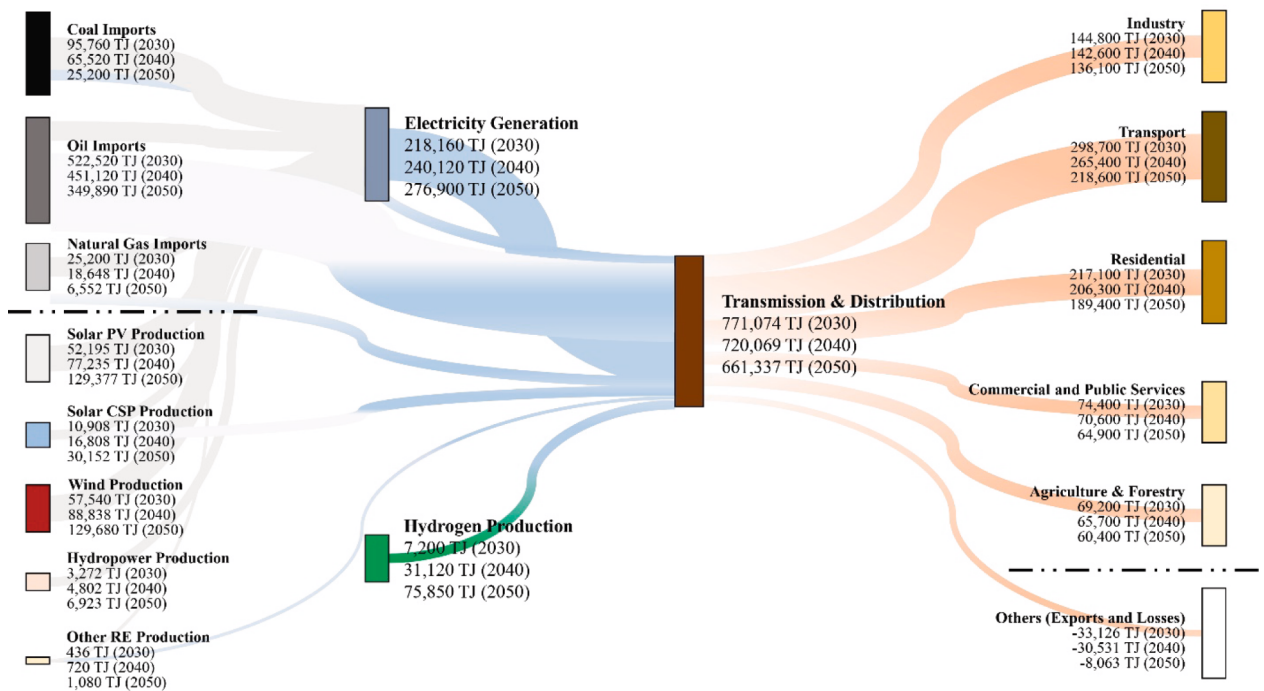
denominator of total generation. Overall, the results demonstrate that Morocco's long-term optimal energy configuration is highly scenario-dependent, with renewable electricity and green hydrogen forming the structural foundation of deep decarbonization pathways, shaped by resource availability, technology performance, and system-level optimization constraints.

Energy balance and distribution

The projected configuration of energy balances and flow distributions across the scenarios shown in Fig. 11 is primarily governed by the fundamental assumptions embedded in the LEAP NEMO optimization structure. These include rapid electrification, improvements in end use efficiency, progressive deployment of renewable technologies, and the

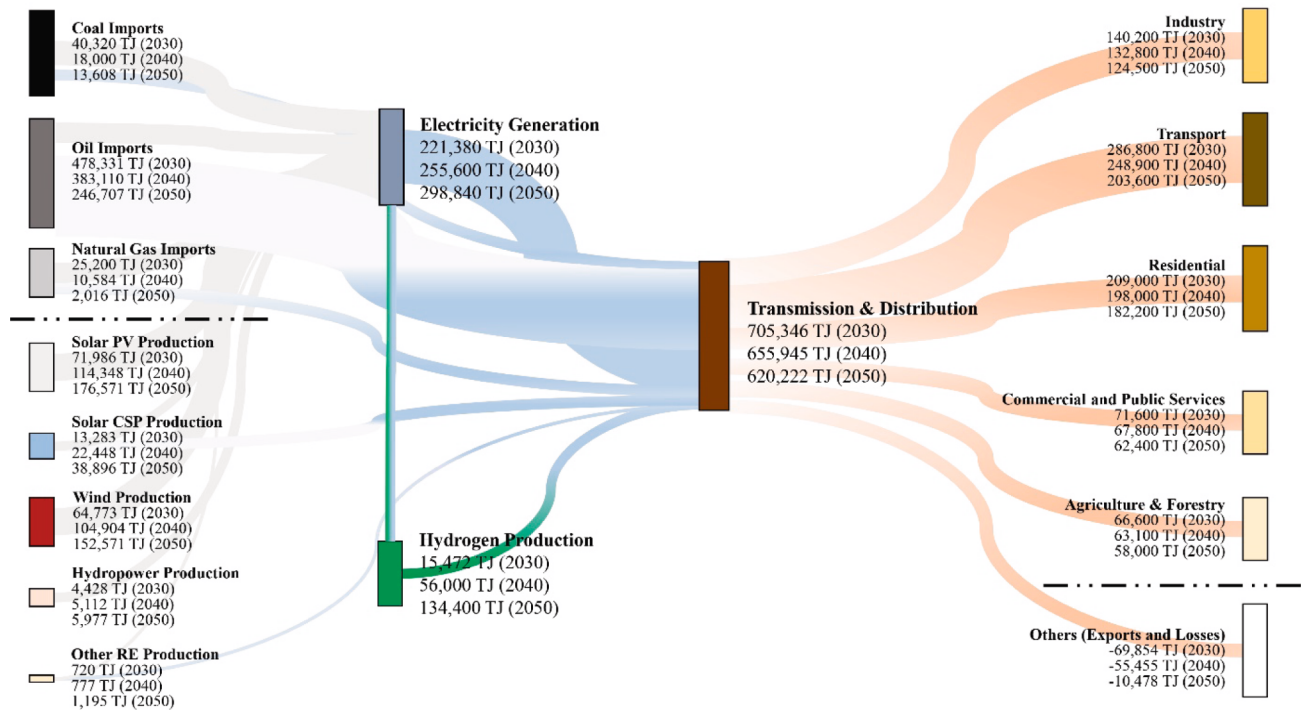


(a) Sankey Diagram - BAU Scenario

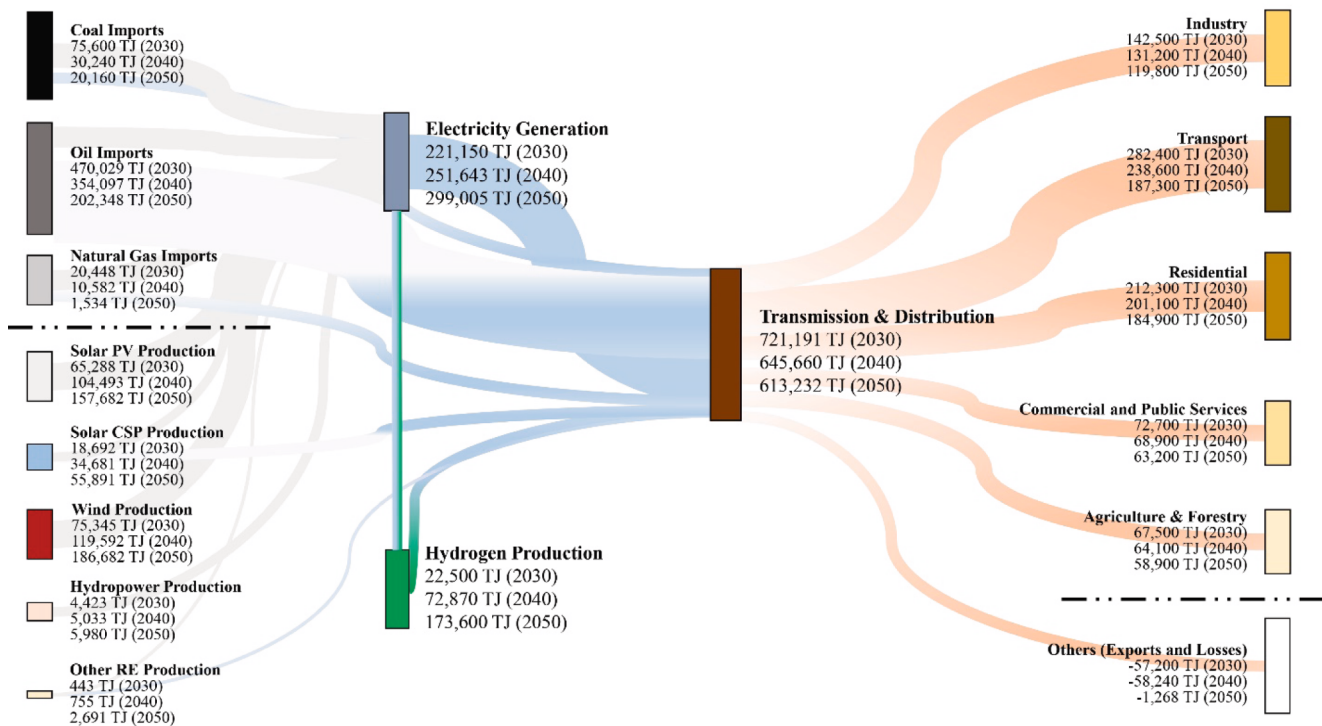


(b) Sankey Diagram - HTCE Scenario

Fig. 11. Projected energy balance and distribution - Sankey diagrams.



(c) Sankey Diagram - NEP-PULP Scenario

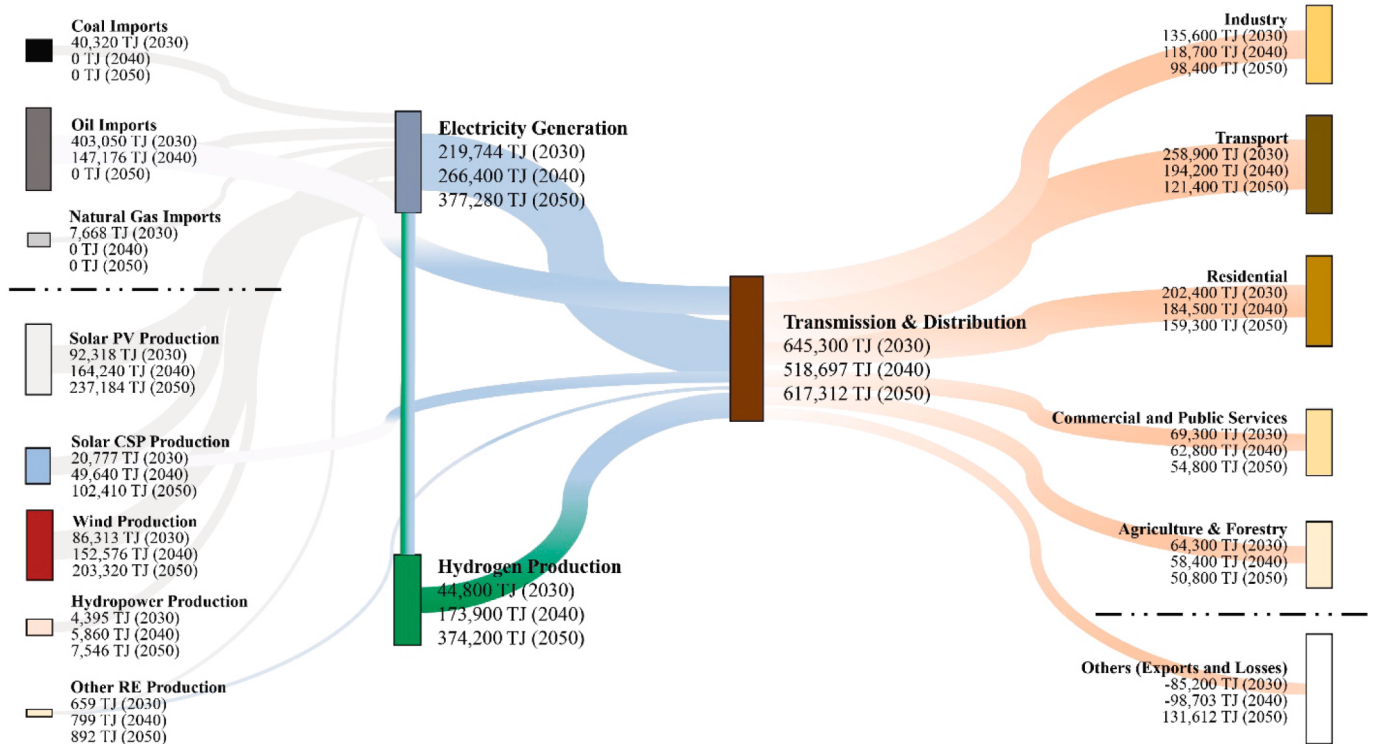


(d) Sankey Diagram - NEP-NEMO Scenario

Fig. 11. (continued).

gradual reduction or complete removal of fossil fuels. Together, these drivers determine the magnitude and composition of imports, domestic electricity generation, renewable dispatch patterns, and the emergence of hydrogen as an additional energy carrier. As discussed earlier, the

transformation of the installed capacity portfolio, especially the rising shares of solar PV, wind, and CSP alongside the phased retirement of coal and oil technologies, directly translates into adjustments in electricity output, fossil fuel import requirements, and renewable supply



(e) Sankey Diagram - NZE Scenario

Fig. 11. (continued).

contributions. As fossil-based generation capacity declines, renewable inputs and hydrogen production progressively reshape the flow architecture depicted in Fig. 11, modifying the relationships among primary supply, sectoral demand, and system losses. The Sankey representations therefore convert the capacity transitions illustrated in Figs. 10a to 10d into an integrated, system wide visualization of optimized energy flows under each policy pathway considered for Morocco.

Under the BAU pathway, the energy balance preserves a fossil intensive configuration throughout 2030–2050, aligning with the limited structural transformation in installed capacity described in Section 3.1. Coal imports rise from 131,040 TJ in 2030 to 141,120 TJ in 2050, while oil imports increase from 590,080 TJ to 653,528 TJ, reinforcing continued dependence on externally sourced fossil fuels. Natural gas imports show moderate variation, declining to 26,208 TJ in 2040 before increasing again to 30,240 TJ by 2050. Renewable generation expands gradually, with the combined output of solar PV, CSP, wind, and hydropower increasing from 81,800 TJ in 2030 to 145,000 TJ in 2050, yet remaining substantially below fossil energy inputs. Electricity production grows from 215,280 TJ to 277,200 TJ, reflecting incremental capacity additions rather than systemic transformation. Hydrogen output remains zero in all years, confirming the absence of electrolysis-based pathways within BAU. Final energy demand continues to be dominated by transport and residential uses, with transport rising from 315,200 to 355,400 TJ and residential consumption increasing from 228,500 to 258,200 TJ between 2030 and 2050. Transmission and distribution flows expand proportionally from 857,835 TJ to 973,147 TJ, indicating persistent inefficiencies associated with limited electrification and sustained reliance on oil-based mobility. Losses and exports reach 16,747 TJ by 2050, reflecting the enlarged scale of the system without meaningful structural change. In summary, BAU represents a fossil locked trajectory characterized by modest renewable growth and limited efficiency improvement, underscoring its misalignment with long term decarbonization objectives.

The HTCE pathway reflects a more advanced yet still incomplete transition. Coal imports decrease markedly from 95,760 TJ in 2030 to 25,200 TJ in 2050, while natural gas imports contract by nearly 75.0% over the same interval. Oil imports remain comparatively elevated, declining from 522,520 TJ to 349,890 TJ, which indicates that fossil fuels continue to meet a considerable portion of final demand, particularly within the transport sector. Renewable generation expands strongly: solar PV rises from 52,195 to 129,377 TJ, CSP from 10,908 to 30,152 TJ, and wind from 57,540 to 129,680 TJ between 2030 and 2050. Electricity output increases from 218,160 TJ to 276,900 TJ in line with expanding solar and wind capacity. Hydrogen production appears for the first time in HTCE, growing from 7,200 TJ in 2030 to 75,850 TJ in 2050, representing roughly a tenfold increase that facilitates decarbonization in industry and heavy transport. Total transmission and distribution flows decline from 771,074 to 661,337 TJ across 2030 to 2050, indicating efficiency improvements and reduced dependence on high loss fossil supply chains. Sectoral demand patterns begin to adjust transport consumption falls from 298,700 to 218,600 TJ, suggesting initial electrification and hydrogen adoption, while residential and commercial demand shows moderate declines linked to efficiency improvements. A distinctive element is the continued negative magnitude reported under “Other Exports and Losses”, which gradually becomes less negative over time, shifting from minus 33,126 TJ in 2030 to minus 8,063 TJ in 2050. Within the Sankey accounting structure, negative Others signifies that imports exceed the sum of exports and system losses, meaning the system functions as a net energy absorber rather than exporter. This outcome aligns with increasing renewable penetration and its coupling with hydrogen production pathways, where surplus renewable electricity during high output periods is redirected toward electrolysis, storage, and reconversion rather than external trade. HTCE therefore represents a substantive but partial decarbonization pathway, characterized by declining fossil dependence, expanding renewables, and scaling hydrogen, while oil remains structurally important.

NEP PULP delivers a significantly deeper transformation of the energy balance, consistent with more ambitious renewable deployment and electrification objectives. Coal imports decline from 40,320 TJ in 2030 to 13,608 TJ in 2050, while oil imports decrease from 478,331 to 246,707 TJ, reflecting accelerated substitution by electricity and hydrogen. Renewable output expands substantially: solar PV increases from 71,986 to 176,571 TJ, CSP from 13,283 to 38,896 TJ, and wind from 64,773 to 152,571 TJ, more than doubling or tripling over the modeling horizon in line with expanded capacity additions. Electricity generation reaches 298,840 TJ by 2050, and hydrogen production grows from 15,472 to 134,400 TJ, underscoring the emergence of hydrogen as a central energy carrier. Transmission and distribution flows decrease steadily from 705,346 to 620,222 TJ, reflecting structural efficiency gains and stronger reliance on electrified end use rather than distributed fossil supply. Transport demand declines from 286,800 to 203,600 TJ, and industrial demand falls from 140,200 to 124,500 TJ, consistent with fuel switching and efficiency measures. The negative magnitude associated with “Other Exports and Losses” narrows considerably, shifting from minus 69,854 TJ in 2030 to 10,478 TJ in 2050, indicating that net imports increasingly align with domestic consumption and reducing imbalances between supply and demand. NEP PULP thus achieves substantial fossil displacement, extensive renewable integration, and strong hydrogen expansion, marking the first fully transitional pathway for Morocco.

NEP-NEMO amplifies the transition dynamics observed in NEP PULP, driven by stronger cost and resource optimization. Coal imports decline from 75,600 TJ to 20,160 TJ and nearly vanish by 2040. Oil imports decrease more sharply than in PULP, falling from 470,029 TJ in 2030 to 202,348 TJ in 2050. Renewable growth is the most pronounced among non NZE pathways: solar PV expands from 65,288 to 157,682 TJ, CSP from 18,692 to 55,891 TJ, and wind from 75,345 to 186,682 TJ, confirming NEP-NEMO as the most renewable intensive NEP configuration as previously illustrated in Fig. 10. Electricity production increases to 299,005 TJ in 2050, and hydrogen reaches 173,600 TJ, the second highest value among all scenarios. Final demand continues to moderate through electrification and efficiency gains, with transport declining from 282,400 to 187,300 TJ and industry from 142,500 to 119,800 TJ between 2030 and 2050. Transmission flows decrease to 613,232 TJ, and the system approaches near equilibrium by 2050, as “Other Exports” and Losses improve from 57,210 TJ in 2030 to 1,268 TJ in 2050. This marked reduction in imbalance indicates close alignment between domestic supply and final demand. NEP-NEMO therefore represents a highly optimized transition pathway characterized by rapid fossil reduction, elevated renewable contribution, and strong hydrogen growth, yielding a tightly integrated energy structure. In comparison with intermediate options, NEP-NEMO surpasses NEP-PULP primarily through higher wind and CSP integration for hydrogen production, accelerated fossil displacement, and more balanced renewable allocation, reflecting efficiency gains from optimization rather than differences in overarching policy ambition alone.

The NZE pathway produces a comprehensive restructuring of the national energy balance, removing fossil fuels entirely by 2050. By 2040, coal and natural gas imports decline to 0 TJ, while oil imports decrease to 147,176 TJ before reaching 0 TJ in 2050. Renewable generation expands sharply: solar PV rises from 92,318 to 237,184 TJ, CSP from 20,777 to 102,410 TJ, and wind from 86,313 to 203,320 TJ. Hydropower and Other renewable sources remain comparatively modest yet stable. Electricity output increases from 219,744 TJ in 2030 to 377,280 TJ in 2050, reflecting the extensive renewable deployment identified earlier. Hydrogen becomes the defining component of NZE, growing from 44,800 TJ in 2030 to 347,200 TJ in 2050, the largest volume across all scenarios, driven by deep decarbonization requirements in industry, freight transport, and export-oriented markets. Transmission and distribution flows decline until 2040 but rise again to 617,312 TJ by 2050 due to the large-scale hydrogen system, which expands energy circulation through electrolysis and synthetic fuel

production. Final demand contracts significantly and becomes more electrified: transport decreases from 258,900 to 121,400 TJ, industry from 135,600 to 98,400 TJ, and residential demand from 202,400 to 159,300 TJ, reflecting substantial efficiency improvements. NZE therefore establishes a fully decarbonized energy configuration reinforced by hydrogen, with structural surplus capacity available for storage or export.

Across all modeled pathways, the energy balance shifts steadily toward greater electrification and reduced fossil reliance as the level of policy ambition intensifies. BAU preserves a fossil dominated structure, whereas HTCE incorporates moderate electrification together with initial hydrogen expansion. NEP PULP and NEP NEMO generate substantially reconfigured systems led by renewable electricity and synthetic fuels, with NEMO approaching an almost optimal internal balance. NZE establishes a fully decarbonized configuration characterized by considerable renewable electricity surpluses and the highest hydrogen volumes. These systemic changes carry direct consequences for land demand: pathways with elevated renewable penetration and large scale hydrogen production, namely NEP PULP, NEP NEMO, and NZE, necessitate extensive areas for solar PV, CSP, wind installations, and electrolysis infrastructure, while most fossil facilities are already in place and involve limited additional land allocation. Consequently, the territorial footprint of the energy transition in Morocco scales in proportion to renewable expansion and hydrogen growth, creating a clear connection between energy balance results and strategic land use planning.

Land requirements

Fig. 12 illustrates the territorial area required for renewable energy deployment and hydrogen production facilities under each scenario, demonstrating how the magnitude and intensity of the energy transition are reflected in physical land use. The general pattern indicates that land demand rises progressively with higher levels of policy ambition, deeper electrification, and greater hydrogen integration, confirming the direct relationship between installed renewable capacity, energy system configuration, and the spatial consequences of large-scale clean energy expansion. In every scenario, solar PV constitutes the largest share of land occupation, followed by wind and CSP, whereas hydropower and other renewable sources retain a comparatively limited spatial presence due to restricted expansion capacity. Hydrogen related facilities emerge as a significant contributor to land demand only in scenarios that strongly emphasize synthetic fuel development including HTCE, NEP PULP, NEP NEMO, and NZE, indicating that advanced decarbonization pathways depend not only on techno economic performance but also on effective spatial planning and sustainable land management considerations.

In the BAU pathway, land demand remains relatively limited and increases gradually over time, reflecting the modest growth of renewable technologies and the persistent reliance on fossil energy supply. Solar PV area expands from 6,458 ha in 2030 to 12,751 ha in 2050, consistent with the restrained capacity additions observed in the generation portfolio. Wind surface requirements grow from 1,550 to 2,661 ha, while CSP rises only slightly from 1,292 to 1,663 ha. Hydropower and other renewable sources remain nearly unchanged, and hydrogen related land demand stays at 0 ha, in line with the absence of Power to X integration under BAU. This pathway therefore represents a low transformation trajectory in which pressures on land systems remain limited, but fossil dependence continues.

The HTCE pathway accelerates renewable expansion and introduces a moderate hydrogen component, leading to a pronounced increase in land demand. Solar PV area more than doubles from 10,439 ha in 2030 to 25,875 ha in 2050, while CSP nearly triples from 2,182 to 6,030 ha. Wind area requirements rise from 2,302 to 5,187 ha, reflecting the faster displacement of fossil generation. A significant change is the appearance of hydrogen infrastructure, which expands from 720 ha in 2030 to 7,585

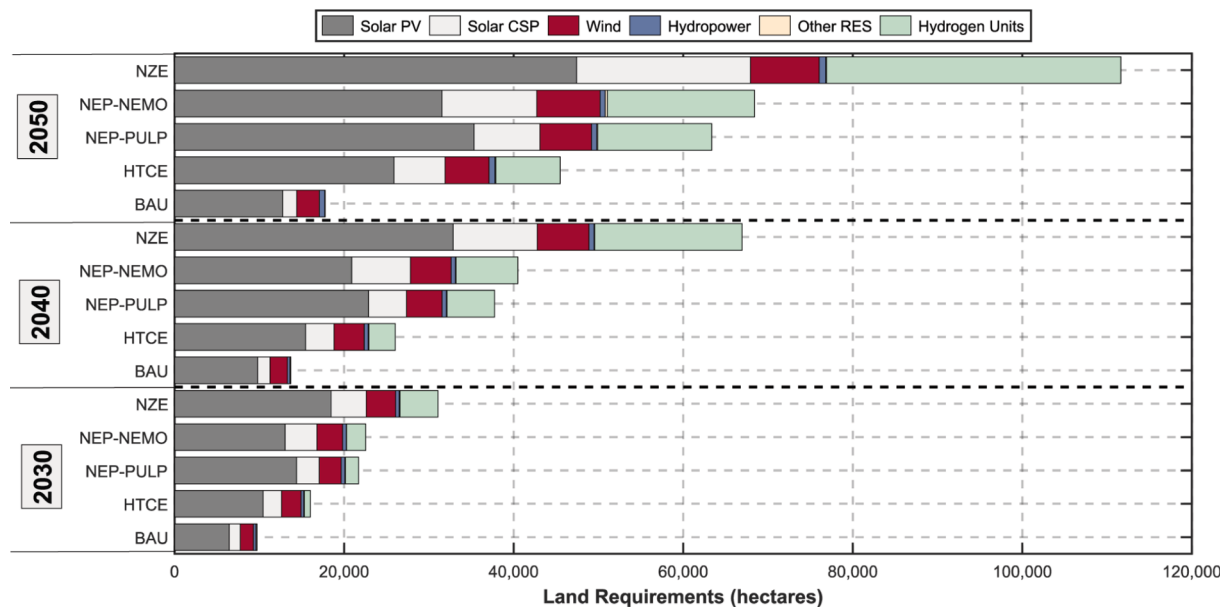


Fig. 12. Projected land requirements for renewable energy deployment and green hydrogen generation.

ha in 2050, utilizing part of the surplus renewable electricity described in Section 3.2. This pattern shows that even intermediate decarbonization approaches substantially increase spatial planning requirements and highlight the importance of zoning mechanisms capable of hosting hydrogen production clusters.

The NEP PULP pathway intensifies these trends through stronger policy alignment, producing markedly higher land requirements across renewable categories. Solar PV increases from 14,397 ha in 2030 to 35,314 ha in 2050, supported by rapid electrification and expanding hydrogen demand. CSP follows a comparable trajectory, rising from 2,657 to 7,779 ha, while wind area grows from 2,591 to 6,103 ha. Hydrogen facilities become a major spatial component, expanding from 1,547 ha to 13,440 ha, consistent with the elevated hydrogen production levels reported in Section 3.2. The scale of land expansion reflects the territorial dimension of NEP aligned energy diversification and emphasizes the need for coordinated spatial planning to address potential land use competition.

The NEP NEMO pathway demonstrates even higher land requirements than NEP PULP, consistent with its more assertive optimization-based deployment strategy. Solar PV expands from 13,058 to 31,536 ha, while CSP increases significantly from 3,738 to 11,178 ha, representing the largest CSP land expansion among all pathways. Wind area rises from 3,014 to 7,467 ha, mirroring its expanded contribution to both electricity generation and hydrogen supply. Hydrogen infrastructure becomes the dominant land component by 2050, growing from 2,250 ha in 2030 to 17,360 ha, the highest among all non NZE pathways. These figures illustrate that an optimization-oriented strategy enhances resource utilization efficiency through high capacity factors and steady hydrogen output, yet at the same time intensifies spatial demand, positioning NEP NEMO as a pathway with considerable land planning implications.

The NZE pathway yields the most land intensive configuration, corresponding to the large scale renewable and hydrogen deployment required for full decarbonization. Solar PV land use increases from 18,464 ha in 2030 to 47,437 ha in 2050, while CSP expands from 4,155 to 20,482 ha, representing the largest CSP footprint among all scenarios. Wind requirements grow from 3,453 to 8,133 ha, consistent with its balancing role within a PV driven hydrogen system. Hydrogen infrastructure becomes the most land demanding element of NZE, expanding from 4,480 ha to 34,720 ha, reflecting very high hydrogen production levels and the transition toward positive net energy exports.

Collectively, these land requirements demonstrate the spatial consequences of a fully renewable and hydrogen based national energy configuration, with NZE emerging as a pathway in which land availability becomes a critical enabling condition for achieving climate neutrality in Morocco.

Overall, land demand across all pathways shows a clear relationship between decarbonization ambition, renewable penetration, hydrogen integration, and spatial extent. While BAU remains comparatively limited in land use, more ambitious pathways, particularly NEP PULP, NEP NEMO, and NZE, generate substantial spatial requirements, with hydrogen production hubs becoming one of the most land intensive components of future energy systems. These results highlight the importance of integrated land use strategies, identification of appropriate renewable development zones, and forward-looking policy instruments to manage environmental, agricultural, and societal tradeoffs.

Energy scenarios evaluation

Fig. 13 synthesizes the multi-criteria performance of all scenarios across 2030–2050, integrating economic, environmental, and energy-system indicators into a unified evaluation heatmap. The results reveal clear divergences in long-term pathway quality, demonstrating that scenarios with higher upfront system costs often achieve superior sustainability outcomes, while low-cost pathways tend to lock in fossil dependence. Critically, the heatmap demonstrates a clear trade-off frontier:

- Lower-cost pathways (BAU, HTCE) deliver weak sustainability outcomes;
- Intermediate-cost pathways (NEP-PULP, NEP-NEMO) achieve significant decarbonization at moderate expense;
- High-cost pathways (NZE) produce transformative system-wide benefits and eliminate long-term climate risks.

Across all years, BAU consistently scores the lowest, with final composite scores decreasing from 42/100 in 2030 to 35/100 in 2050. This scenario retains the lowest levelized cost of energy (26–28 MAD/GJ), but this economic advantage is counterbalanced by persistently high fossil dependence (85.0–87.0%) and emission factors remaining above 70 tCO_{2e}/TJ. These outcomes demonstrate that small gains in system cost cannot offset the environmental and strategic risks

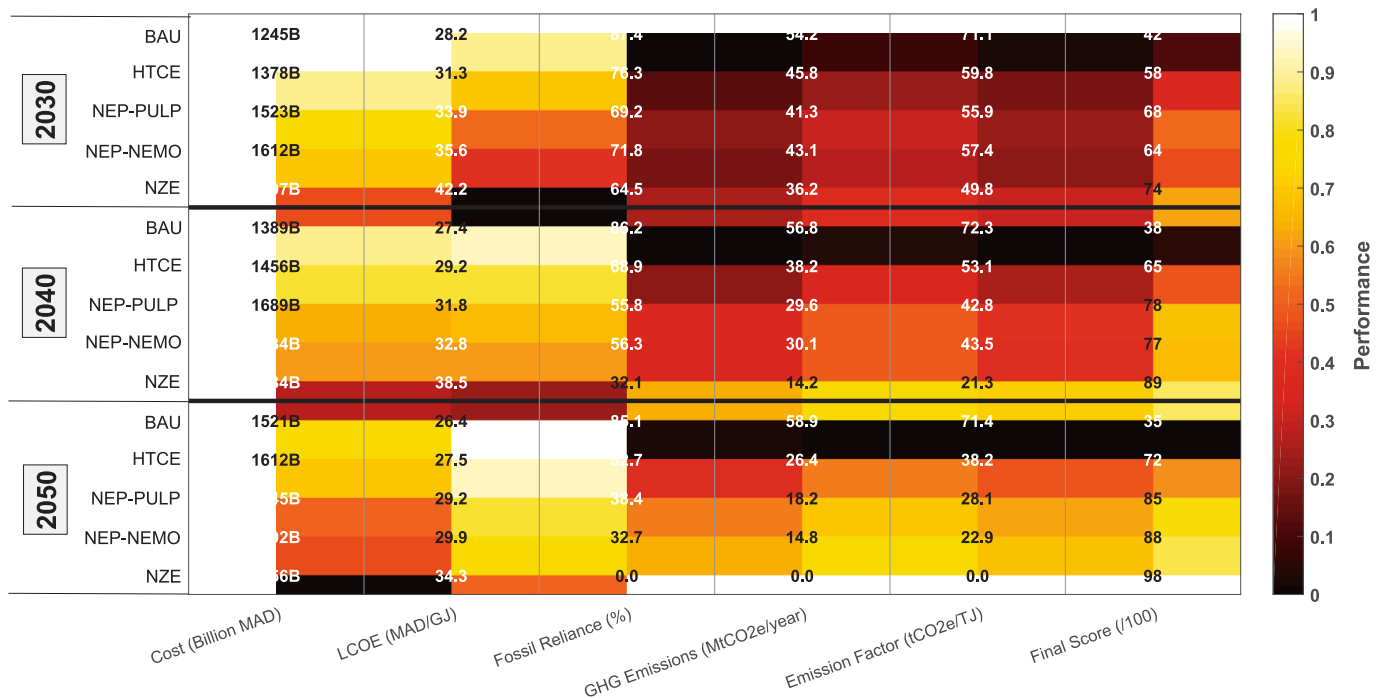


Fig. 13. Heatmap evaluation of energy scenario for future energy planning of Morocco.

associated with delayed decarbonization, confirming BAU's misalignment with long-term climate commitments of Morocco.

HTCE introduces accelerated deployment of clean energy technologies and produces a more balanced profile. While system costs remain comparable to BAU (1,378–1,612 billion MAD Net Present Value), fossil dependence declines sharply from 76.0% in 2030 to 53.0% by 2050, and GHG emissions are reduced by nearly 50.0% over the same horizon. As a result, the composite performance score rises from 58/100 in 2030 to 72/100 in 2050, indicating that moderate technological progress combined with incremental decarbonization yields tangible improvements, though still insufficient to achieve deep-transition objectives.

NEP-PULP shows stronger system transformation through policy-driven electrification and renewable expansion. Fossil dependence drops from 69.0% in 2030 to 38.0% by 2050, while GHG emissions fall to 18.2 MtCO_{2e}/year. These changes raise the scenario's final score to 85/100 in 2050. However, this pathway incurs higher net present costs (up to 1,845 billion MAD) and increasing levelized cost of energy, reflecting the capital intensity of rapid deployment of renewables and hydrogen infrastructure.

NEP-NEMO achieves an even more optimal transition profile. Despite similar NPV costs to NEP-PULP, it outperforms across all environmental metrics: fossil dependence decreases to 32.7% by 2050 and the emission factor declines to 22.9 tCO_{2e}/TJ. Its performance scores increase steadily from 64 in 2030 to 88 in 2050, positioning NEP-NEMO as a high-efficiency, high-impact decarbonization pathway. Superior performance relative to NEP-PULP demonstrates that diversified generation, flexible operation, and optimized electrification scheduling produce cleaner energy outcomes without additional economic burden.

NZE represents the only pathway achieving full decarbonization by 2050, with fossil reliance and GHG emissions reduced to zero. While it incurs the highest economic cost (total investment reaching 2,456 billion MAD and cost of energy exceeding 34 MAD/GJ), this corresponds to roughly 3.1% of projected 2050 GDP ($\approx$ 79,000 billion MAD) [6], representing a substantial yet feasible share of national energy expenditure within the fiscal context.

Overall, scenario evaluation indicates that the strategic decision is not a trade-off between economy and environment but between low-cost stagnation and high-value transformation. The most resilient long-term

pathways (NEP-NEMO and NZE) demonstrate that optimized electrification, expansion of renewable generation, and large-scale hydrogen integration are central to developing a robust and sustainable Moroccan energy system.

Results takeaways and discussion

The multi-scenario optimization produces a consistent and quantitatively detailed overview of energy transition pathways in Morocco. Scenarios with progressively higher policy ambition (BAU $\rightarrow$ HTCE $\rightarrow$ NEP-PULP/NEP-NEMO $\rightarrow$ NZE) show sequential improvements in decarbonization, renewable penetration, and hydrogen deployment, accompanied by increasing system investment and levelized cost of energy. Technically, the installed-capacity and generation outcomes depict a transition from a fossil-dominated electricity system under BAU, where coal and oil remain significant through 2050, to renewable- and Power-to-X-dominated systems in NEP and NZE, with renewable shares exceeding 80.0% in NEP scenarios and substantial combined renewable plus hydrogen output in NZE. Energy balances mirror these structural shifts: BAU maintains high fossil imports (coal $\sim$ 131,000 $\rightarrow$ 141,000 TJ; oil $\sim$ 590,000 $\rightarrow$ 654,000 TJ 2030 $\rightarrow$ 2050) with negligible hydrogen, whereas NEP and NZE pathways replace imports with domestic PV, CSP, and wind generation alongside rapidly scaling electrolysis (hydrogen reaching 134,000–173,000 TJ in NEP and $\sim$ 347,000 TJ in NZE by 2050). Land-use analysis underscores the magnitude of physical transition: solar PV footprint rises from a few thousand hectares in BAU to tens of thousands in NEP and NZE (e.g., NZE PV $\approx$ 47,437 ha in 2050), and hydrogen units emerge as a major land category (up to 34,720 ha in NZE). Multi-criteria evaluation quantifies these trade-offs: net present value and levelized cost of energy increase with ambition (up to 2,456 MAD and 34.3 MAD/GJ in 2050 for NZE), while environmental metrics improve substantially, with fossil reliance falling to zero and composite score reaching 98/100 for NZE in 2050. Although carbon reductions dominate, non-carbon impacts, such as ecosystem disturbance from large PV and wind installations, CSP social acceptance, and potential land-use conflicts, are discussed qualitatively. Future research should incorporate quantitative measures for biodiversity, soil degradation, and community engagement to broaden sustainability

assessment. Overall, modeling produces internally consistent trajectories across capacity, energy flows, spatial footprints, and economic and environmental performance.

Detailed cross-checks confirm strong internal consistency and technical validation across model outputs. Capacity expansion directly drives energy balance outcomes: increases in PV, wind, and CSP capacities lead to higher electricity production and hydrogen flows while reducing fossil imports, reflected in Sankey balances and hydrogen time-series (e.g., NEP-NEMO's higher wind and CSP shares explain 173,600 TJ hydrogen output by 2050). Land metrics align predictably with technology-specific ha/MW coefficients, with scenarios featuring larger PV and CSP expansions showing correspondingly higher land requirements. Economic indicators support the physical plans: capital-intensive, low-carbon pathways (NEP and NZE) result in higher net present value and cost of energy because the model includes CAPEX, storage, electrolyzer costs, and long-term system flexibility, but these higher costs are offset by drastic reductions in fossil reliance and emissions (emission factors fall from ~ 71 tCO_{2e}/GJ in BAU to near 0 in NZE by 2050). Temporal trends are coherent: transitional scenarios (HTCE) show intermediate reductions in fossil imports and moderate hydrogen growth, while optimization-driven scenarios (NEP-PULP, NEP-NEMO) exhibit more aggressive restructuring and improved system balance, evidenced by decreasing net import/loss "Others" and near-zero imbalance for NEP-NEMO around 2050. Sensitivity analysis is consistent: cost-minimization under policy constraints (NEP-PULP) produces different technology mixes than dispatch-aware optimization (NEP-NEMO), which favors resources that enhance firm capacity and stable hydrogen production (hence higher CSP and wind shares). Overall, cross-consistency among capacity, dispatch, flows, land footprint, and multi-criteria scoring demonstrates robust internal validation, with results emerging logically from techno-economic assumptions (degradation, capacity factors, learning rates, electrolyzer constraints, transmission and distribution losses) and scenario boundaries (renewable targets, fossil phase-down). Trade-offs (higher cost vs. near-zero emissions) are explicitly quantified, enabling transparent policy evaluation.

The findings have direct policy relevance. First, achieving mid-century decarbonization efficiently and securely requires a phased approach combining (i) accelerated utility-scale PV and wind procurement to capture low-marginal-cost energy (annual additions aligned with NEP trajectories, cumulatively tens of GW by 2050, corresponding to PV land of $\sim 30,000$ – $50,000$ ha and wind area $\sim 7,000$ – $8,000$ ha in the most ambitious scenarios), (ii) investment in dispatchable CSP and long-duration storage to reduce curtailment and support electrolytic hydrogen (CSP area up to nearly 20,000 ha in the deepest decarbonization pathway), and (iii) national electrolyzer deployment sized to hydrogen demand (capacity sufficient for 134,000–347,000 TJ of green hydrogen by 2050 across NEP→NZE paths, with hydrogen-unit land of 13,000–35,000 ha). Second, economic trade-offs highlight the role of targeted public finance and blended instruments: NZE's higher net present value (2,456 MAD) and levelized cost of energy (34.3 MAD/GJ) can be mitigated through guarantees, concessional finance, and risk-sharing for large electrolyzer and storage projects, potentially lowering effective discount rates to $\sim 4.0\%$ and improving cost competitiveness. Third, spatial governance must become a national priority: NEP/NZE land footprints require proactive designation of

renewable energy zones, environmental screening, and frameworks balancing agricultural and energy land use to prevent conflicts. Fourth, system operation and market reforms are essential—dynamic dispatch markets, flexible demand programs, and hydrogen offtake agreements are needed to convert surplus renewable electricity into tradable hydrogen products and realize reductions in "Others" imbalance. Finally, an adaptive policy pathway combining near-term NEP actions (rapid renewable scale-up and early electrolyzer deployment) with a credible NZE commitment (to mobilize capital and signal long-term hydrogen demand) optimizes affordability, energy security, and climate goals. Quantitatively, this implies accepting moderately higher near-term net present value and levelized cost of energy to avoid long-term fossil lock-in, achieving reductions in fossil reliance from $\sim 85.0\%$ (BAU) to 0.0% (NZE) and emission factor declines from 71 tCO_{2e}/TJ to zero by 2050.

The multi-criteria evaluation of the modeled scenarios is summarized qualitatively in Table 4, providing a consolidated view of their relative strengths and weaknesses across energy demand, economic efficiency, environmental impact, land use requirements, and final composite performance scores. As shown, the BAU scenario exhibits low energy demand growth and high economic efficiency due to minimal upfront investments but performs very poorly in environmental impact and maintains a low spatial footprint, resulting in an overall low final score. The HTCE scenario represents a moderate transition pathway, balancing incremental decarbonization with manageable land requirements, achieving medium scores across all criteria. In contrast, the NEP-PULP and NEP-NEMO scenarios reflect aggressive policy-driven and optimization-based decarbonization strategies. Both achieve high environmental performance, with NEP-NEMO slightly outperforming NEP-PULP due to superior system flexibility and hydrogen integration, yet they impose substantial land demands and higher system costs. The NZE scenario represents the most ambitious pathway, achieving near-complete decarbonization and maximizing environmental benefits, but at the expense of the highest energy demand and economic costs, as well as very large land-use requirements due to extensive deployment of PV, CSP, wind, and hydrogen infrastructure. These modeling outcomes translate into concrete policy guidance: (i) deployment timelines suggest achieving 50.0% of NEP-NEMO renewable capacity by 2040 to secure system integration and avoid bottlenecks; (ii) regulatory priorities include incentivizing flexible dispatchable resources and grid-scale storage to manage surplus renewable energy; (iii) investment sequencing should first target PV and wind scale-up, followed by CSP and electrolyzer deployment aligned with hydrogen demand trajectories, ensuring cost-effective decarbonization while preserving system reliability.

Conclusion and perspectives

The introductory context established that the energy transition in Morocco sits at the intersection of urgent national challenges and emerging global opportunities, as documented extensively in the literature on decarbonization strategies, renewable integration, and energy security in resource-constrained economies. Existing studies converge on the structural dependence of Morocco on imported fossil fuels, historically above 90.0% of total supply, and the resulting exposure to macroeconomic volatility, high trade deficits, and escalating GHG

Table 4
Qualitative evaluation of energy scenarios for Morocco (2030–2050).

Scenario	Electricity Demand	Economic Efficiency	Environmental Impact	Land Requirements	Final Score
BAU	Low	Medium-High	Very High	Low	Low
HTCE	Medium	Medium	Medium	Medium	Medium
NEP-PULP	Medium-High	Medium-Low	Medium-Low	High	High
NEP-NEMO	Medium-High	Medium-Low	Medium-Low	Very High	Very High
NZE	High	Low	Very Low	Very High	Very High

emissions. Parallel works on renewable energy expansion highlight the country's exceptional technical potential, particularly for solar and wind, supported by decade-long trends of declining levelized cost of energy and large-scale rollouts under the national energy strategy. However, despite these contributions, the literature remains fragmented in its analytical coverage: most prior assessments rely on deterministic projections, partial-equilibrium modeling, or single-scenario explorations that do not fully capture the complex trade-offs between energy security, long-term system cost, land requirements, and decarbonization imperatives. This gap is particularly pronounced in the absence of integrated, multi-scenario, optimization-based frameworks capable of simultaneously addressing techno-economic performance, spatial constraints, and policy-driven decarbonization targets within a unified modeling environment. The present research addresses this gap by introducing a novel, holistic optimization approach that couples system cost minimization, emissions reduction trajectories, and land-use feasibility across multiple policy and technology futures. By embedding uncertainty, sectoral interdependencies, and renewable-hydrogen synergies into a dynamic scenario architecture, the study advances the methodological landscape of Moroccan energy planning and provides a level of analytical rigor not previously achieved in national or regional assessments. While the modeling demonstrates technical feasibility within the assumptions and data constraints applied, it does not imply that full-scale implementation is immediately practical; real-world deployment will require overcoming regulatory, social, and logistical hurdles.

The methodological framework underpinning this study integrates a rigorous, multi-layered approach that combines data-driven modeling, scenario design, and mixed-integer optimization to evaluate the long-term energy transition pathways of Morocco. Historical energy demand, supply, techno-economic, and cost datasets (2000–2022) were systematically compiled and standardized using international energy-system standards to ensure consistency across modeling environments. These inputs were operationalized within a hybrid LEAP-NEMO architecture, where LEAP managed narrative scenario formulation, energy accounting, and emission baselines, while NEMO executed multi-period MILP optimization to determine least-cost and least-emission configurations under evolving policy and technology constraints. The scenarios, ranging from Business-as-Usual to Net Zero Emissions, were constructed to capture divergent policy ambitions, renewable adoption trajectories, and electrification strategies, allowing comprehensive exploration of cost, reliability, land requirements, and decarbonization outcomes. Through iterative LEAP-NEMO exchanges and the use of auxiliary analytical tools (Excel and MATLAB) for preprocessing and post-processing, the framework ensured a transparent, reproducible, and technically robust assessment of the future energy mix for Morocco under multiple strategic futures. Although the framework is calibrated specifically to the Moroccan energy system, the underlying methodology, scenario construction, and optimization principles are transferable to other countries with similar renewable energy potential and fossil import dependence, provided contextual adjustments are applied.

The results demonstrate a sharply differentiated evolution of Morocco's energy system across scenarios, with quantitative shifts in capacity expansion, energy flows, land occupation, and multi-criteria performance. Installed capacity trajectories show that BAU reaches only 18–20 GW of renewables by 2050, whereas HTCE and NEP-PULP surpass 35–45 GW, and NEP-NEMO and NZE exceed 55 GW and 70 GW, respectively, driven by large-scale solar PV additions (up to 47,437 ha in NZE by 2050) and the rapid integration of wind resources (up to 8,133 ha). Energy balance results further highlight structural divergence: fossil fuel reliance remains above 85.0% in BAU by 2050, decreases to 53.0% in HTCE, and falls below 40.0% in NEP-PULP and 33.0% in NEP-NEMO, before reaching 0.0% in NZE. Hydrogen deployment emerges as a key differentiator is absent in BAU but expanding to 7,585 ha in HTCE, 13,440 ha in NEP-PULP, 17,360 ha in NEP-NEMO, and 34,720 ha in NZE by 2050, reflecting rising excess-renewable

conversion and export potential. Multi-criteria scenario evaluation confirms the same hierarchy: BAU's final score deteriorates to 35/100 by 2050, whereas HTCE improves to 72/100, NEP-PULP and NEP-NEMO reach 85/100 and 88/100, respectively, and NZE attains 98/100, driven by a complete elimination of the 58.9 MtCO_{2e} emitted in BAU and the reduction of the system-wide emission factor from 71 tCO_{2e}/GJ to nearly zero. While carbon emissions reductions dominate the environmental assessment, non-carbon impacts (such as ecosystem disturbance from large-scale PV and wind deployment, social acceptance of CSP plants, and potential land-use conflicts) are treated qualitatively. Future work should incorporate quantitative indicators for biodiversity impact, soil degradation, and community engagement to enhance comprehensiveness of sustainability evaluation.

Despite its comprehensive scope, this study is subject to several methodological and data-related limitations that warrant careful interpretation of the results. First, the techno-economic and resource data are subject to uncertainty, with CAPEX variations of $\pm 15.0\%$, electrolyzer efficiency variability from 65.0–75.0%, and PV/CSP capacity factors fluctuating $\pm 10.0\%$ due to climate variability. Second, institutional and regulatory constraints, including permitting delays, grid-access limitations, and sectoral coordination challenges, could extend project timelines by 3–7 years relative to model assumptions. Third, the spatial land allocation, while informed by ha/MW coefficients, may face conflicts with protected areas, urban expansion, or agricultural zones, potentially requiring 10.0–15.0% of modeled land to be substituted or relocated. Fourth, operationally, the LEAP-NEMO framework approximates hourly renewable variability, meaning high-penetration scenarios like NEP-NEMO and NZE may require additional reserve margins (5.0–15.0% of peak load) in real-world implementation. Collectively, these limitations underscore that while the study confirms technical feasibility and optimal pathways on a model basis, actual deployment will be conditioned by data precision, institutional readiness, and on-the-ground logistical constraints. These caveats also justify the paper's length and technical density: a rich dataset, multiple scenario comparisons, and detailed techno-economic-environmental analyses are necessary to convey the full scope of findings while maintaining transparency and scientific rigor.

Future research can build upon this work in several directions to expand analytical resolution and strengthen policy applicability. Integrating monthly or yearly modeling into the LEAP-NEMO workflow would allow explicit treatment of intermittency, ramping constraints, and flexibility requirements, especially important for optimistic demanding scenarios. A second pathway is the development of a spatially explicit energy-land-grid co-optimization model that incorporates real geospatial layers (topography, protected areas, grid corridors, urban expansion), enabling a more realistic assessment of siting potential and transmission expansion needs. Future studies could also embed hydrogen value-chain modeling, including electrolyzer clustering, pipeline routing, storage typologies, and export logistics, to evaluate the techno-economic viability of Morocco's emerging role as a green hydrogen exporter under NZE and NEP-NEMO conditions. Additionally, linking the energy system optimization with macroeconomic or general-equilibrium models would allow quantification of employment, GDP, and trade impacts, particularly relevant for scenarios that reduce fossil imports by 100.0% and generate positive energy exports by 2050. Finally, future work should include a detailed timeline and budgetary analysis for scenario implementation, quantifying expected capital requirements, operational expenditures, and policy-driven financing needs to guide realistic rollout planning. This would allow the strategic assessment of each scenario's cost-benefit performance, as demonstrated in comparable studies on green electricity transitions in Libya [29]. Finally, expanding the analysis to include stochastic optimization, climate-risk assessment (solar irradiation variability or wind resource shifts), and cross-border energy integration with the EU and West Africa would provide deeper insights into long-term system resilience and strategic positioning of Morocco in regional energy markets. These

extensions would not only refine the robustness of the current findings but also establish a next-generation research agenda for strategic energy planning in Morocco.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Writefull and Grammarly in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRedit authorship contribution statement

Hamza El Hafdaoui: Writing – original draft, Software, Resources, Investigation, Formal analysis, Data curation. **Mohammed Abbas Hattati:** Software, Methodology, Conceptualization. **Nouriddine Houran:** Writing – review & editing, Visualization, Validation. **Oumaima Mahir:** Writing – review & editing, Visualization, Validation. **Abdelilah Rochd:** Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data generated or analyzed during this study are included in this published article.

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